

RHYME

Algorithm Specifications And Supporting Documentation

Zhongxiang Zheng, Anyu Wang, Chunhuan Zhao, Guangwu Xu,
Sibo Feng, Zhichen Yan, Yuhang Sun, Zhengtao Jiang, Congming
Wei, Shuang Sun, Chen Jia, Shuaigang Li

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1 Introduction

Rhyme is a lattice-based digital signature that follows the Fiat–Shamir framework and enjoys (s)EU-CMA security. To achieve security, current Fiat–Shamir lattice-based signature schemes commonly rely on techniques such as rejection sampling and convolution theorems to make the statistical distribution of signatures indistinguishable from a fixed distribution. The key difference in the Rhyme scheme is that it replaces these approaches with *Convolution of Constant Count Sampling* (3C Sampling), a deterministic linear transform derived from the secret key that acts on a *constant* number of independent discrete Gaussian noises. Crucially, the number of noise components is independent of the security parameter, and no floating-point sampling from a varying covariance matrix is ever required. The remaining signature distribution is then made publicly simulatable by publishing a Gram matrix (or a submatrix thereof) as auxiliary information. In this way, Rhyme simultaneously achieves compact bandwidth, structural simplicity, and implementation friendliness. The security of Rhyme rests on the *Gram-Aided Module-LWE* (GA-MLWE) assumption, a new computational problem that enriches the standard Module-LWE formulation by additionally providing an adversary with a Gram matrix (or a submatrix thereof) of a short unimodular matrix whose first column encodes the MLWE secret. Informally, GA-MLWE asserts that an MLWE sample remains pseudorandom even when such auxiliary information is available. This auxiliary information directly enables the simulator to reproduce the correct signature distribution without knowing the secret key, closing the zero-knowledge gap that 3C Sampling introduces. The algebraic structure of the Gram matrix makes GA-MLWE closely related to the module Lattice Isomorphism Problem (module-LIP) studied in Hawk, yet it is a strictly weaker assumption because only partial Gram information is revealed. This combination of a novel sampling paradigm, a flexible design space, and a rigorous security foundation distinguishes Rhyme from prior lattice-based signatures.

We provide a full security proof in the (quantum) random oracle model and present concrete parameter sets for 128-, 256-, 384-, and 512-bit classical security that meet the requirements of NGCC [32]. A reference implementation demonstrates that Rhyme achieves a combined public-key and signature size 30%–40% smaller than Dilithium at equivalent security levels. In summary, Rhyme offers several advantages:

- **Compactness:** By leveraging 3C sampling, Rhyme achieves a significantly smaller combined public key and signature size (bandwidth) compared to prior Fiat-Shamir schemes, as demonstrated in Table 1.
- **Simplicity:** Signing in Rhyme is simply a deterministic linear transform of independent discrete Gaussian noises. The Gaussian noise ensures that

the signature distribution is fully captured by the Gram matrix of the secret. This means honest-verifier zero-knowledge can be realized by merely publishing the Gram matrix (or a submatrix thereof) as auxiliary public information, without rejection sampling or convolution theorems. Conversely, recovering the secret from the public key and the Gram matrix is provably as hard as the *Gram-Aided Module-LWE* problem, a natural extension of Module-LWE. This clean separation between simulation (via the Gram matrix) and security (via GA-MLWE) makes the scheme transparent, modular, and easy to implement.

- **Implementation Friendly:** Many lattice-based signature with compact size needs to use floating-point sampling, which is hard to be realized securely. Our scheme relies solely on integer arithmetic over bounded domains, similar to Dilithium which simplifies both software and hardware implementation.

1.1 Design Rationale

The Rhyme signature scheme is designed with three primary goals in mind: structural simplicity, bandwidth efficiency, and implementation friendliness. Despite the considerable number of lattice-based digital signature schemes proposed in recent years, existing designs consistently exhibit a fundamental trade-off: schemes that achieve low bandwidth tend to rely on complex structures that complicate implementation, such as high-dimensional floating-point Gaussian sampling or hyperball sampling over intricate domains [13, 17, 24, 35]; conversely, structurally simple schemes often pay the price of inflated signatures [33]. Hawk, a signature scheme based on the Lattice Isomorphism Problem (LIP) [20], stands out as a notable exception that successfully reconciles implementation friendliness with low bandwidth. However, its security rests on the smLIP and omSVP assumptions—the latter, in particular, is a relatively new hardness conjecture that still warrants further cryptanalytic scrutiny.

Motivated by this landscape, we designed the Rhyme signature scheme to occupy a favourable position in the design space. Rhyme features a structurally simple algorithm that requires no high-dimensional floating-point sampling whatsoever, yet achieves compact bandwidth comparable to the most efficient lattice-based schemes. On the security front, Rhyme builds upon the well-studied MLWE and MSIS problems, and supplements them with a module-LIP component. Crucially, whereas Hawk instantiates LIP at rank 2, we adopt a more conservative design strategy by requiring the module-LIP instances in our GA-MLWE assumption to have rank at least 4. This choice substantially mitigates the risk of algebraic attacks that could exploit low-rank structures, and places our security assumptions on a more robust footing.

1.2 Feature Statements

- **Security foundation.** Rhyme relies on the hardness of the MLWE and MSIS problems, two of the most extensively studied lattice assumptions, together with a module-LIP component that has been used in Hawk and related designs. The resulting GA-MLWE assumption inherits the cryptanalytic confidence of both families, while the conservative choice of $\text{rank} \geq 4$ for the underlying LIP instance provides an additional safety margin against structural attacks.
- **Compactness.** By leveraging the key-independent property of 3C Sampling, supplemented only by a lightweight coefficient-wise rejection step, Rhyme achieves a small combined public key and signature size (bandwidth) among known Fiat-Shamir schemes, as demonstrated in Table 1 (where FS and HtS denote Fiat-Shamir and Hash-then-Sign schemes, respectively).
- **Simplicity and efficiency.** Unlike schemes that require high-precision Gaussian or hyperball sampling (e.g., G+G, HAETAE), Rhyme relies solely on integer arithmetic over bounded domains, similar in spirit to CRYSTALS-Dilithium. This design choice simplifies both software and hardware implementations, as evidenced by our reference implementation, which shows that the signing and verification efficiency of Rhyme is comparable to that of Dilithium.

2 Preliminaries

2.1 Notations

Matrices are denoted in bold font and uppercase letters (e.g., $\mathbf{A}$), while vectors are denoted in bold font and lowercase letters (e.g., $\mathbf{y}$ or $\mathbf{z}_1$). The i -th component of a vector is denoted with subscript i (e.g., y_i for the i -th component of $\mathbf{y}$). We denote concatenation between vectors by putting the rows below as $(\mathbf{u}, \mathbf{v})$ and the columns on the right as $(\mathbf{u}|\mathbf{v})$. We naturally extend the notation to concatenations between matrices (e.g., $(\mathbf{A}, \mathbf{B})$ or $(\mathbf{A}|\mathbf{B})$). We let $\mathcal{R} = \mathbb{Z}[x]/(x^n + 1)$ be a polynomial ring where n is a positive integer and for any positive integer q the quotient ring $\mathcal{R}_q = \mathbb{Z}[x]/(q, x^n + 1) = \mathbb{Z}_q[x]/(x^n + 1)$. We identify $\mathcal{R}_2$ with the set of polynomials in $\mathcal{R}$ whose coefficients are in $\{0, 1\}$. Given $\mathbf{y} = (\sum_{0 \leq i < n} y_i x^i, \dots, \sum_{0 \leq i < n} y_{nk-n+i} x^i)^\top \in \mathcal{R}^k$, we define its l_∞ -norm as the l_∞ -norm of the corresponding “flattened” vector $\|\mathbf{y}\|_\infty = \|(y_0, \dots, y_{nk-1})^\top\|_\infty$. For a positive integer α , we define $r \bmod^+ \alpha$ as the unique integer r' in the range $[0, \alpha)$ that satisfies $r \equiv r' \bmod \alpha$.

Table 1. Comparison of bandwidth ($|\text{PK}| + |\text{SIG}|$ in bytes) between Rhyme and other lattice-based signatures.

Scheme	Level	Frame	Assumption	$ \text{PK} $	$ \text{SIG} $	Bandwidth
GPV with WS [42]	1	HtS	NTRU	640	474	1114
Falcon-512 [35]		HtS	NTRU	897	666	1563
Hawk-512 [20]		HtS	LIP	1024	555	1579
IRS with rejection [24]		FS	NTWE	928	775	1703
NTRU G+G-512 [17]		FS	NTRU	1021	992	2013
Rhyme-128 (This work)		FS	GAMLWE	800	1483	2283
HAETAE-120 [13]		FS	MLWE	992	1474	2466
G+G-120 [17]		FS	MLWE	1472	1677	3149
ML-DSA-2 (Dilithium2) [33]		FS	MLWE	1312	2420	3732
Aigis-sig-1 [41]		FS	MLWE	1312	2445	3757
GPV with WS [42]	5	HtS	NTRU	1408	979	2387
Falcon-1024 [35]		HtS	NTRU	1793	1280	3073
IRS with rejection [24]		FS	NTWE	1568	1694	3262
Hawk-1024 [20]		HtS	LIP	2440	1221	3661
HAETAE-260 [13]		FS	MLWE	2080	2948	5028
Rhyme-256 (This work)		FS	GAMLWE	1824	3258	5082
G+G-260 [17]		FS	MLWE	2336	2804	5140
ML-DSA-5 (Dilithium5) [33]		FS	MLWE	2592	4627	7219

2.2 Lattice

An m -dimensional *lattice* is a discrete additive subgroup in $\mathbb{R}^m$ which can be represented as the set of integer linear combination of n linearly independent vectors $\{\mathbf{b}_1, \dots, \mathbf{b}_n\}$, i.e.

$$\mathcal{L}(\mathbf{B}) = \left\{ \sum_{i=1}^n x_i \mathbf{b}_i \mid x_i \in \mathbb{Z}, \forall i \in [1, n] \right\},$$

where $\mathbf{B} = [\mathbf{b}_1, \dots, \mathbf{b}_n]$ is called a *basis* of $\mathcal{L}$ which is not unique, $n(n \leq m)$ is the *rank* of the lattice, a lattice is called full-rank if $m = n$. The determinant of $\mathcal{L}$ is defined as

$$\det(\mathcal{L}) = \sqrt{\det(\mathbf{B}^\top \mathbf{B})}.$$

The quantity $\det(\mathcal{L})$ is invariant regardless of the choice of $\mathbf{B}$. The *dual lattice* $\mathcal{L}^*$ is defined as

$$\mathcal{L}^* = \{\mathbf{w} \in \mathbb{R}^m \mid \forall \mathbf{v} \in \mathcal{L}, \langle \mathbf{w}, \mathbf{v} \rangle \in \mathbb{Z}\}.$$

q -ary lattice As a kind of important lattices in lattice-based cryptography, a q -ary lattice refers to the lattice such that $q\mathbb{Z}^n \subseteq \mathcal{L} \subseteq \mathbb{Z}^n$ where q is an integer.

Two types of q -ary lattices frequently used in lattice cryptography are defined as follows with respect to an $n \times m$ matrix $\mathbf{B} \in \mathbb{Z}_q^{n \times m}$,

$$\begin{aligned} \mathcal{L}_q(\mathbf{B}) &= \{\mathbf{y} \in \mathbb{Z}^m \mid \mathbf{y} = \mathbf{B}^\top \mathbf{x} \bmod q, \mathbf{x} \in \mathbb{Z}^n\}, \\ \mathcal{L}_q^\perp(\mathbf{B}) &= \{\mathbf{y} \in \mathbb{Z}^m \mid \mathbf{B}\mathbf{y} = \mathbf{0} \bmod q\}. \end{aligned}$$

2.3 LWE Problem

Learning with Errors (LWE) problem was proposed by Regev [36] in 2005 and has been widely used in the construction of lattice-based cryptography. We first introduce some definitions in order to describe LWE problems.

Definition 2.1 (LWE distribution). Let $n \geq 1$, $q \geq 2$ and χ be an error distribution over $\mathbb{Z}_q$, given a secret vector $\mathbf{s} \in \mathbb{Z}_q^n$, the LWE distribution $L_{\mathbf{s}, \chi}$ over $\mathbb{Z}_q^n \times \mathbb{Z}_q$ is sampled by choosing $\mathbf{a} \leftarrow U(\mathbb{Z}_q^n)$ and $e \leftarrow \chi$, and outputting $(\mathbf{a}, b = \langle \mathbf{a}, \mathbf{s} \rangle + e \bmod q)$.

The LWE problem has a search version and a decision version, which are defined as follows.

Definition 2.2 (Search-LWE). Given m independent samples $(\mathbf{a}_i, b_i) \in \mathbb{Z}_q^n \times \mathbb{Z}_q$ drawn from $L_{\mathbf{s}, \chi}$ for a fixed $\mathbf{s}$, the goal is to recover $\mathbf{s}$.

In the following part of this paper, we denote $\mathbf{A} \in \mathbb{Z}_q^{m \times n}$ to be the matrix formed by m rows $\{\mathbf{a}_i\}_{i=1}^m$ and $\mathbf{b} = (b_1, b_2, \dots, b_m)^\top \in \mathbb{Z}_q^m$, where $b_i = \langle \mathbf{a}_i, \mathbf{s} \rangle + e_i \bmod q$.

Definition 2.3 (Decision-LWE). Given m independent samples $(\mathbf{A}, \mathbf{b}) \in \mathbb{Z}_q^{m \times n} \times \mathbb{Z}_q^m$, the goal is to distinguish whether they are drawn from $L_{\mathbf{s}, \chi}$ (for some fixed but unknown $\mathbf{s} \leftarrow \chi^n$) or from the uniform distribution over $\mathbb{Z}_q^{m \times n} \times \mathbb{Z}_q^m$.

We denote the LWE problem as $\text{LWE}_{m,n,\chi,q}$ where the secret vector $\mathbf{s} \leftarrow \chi^n$ and the error vector $\mathbf{e} \leftarrow \chi^m$ are both drawn independently from the same distribution χ . Applebaum et al. [3] proved that this variant is as hard as the standard LWE problem (where $\mathbf{s}$ is uniform). Besides, to make LWE more practical in cryptography, variants of LWE problems (e.g., Ring-LWE and Module-LWE) have been investigated. More details of these variants can be found in [29]. We remark that any $\text{MLWE}_{n,q,k,l,\chi}$ instance can be viewed as an $\text{LWE}_{nk,nl,\chi,q}$ instance, where n denotes the dimension of the polynomial ring.

Definition 2.4 (Binary-LWE [10]). Let $n \geq 1$, $q \geq 2$ and σ_{bin} denote the uniform distribution in $\{0, 1\}$, the binary-LWE distribution $L_{\mathbf{s}, \sigma_{\text{bin}}}$ over $\mathbb{Z}_q^n \times \mathbb{Z}_q$ is sampled by choosing $\mathbf{a} \leftarrow U(\mathbb{Z}_q^n)$ and $s_i, e \leftarrow \sigma_{\text{bin}}$, and outputting $(\mathbf{a}, b = \langle \mathbf{a}, \mathbf{s} \rangle + e \bmod q)$.

2.4 Reduction from $\text{LWE}_{m,n,\sigma_{\text{bin}},q}$ to $\text{LWE}_{m,n,\sigma_k,q}$

In Construction 1, we choose secret and error uniformly from $\{-1, 1\}$, denoted as σ_1 , and it is not the usual case for LWE designs. To argue the security of our construction, we demonstrate the classical polynomial-time equivalence between the $\text{LWE}_{m,n,\sigma_k,q}$ problem (where secrets and errors are uniform in $\{\pm k\}$, and we denote this distribution as σ_k) and the well-studied Binary-LWE problem $\text{LWE}_{m,n,\sigma_{\text{bin}},q}$ (where secrets and errors are uniform in $\{0, 1\}$) for prime modulus q .

Lemma 2.1 (Reduction from Binary-LWE to σ_k -LWE). Let q be a prime and $k \in \mathbb{Z}_q^*$ be a non-zero integer. Given an oracle $\mathcal{O}$ that solves $\text{LWE}_{m,n,\sigma_k,q}$, one can construct an efficient adversary $\mathcal{A}$ that solves $\text{LWE}_{m,n,\sigma_{\text{bin}},q}$ with the same success probability.

Proof. Given a Binary-LWE instance $(\mathbf{A}, \mathbf{b} = \mathbf{A}\mathbf{s} + \mathbf{e} \bmod q)$ where $\mathbf{s} \leftarrow \sigma_{\text{bin}}^n, \mathbf{e} \leftarrow \sigma_{\text{bin}}^m$, the adversary $\mathcal{A}$ constructs a new instance as follows:

1. Compute $\mathbf{b}' = 2k \cdot \mathbf{b} \bmod q$. Since $\mathbf{b} = \mathbf{A}\mathbf{s} + \mathbf{e} \bmod q$, we have $\mathbf{b}' = \mathbf{A}(2k\mathbf{s}) + (2k\mathbf{e}) \bmod q$.
2. Let $\mathbf{s}' = 2k\mathbf{s}$ and $\mathbf{e}' = 2k\mathbf{e}$. Note that $\mathbf{s}'$ and $\mathbf{e}'$ are uniform in $\{0, 2k\}$.
3. Compute $\tilde{\mathbf{b}} = \mathbf{b}' - k \cdot \mathbf{A}\mathbf{1} - k \cdot \mathbf{1} \bmod q$, where $\mathbf{1}$ denotes the all-ones vector of appropriate dimension. Substituting $\mathbf{b}'$, we get:

$$\tilde{\mathbf{b}} = \mathbf{A}(2k\mathbf{s} - k\mathbf{1}) + (2k\mathbf{e} - k\mathbf{1}) = \mathbf{A}\tilde{\mathbf{s}} + \tilde{\mathbf{e}} \bmod q,$$

where $\tilde{\mathbf{s}} = 2k\mathbf{s} - k\mathbf{1}$ and $\tilde{\mathbf{e}} = 2k\mathbf{e} - k\mathbf{1}$.

4. Observe that since the coefficients of $\mathbf{s}, \mathbf{e}$ are uniformly distributed in $\{0, 1\}$, the transformed vectors satisfy $(\tilde{\mathbf{s}}, \tilde{\mathbf{e}}) \in \{\pm k\}^n \times \{\pm k\}^m$. Moreover, because $k \not\equiv 0 \pmod q$ and q is prime, the mapping $(\mathbf{s}, \mathbf{e}) \mapsto (\tilde{\mathbf{s}}, \tilde{\mathbf{e}})$ is a bijection between $\{0, 1\}^{n+m}$ and $\{\pm k\}^{n+m}$. Hence, $(\tilde{\mathbf{s}}, \tilde{\mathbf{e}})$ are uniformly distributed in $\{\pm k\}^{n+m}$.
5. Submit the instance $(\mathbf{A}, \tilde{\mathbf{b}})$ to the $\text{LWE}_{m,n,\sigma_k,q}$ oracle $\mathcal{O}$, which returns a solution $\tilde{\mathbf{s}}$ (or $\tilde{\mathbf{e}}$) with the oracle's advertised advantage.
6. Recover the original Binary-LWE secret $\mathbf{s}$ via the inverse transform: $\mathbf{s} = (2k)^{-1}(\tilde{\mathbf{s}} + k \cdot \mathbf{1}) \pmod q$.

Thus, any efficient solver for $\text{LWE}_{m,n,\sigma_k,q}$ yields an efficient solver for Binary-LWE problem $\text{LWE}_{m,n,\sigma_{bin},q}$. $\square$

A converse reduction (from σ_k -LWE to Binary-LWE) can be constructed analogously by applying the inverse linear transformation. Therefore, for prime q , the problems $\text{LWE}_{m,n,\sigma_k,q}$ and $\text{LWE}_{m,n,\sigma_{bin},q}$ are polynomial-time equivalent.

Consequently, the security of our scheme, which relies on $\text{LWE}_{m,n,\sigma_1,q}$, can be based on the conjectured hardness of the Binary-LWE problem. The hardness of Binary-LWE has been extensively analyzed in the literature [5, 9, 10, 14, 31, 38]. As a result, we will estimate the security of $\text{LWE}_{m,n,\sigma_1,q}$ by $\text{LWE}_{m,n,\sigma_{bin},q}$ in this paper.

2.5 SIS Problem

The Short Integer Solution (SIS) problem, first introduced by Ajtai [1], acts as the dual counterpart to the LWE problem. It serves as the foundation for the security of many lattice-based cryptographic primitives, particularly digital signatures and collision-resistant hash functions.

Definition 2.5 (SIS). *Let $n, m, B > 0$ and $q \geq 2$ be a modulus. Given a uniformly random matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$, the Short Integer Solution problem $\text{SIS}_{n,m,q,B}$ asks to find a non-zero vector $\mathbf{z} \in \mathbb{Z}^m \setminus \{\mathbf{0}\}$ such that:*

$$\mathbf{A}\mathbf{z} = \mathbf{0} \pmod q \quad \text{and} \quad \|\mathbf{z}\|_\infty \leq B.$$

Similar to LWE, the SIS problem can be generalized over rings. Also, any $\text{MSIS}_{n,q,k,l,B}$ can be viewed as a $\text{SIS}_{nk,nl,B,q}$ instance, where n denotes the dimension of the polynomial ring.

2.6 LIP Problem

Definition 2.6 (Search Module Lattice Isomorphism Problem, smLIP [21]). *Let $K = \mathbb{Q}(\zeta_{2^k})$ be a power-of-two cyclotomic field, and let $\mathcal{R} = \mathcal{O}_K$ be its ring of integers. Let $n = \varphi(2^k)$ be the degree of K (i.e., $K \cong \mathbb{Q}[X]/(X^n + 1)$).*

Consider a rank-2 free $\mathcal{R}$ -module $M \subset \mathbb{C}^{2n}$ (after the canonical embedding), represented by an $\mathcal{R}$ -basis $B = (\mathbf{b}_1, \mathbf{b}_2) \in (\mathbb{C}^{2n})^2$. The corresponding Gram matrix is $G = B^*B \in \mathcal{H}_2(\mathbb{C})$, where $\mathcal{H}_2(\mathbb{C})$ denotes the set of 2×2 positive definite Hermitian matrices.

Given two Hermitian matrices $G, G' \in \mathcal{H}_2(\mathbb{C})$ belonging to the same isomorphism class, i.e., there exists a (unknown) unimodular matrix $U \in \text{GL}_2(\mathcal{R})$ such that

$$G' = U^*GU,$$

the search Module Lattice Isomorphism Problem (smLIP) asks to find any matrix $U \in \text{GL}_2(\mathcal{R})$ satisfying the above equation.

In the specific context of HAWK, the base form is $G = I_2(K)$ (the identity matrix), and the given instance G' is the public key generated by HawkKeyGen. Thus, the problem reduces to finding a unimodular U such that $I = U^*G'U$, i.e., U is an orthonormalizer of the lattice defined by G' .

Definition 2.7 (Bounded Search Module Lattice Isomorphism Problem, B-smLIP). Let χ be the distribution over $\mathcal{R}^2$ induced by HawkKeyGen (i.e., coefficients of f, g are drawn from $\text{Bin}(\eta)$ and then conditioned on the NTRU solvability and norm lower bound). Let $\beta > 0$ be a norm bound. Given $G' = U^*GU$, where the unknown matrix $U \in \text{GL}_2(\mathcal{R})$ additionally satisfies the norm constraint

$$\|U\| \leq \beta,$$

the Bounded search Module Lattice Isomorphism Problem (B-smLIP) asks to find the unique matrix U satisfying both the isomorphism equation and the norm bound.

In HAWK, the norm $\|\cdot\|$ is the Euclidean operator norm (or entry-wise ℓ_2 -norm). The boundedness ensures uniqueness: the private key B is the unique short solution to the NTRU equation. However, unlike earlier works, HAWK does not rely on a radially symmetric average-case distribution; instead, the instances are taken from the specific output distribution of HawkKeyGen.

Remark 2.1. Change from earlier versions. In prior definitions (e.g., [DvW22, DPPvW22a]), the average-case hardness of smLIP was formalized via distributions $\mathbf{AC}_\sigma([Q])$ that are radial, enabling worst-case to average-case reductions. However, HAWK (NIST Version 1.1) adopts a centred binomial distribution for sampling f, g , which is not radial. Consequently, as explicitly stated in Section 6.7 of the HAWK specification, “we cannot claim any worst-case to average-case reduction for the problem of recovering the secret key”. The hardness of B-smLIP in HAWK is thus grounded on the *practical average-case* hardness of the concrete instance distribution produced by HawkKeyGen, supported by extensive lattice-reduction experiments (Section 5).

Moreover, the norm constraint on U is a key security feature: the private key B is guaranteed to satisfy $\|(f, g)\|^2 > 2n\sigma_{\text{sec}}^2$ (lower bound to prevent “dimensions-for-free” attacks), while the boundedness $\|U\| \leq \beta$ ensures that among the infinitely many unimodular solutions, the short one is unique and hard to find.

2.7 Gaussian Distribution and Smoothing Parameter

Let $k > 0, \mathbf{c} \in \mathbb{R}^k$ and $\Sigma \in \mathbb{R}^{k \times k}$ be a positive definite symmetric matrix. The Gaussian function with parameter Σ and center parameter $\mathbf{c}$ is defined as $\rho_{\sqrt{\Sigma}, \mathbf{c}} : \mathbf{x} \rightarrow \exp(-\pi(\mathbf{x} - \mathbf{c})^\top \Sigma^{-1}(\mathbf{x} - \mathbf{c}))$.

The Gaussian distribution over the lattice $\Lambda \subseteq \text{span}(\Sigma)$ with covariance parameter Σ and center parameter $\mathbf{c}$ is the distribution with support Λ and probability mass function $D_{\Lambda, \sqrt{\Sigma}, \mathbf{c}} : \mathbf{x} \rightarrow \frac{\rho_{\sqrt{\Sigma}, \mathbf{c}}(\mathbf{x})}{\sum_{\mathbf{y} \in \Lambda} \rho_{\sqrt{\Sigma}, \mathbf{c}}(\mathbf{y})}$. If $\Sigma = \sigma^2 \mathbf{Id}_k$, we write $\rho_{\sigma, \mathbf{c}}$ and $D_{\Lambda, \sigma, \mathbf{c}}$. We omit $\mathbf{c}$ when it is 0.

Given a k -dimensional lattice $\Lambda \subseteq \mathbb{R}^k$, its dual lattice Λ^* is defined as the set $\Lambda^* = \{\mathbf{x} \in \text{span}(\Lambda) \mid \mathbf{x}^\top \mathbf{y} \in \mathbb{Z}, \forall \mathbf{y} \in \Lambda\}$. Given a lattice $\Lambda \subseteq \mathbb{R}^k$ and $\varepsilon > 0$, the smoothing parameter $\eta_\varepsilon(\Lambda)$ of the lattice Λ is defined as the smallest σ such that $\rho_{1/\sigma}(\Lambda^* \setminus \{0\}) \leq \varepsilon$. The smoothing parameter satisfies the following two properties.

Lemma 2.2 (Upper Bound of Smoothing Parameter [43]). *Let $k > 1$ and $\varepsilon < 0.086k$. Let $\Lambda \subseteq \mathbb{R}^k$ be a full-rank lattice with basis $\mathbf{B} = (\mathbf{b}_1, \dots, \mathbf{b}_k)$. It holds that*

$$\eta_\varepsilon \leq \sqrt{\frac{\ln(k-1 + 2k/\varepsilon)}{\pi}} \cdot \max_{i \leq k} \|\mathbf{b}_i\|.$$

Lemma 2.3 (Linear Transformation Lemma [19]). *Let $\mathbf{A} \in \mathbb{Z}^{n \times m}$ such that $\mathbf{A} \cdot \mathbb{Z}^m = \mathbb{Z}^n$. Let $\Sigma = \mathbf{A}\mathbf{A}^\top$. For $\varepsilon \in (0, 1/2)$, if $\sigma \geq \eta_\varepsilon(\Lambda^\perp(\mathbf{A}))$, then the max-log distance between the distribution of $\mathbf{y} = \mathbf{A}\mathbf{x}$ where $\mathbf{x} \leftarrow D_{\mathbb{Z}^m, \sigma}$ and $D_{\mathbb{Z}^n, \sigma\sqrt{\Sigma}}$ is at most $4\varepsilon + O(\varepsilon^2)$.*

2.8 Security Definition for Signature

In this section, we formally present precise definitions for the aforementioned security properties of digital signature schemes.

Definition 2.8 (BimodalSelfTargetMSIS _{H, n, q, k, l, B}). *Suppose that $H : \{0, 1\}^* \times M \rightarrow \mathcal{R}_2$ is a cryptographic hash function. For positive integers q, k, l , a positive real number B and the dimension n of $\mathcal{R}$, the advantage of an adversary $\mathcal{A}$ solving the search-BimodalSelfTargetMSIS _{H, n, q, k, l, B} problem with respect to $\mathbf{j} \in$*

$\mathcal{R}_2^k \setminus \{0\}$ is defined as:

$$\text{Adv}_{H,n,q,k,l,B}^{\text{BimodalSelfTargetMSIS}}(\mathcal{A}) = \Pr \left[\begin{array}{l} 0 < \|\mathbf{z}\|_\infty \leq B \wedge \\ H(\mathbf{A}\mathbf{z} - qc\mathbf{j} \bmod 2q, M) = c \end{array} \middle| \begin{array}{l} (\mathbf{A}_0, \mathbf{b}) \leftarrow \mathcal{R}_q^{k \times (l-1)} \times \mathcal{R}_q^k; \\ \mathbf{A} = (-2\mathbf{b} + q\mathbf{j} | 2\mathbf{A}_0 | 2\mathbf{Id}_k) \bmod 2q; \\ (\mathbf{z}, c, M) \leftarrow \mathcal{A}^{H(\cdot)}(\mathbf{A}) \end{array} \right].$$

In the ROM (resp. QROM), the adversary is given classical (resp. quantum) access to H .

According to [13], there is a classical reduction from $\text{MSIS}_{H,n,q,k,l,B}$ to $\text{BimodalSelfTargetMSIS}_{H,n,q,k,l,B/2}$. Besides, since the reduction relies on the forking lemma, it cannot be directly extended to a quantum reduction in the QROM. So we will then introduce the definition of Existential Unforgeability under No Message Attacks (EU-NMA) which will be used in our security proof.

Definition 2.9 (Existential Unforgeability under No Message Attacks (EU-NMA)). In the EU-NMA, the adversary attempts to forge a signature without access to any valid message-signature pairs beforehand, relying solely on the public key for the attack. For a signature scheme $\mathcal{S} = (\text{KeyGen}, \text{Sign}, \text{Verify})$, the advantage of an EU-NMA adversary $\mathcal{A}$ is defined as:

$$\text{Adv}_S^{\text{EU-NMA}}(\mathcal{A}) = \Pr[\text{Verify}(vk, M, \sigma) = 1 \mid (sk, vk) \leftarrow \text{KeyGen}; (M, \sigma) \leftarrow \mathcal{A}(vk)].$$

Definition 2.10 (Existential Unforgeability under Chosen Message Attacks (EU-CMA)). In EU-CMA, the adversary $\mathcal{A}$ can adaptively choose messages and obtain their corresponding signatures but cannot forge a valid signature for a new message that has never been signed before. Let $\mathcal{S} = (\text{KeyGen}, \text{Sign}, \text{Verify})$ be a signature scheme and let (M_i, σ_i) be the set of message-signature pairs queried by the adversary and generated by the signer, for $i = 1, 2, \dots, Q$. The adversary ultimately attempts to forge and outputs (M^*, σ^*) , while the advantage of $\mathcal{A}$ is defined as:

$$\text{Adv}_S^{\text{EU-CMA}}(\mathcal{A}) = \Pr \left[\begin{array}{l} M^* \notin \{M_i\}_{i \in [Q]} \wedge \\ \text{Verify}(vk, M^*, \sigma^*) = 1 \end{array} \middle| \begin{array}{l} (sk, vk) \leftarrow \text{KeyGen}; \\ (M^*, \sigma^*) \leftarrow \mathcal{A}^{\text{Sign}(sk, \cdot)} \end{array} \right].$$

Definition 2.11 (Strong Existential Unforgeability under Chosen Message Attacks (sEU-CMA)). Compared to the EU-CMA, the sEU-CMA imposes stricter constraints on the adversary's forgery attempts: the forged signature-message pair must not replicate any previously obtained pairs. In other words, the adversary could succeed merely by finding a different valid signature for a previously queried message. Let $\mathcal{S} = (\text{KeyGen}, \text{Sign}, \text{Verify})$ be a signature scheme. An sEU-CMA adversary $\mathcal{A}$ has access to the verification key and a

signing oracle to make adaptive queries. Let the queried messages and the received signatures be (M_i, σ_i) for $i = 1, 2, \dots, Q$. At the end of the experiment, it outputs a message-signature pair (M^*, σ^*) . Then the advantage of $\mathcal{A}$ is defined as:

$$\text{Adv}_S^{\text{sEU-CMA}}(\mathcal{A}) = \Pr \left[(M^*, \sigma^*) \notin \{(M_i, \sigma_i)\}_{i \in [Q]} \mid (sk, vk) \leftarrow \text{KeyGen}; \right. \\ \left. \wedge \text{Verify}(vk, M^*, \sigma^*) = 1 \mid (M^*, \sigma^*) \leftarrow \mathcal{A}^{\text{Sign}(sk, \cdot)} \right].$$

2.9 Security Definition for Identification Protocol

To refine our reduction process, we define a set of properties for Fiat-Shamir identification protocols and analyze their roles in assisting the security proof of our signature scheme.

In the identification scheme, we require a sufficiently high probability for the honest entity to successfully pass the authentication.

Definition 2.12 (Identification Scheme). An identification scheme is a tuple of PPT algorithms $\text{ID} = (\text{Igen}, \text{P}, \text{V})$ such that:

- Igen : On input the security parameter 1^λ , algorithm Igen outputs a verification key vk and a signing key sk . We assume that vk defines the challenge space $\mathcal{C}$.
- P : The prover $\text{P} = (\text{P}_1, \text{P}_2)$ is split into two algorithms: given sk , algorithm P_1 produces a commitment w (the first message sent to the verifier) and a state st ; algorithm P_2 , on input (sk, w, st) and a uniformly random challenge $c \in \mathcal{C}$ sent by the verifier in response to commitment w , outputs an answer z .
- V : On input (vk, w, c, z) , the deterministic verifier V outputs 1 or 0.

Definition 2.13 (Completeness and Commitment-recoverability). An identification scheme $\text{ID} = (\text{Igen}, \text{P}, \text{V})$ is ϵ -complete for some $\epsilon > 0$ if for any $(vk, sk) \leftarrow \text{Igen}(1^\lambda)$, for any challenge $c \in \mathcal{C}$, we have:

$$\Pr[\text{V}(vk, w, c, z) = 0 \mid (w, c, z) \leftarrow (\text{P}(sk, vk) \leftrightarrow \text{V}(vk))] \leq \epsilon$$

where the randomness is taken over the random coins of P .

In addition, we need to ensure the commitment's recoverability, meaning that for any public key, challenge and answer, there is at most one commitment such that the transcript is valid. Also there exists a PPT algorithm that can recover the commitment using the answer and challenge such that $w = \text{Rec}(vk, c, z)$.

For an identification scheme ID and a hash function H (modeled as a random oracle in the analysis), we let $\text{FS}[\text{ID}, \text{H}]$ denote the resulting signature scheme.

We then recall the definitions of honest-verifier zero-knowledge (HVZK) and commitment min-entropy, which allow to reduce EU-CMA security of $\text{FS}[\text{ID}, \text{H}]$ to its EU-NMA security.

Definition 2.14 (HVZK and Commitment Min-entropy). For an identification scheme $ID = (\text{Igen}, P, V)$, it is *Honest-Verifier Zero-Knowledge* if there exists a PPT simulator **Sim** such that, for any honest verifier V , the following distributions are computationally indistinguishable:

- Real transcription $(w, c, z) \leftarrow (P(sk, vk) \leftrightarrow V(vk))$.
- Simulated transcription $(w, c, z) \leftarrow \text{Sim}(c, vk)$.

An identification scheme satisfies α - Min Entropy or has α bits of commitment min-entropy if for any (vk, sk) in the range of Igen :

$$H_\infty(w \mid (w, c, z) \leftarrow (P(sk, vk) \leftrightarrow V(vk))) \geq \alpha,$$

where H_∞ denotes information entropy.

Definition 2.15 (Lossy Identification Scheme). We say that an identification scheme $ID = (\text{Igen}, P, V)$ is *lossy* if it satisfies key-indistinguishability and ε_{ls} -lossy-soundness.

- *key-indistinguishability*: there exists a PPT lossy key generation algorithm LossyIgen that, on input a security parameter, outputs a verification key vk_{ls} , such that for any (possibly quantum) adversary $\mathcal{A}$

$$\text{Adv}^{\text{key-ind}}(\mathcal{A}) := |\Pr(\mathcal{A}(vk_{ls}) = 1) - \Pr(\mathcal{A}(vk) = 1)| = \text{negl}(\lambda),$$

where $vk_{ls} \leftarrow \text{LossyIgen}(1^\lambda)$ and $(vk, sk) \leftarrow \text{Igen}(1^\lambda)$ and the probability is taken over the random coins of $\mathcal{A}$, LossyIgen and Igen .

- ε_{ls} -lossy-soundness for some $\varepsilon_{ls} > 0$: for any (unbounded) P^* interacting with V , we have:

$$\Pr[V(vk_{ls}, w, c, z) = 1 \mid (w, c, z) \leftarrow (P^*(vk_{ls}) \leftrightarrow V(vk_{ls}))] \leq \varepsilon_{ls}.$$

If we only consider classical adversaries, EU-NMA security of $\text{FS}[ID, H]$ can be argued by relying on the simpler notion of lossy special soundness.

Definition 2.16 (Lossy Special Soundness). Let $ID = (\text{Igen}, P, V)$ be an identification scheme with key-indistinguishability and LossyIgen be the corresponding lossy key generation algorithm. The scheme is *lossy-special-sound* if for any PPT adversary $\mathcal{A}$, the quantity:

$$\Pr[c_0 \neq c_1 \wedge V(vk, w, c_0, z_0) = V(vk, w, c_1, z_1) = 1 \mid (w, c_0, c_1, z_0, z_1) \leftarrow \mathcal{A}(vk)]$$

is $\text{negl}(\lambda)$, where the probability is over the choice of $vk \leftarrow \text{LossyIgen}(1^\lambda)$ and the coins of $\mathcal{A}$.

An identification scheme has unique response if, for any (w, c) , there is at most one z such that the transcript (w, c, z) is accepted by the verifier. While this is not true by design for this scheme, we follow the proof idea of G+G by relaxing this property to computational unique response (CUR), which states that it is difficult to come up with two different responses for (w, c) .

Definition 2.17 (Computational Unique Responses). A Fiat-Shamir identification scheme is said to have computational unique responses if, for any fixed commitment w and c , there is at most one valid response z that will be accepted by the verifier, except with negligible probability:

$$\Pr [z_0 \neq z_1 \wedge V(vk, w, c, z_0) = V(vk, w, c, z_1) = 1 \mid (w, c, z_0, z_1) \leftarrow \mathcal{A}(vk)]$$

is $\text{negl}(\lambda)$, where the probability is over the choice of vk and the coins of $\mathcal{A}$.

2.10 Range Asymmetric Numeral Systems Entropy Coding

Arithmetic coding encodes the entire vector as a single real number. The recent range Asymmetric Numeral Systems (rANS) [22] entropy coding encodes states to natural numbers for efficient hardware implementation. The rANS supports stream processing and enables finite-precision integer arithmetic via renormalization, reducing hardware implementation complexity. In addition, it can be transformed into a lookup table-based implementation (tANS), eliminating arithmetic operations entirely to accelerate computation.

Definition 2.18 (Range Asymmetric Numeral System). Let $S \subseteq [0, 2^n - 1]$, $n \in \mathbb{N}$ and we define a function $g : S \rightarrow \mathbb{Z} \cap (0, 2^n]$, satisfying $\sum_{s \in S} g(s) \leq 2^n$ and $g(s) = 0$ for all $s \notin S$.

Then, we define the following: 1) CDF: $S \rightarrow \mathbb{Z}$, defined as $\text{CDF}(s) = \sum_{i=0}^{s-1} g(i)$ with $\text{CDF}(0) = 0$. 2) symbol: $\mathbb{Z} \rightarrow S$, defined as $s \in S$ satisfying $\text{symbol}(x) = s \iff \text{CDF}(s) \leq x < \text{CDF}(s+1)$. Finally, the function C is defined as:

$$C(x, s) = \left\lfloor \frac{x}{g(s)} \right\rfloor \cdot 2^n + (x \bmod^+ g(s)) + \text{CDF}(s).$$

The rANS encoding and decoding functions for the set S and frequency $g/2^n$ are defined as follows in Algorithm 1 and 2:

Algorithm 1 rANSEncode

Input: $\mathbf{z} = (z_1, \dots, z_m)$;

Output: x ;

```

1:  $x \leftarrow 0$ 
2: for  $i = m$  downto 1 do
3:    $s \leftarrow \mathbf{z}[i]$ 
4:    $x \leftarrow \left\lfloor \frac{x}{g(s)} \right\rfloor \cdot 2^n + (x \bmod^+ g(s)) + \text{CDF}(s)$ 
5: end for
6: return  $x$ 
```

Algorithm 2 rANSDecode

Input: x, m ;**Output:** $S = (t_1, \dots, t_m)$;

```
1:  $y \leftarrow x$ 
2:  $M \leftarrow 2^n$ 
3: for  $i = 1, \dots, m$  do
4:    $s \leftarrow y \bmod^+ M$ 
5:    $t_i \leftarrow \text{symbol}(s)$ 
6:    $S[i] \leftarrow t_i$ 
7:    $y \leftarrow g(t_i) \cdot \lfloor \frac{y}{M} \rfloor + s - \text{CDF}(t_i)$ 
8: end for
9: return  $S$ 
```

Lemma 2.4. *The rANS coding enables lossless compression, where any sequence s can be perfectly reconstructed after encoding and decoding. The encoded size of rANS asymptotically equals the Shannon entropy of symbols, indicating that rANS coding approaches optimal compression efficiency. For any positive x and any probability distribution q over S , the following holds*

$$\sum_{s \in S} q(s) \log(C(x, s)) \leq \log(x) + \sum_{s \in S} q(s) \log\left(\frac{g(s)}{2^n}\right) + \frac{2^n}{x}.$$

Finally, the cost of encoding the first symbol is bounded by n . Specifically, for any $x \in S$, the logarithmic value of the state transition function $C(0, s)$ satisfies $\log(C(0, s)) \leq n$, confirming the upper bound of the encoding complexity for the initial symbol.

3 Signature Scheme

3.1 Overview

The Rhyme signature scheme is designed with the primary goals of achieving structural simplicity, bandwidth efficiency and implementation friendliness. Our design builds upon the Fiat-Shamir paradigm but introduces a novel sampling technique—Convolution of Constant Count Sampling (3C Sampling)—to efficiently prevent secret key leakage without relying on costly rejection sampling or imposing a complex high-dimensional floating-point architecture in Gaussian convolution approaches. Therefore, in this section, we first introduce the general methods used in our algorithm, and then present detailed explanations for the core innovation of Rhyme.

Foundation: Fiat-Shamir Signature and Bimodal Distributions. Fiat-Shamir signature is a well-known signature structure, which is generalized from

Schnorr’s identification protocol. The public key is a pair of matrices $(\mathbf{A}, \mathbf{T} = \mathbf{A}\mathbf{S} \bmod q)$, where $\mathbf{A} \in \mathbb{Z}_q^{k \times m}$ is a uniform matrix modulo some integer q and $\mathbf{S} \in \mathbb{Z}_q^{m \times n}$ is a small-magnitude matrix that makes up the secret key. A signature for a message M is comprised of an integer vector $\mathbf{z} \in \mathbb{Z}_q^m$ of the form $\mathbf{y} + \mathbf{S}\mathbf{c}$, for some random small-magnitude $\mathbf{y} \in \mathbb{Z}_q^m$ and some small-magnitude challenge $\mathbf{c} = H(\mathbf{w} = \mathbf{A}\mathbf{y} \bmod q, M) \in \mathbb{Z}_q^n$. At last, the verifier checks that the vector $\mathbf{z}$ is short and that $\mathbf{c} = H(\mathbf{A}\mathbf{z} - \mathbf{T}\mathbf{c} \bmod q, M)$. An important point of designing signature with Fiat-Shamir structure is to deal with the leakage of the secret information in $\mathbf{z} = \mathbf{y} + \mathbf{S}\mathbf{c}$. For example, flooding, rejection sampling and Gaussian convolution technique are several ways for solving this problem with different properties. However, each of these methods has its own disadvantages. Flooding leads to very large signature size, rejection sampling reduces efficiency and using Gaussian convolution method increases the difficulty of implementations and faces the challenge of the ratio attack [39].

In the design of Fiat-Shamir signatures, bimodal distribution technique is another important technique to achieve compact sizes, which was first introduced in [18]. More specifically, the key generation process forces the equality $\mathbf{A}\mathbf{S} = q\mathbf{I} \bmod 2q$ where $\mathbf{I}$ is a matrix consisting of an identity block stacked on top of a zero block, then the signature is now of the form $\mathbf{z} = \mathbf{y} + (-1)^b \mathbf{S}\mathbf{c}$, where each y_i is sampled from a fixed distribution and $b \in \{0, 1\}$ is sampled uniformly. To make sure that the verification test passes, computations are performed modulo $2q$ because $\mathbf{A}\mathbf{z} = \mathbf{w} + q\mathbf{I}\mathbf{c} = \mathbf{w} - q\mathbf{I}\mathbf{c} \bmod 2q$ for any $\mathbf{c} \in \{0, 1\}^n$. This bimodal technique inherently reduces signature size compared to the unimodal counterpart. And based on this structure, many improvements have been proposed. For example, HAETA [13] uses uniform distributions over hyperballs to reduce the size of signatures instead of the discrete Gaussian distributions used in the original version. And G+G [17] also uses the property of $\mathbf{A}\mathbf{S} = q\mathbf{I} \bmod 2q$ to construct one of the Gaussian distributions used in Gaussian convolution technique without affecting the verification process. One way to generate a key pair $(\mathbf{A}, \mathbf{S})$ that satisfies $\mathbf{A}\mathbf{S} = q\mathbf{I} \bmod 2q$ is to first generate an LWE sample $\mathbf{B}_{gen} = \mathbf{A}_{gen}\mathbf{S}_{gen} + \mathbf{E}_{gen} \bmod q$ and then set $\mathbf{A} = (-2\mathbf{B}_{gen} + q\mathbf{I}|2\mathbf{A}_{gen}|2\mathbf{Id}) \bmod 2q$, $\mathbf{S} = (\mathbf{Id}, \mathbf{S}_{gen}, \mathbf{E}_{gen})$. And it is easy to check $\mathbf{A}\mathbf{S} = q\mathbf{I} \bmod 2q$.

Core Innovation: 3C Sampling. Many Fiat-Shamir lattice signatures rely on flooding, rejection sampling, or Gaussian convolutions to mask the secret information in $\mathbf{z} = \mathbf{y} + \mathbf{S}\mathbf{c}$. However, none of these methods simultaneously achieves compact bandwidth and structural simplicity. To overcome this, we introduce *Convolution of Constant Count Sampling* (3C Sampling). The guiding principle is simplicity and elegance: the final output $\mathbf{z}$ is obtained by applying a *deterministic* integer matrix (derived from the secret key) to a vector of independent discrete Gaussian noises, whose number is *constant* and independent of the security parameter. Crucially, no floating-point sampling from a varying covariance

matrix is ever required, which is the key to both efficiency and implementation friendliness. The remaining challenge — achieving zero-knowledge despite the deterministic, secret-dependent transform — is addressed by publishing a Gram matrix (or part thereof) as auxiliary public information, from which a simulator can reproduce the correct signature distribution without knowing the secret key. The three constructions that follow illustrate progressively more practical instantiations of this paradigm and are all based on a parametrised family of *Gram-Aided Module-LWE* (GA-MLWE) assumptions defined in Section 3.2.

Optimizations for Compactness and Efficiency. To further enhance practicality, Rhyme incorporates several optimizations:

- **Scalar Challenge:** Following HAETAE and G+G, we define the challenge c as a ring element in $\mathcal{R}_2$ rather than a vector, reducing the secret $\mathbf{S}$ to a vector $\mathbf{s}$ and simplifying the key relation to $\mathbf{A}\mathbf{s} = q\mathbf{j}$ where $\mathbf{j}$ is the vector starting with 1 and then continuing with 0. The number of nonzero coefficients in c is τ , so the challenge space has size $\binom{n}{\tau}$. To sample such challenges we rely on the (binary version of) SampleInBall algorithm from Dilithium with slight modification because now $c_i \in \{0, 1\}$ instead of $c_i \in \{-1, 0, 1\}$.
- **Small Prime Modulus:** In Rhyme, we set q as a prime satisfying $n \mid (q-1)$, enabling efficient Number Theoretic Transform (NTT) operations.
- **Entropy Coding of Signature:** We apply entropy coding (specifically, the Asymmetric Numeral Systems [22]) to the signature as suggested in [23]. Since the signature’s distribution has low entropy, this technique reduces its size to nearly the theoretical minimum, improving bandwidth as defined in Algorithm 1 and 2. It should be noted that there are two types of distributions for the elements in our signatures as specified in the following subsection, which need two independent coding parameters.

3.2 Security Assumptions

Our schemes are based on the hardness of the Module-LWE (MLWE) and Module-SIS (MSIS) problems, two extensively studied lattice problems that underpin the NIST standard FIPS 204. To capture the additional algebraic information leaked by our constructions, we introduce a parametrised family of *Gram-Aided Module-LWE* (GA-MLWE) assumptions.

The GA-MLWE problem. Let $\mathcal{R}_q = \mathbb{Z}_q[x]/(x^n + 1)$ with $n = 2^\kappa$ and $q \equiv 1 \pmod{2n}$ a prime. Fix dimensions $k, l \geq 1$ and set $m = k + l$. Let χ be a distribution over $\mathcal{R}_q$ with small coefficients (e.g. the uniform distribution over $\{\pm 1\}$ or a centred binomial distribution). Let $\mathbf{s} = (1, \mathbf{s}_{\text{gen}}, \mathbf{e}_{\text{gen}}) \in \mathcal{R}_q^m$ be a secret vector with $\mathbf{s}_{\text{gen}} \leftarrow \chi^{l-1}$ and $\mathbf{e}_{\text{gen}} \leftarrow \chi^k$, and denote its tail by $\mathbf{s}_{\text{tail}} = (\mathbf{s}_{\text{gen}}, \mathbf{e}_{\text{gen}}) \in \mathcal{R}_q^{m-1}$. Let $\mathcal{F} : \mathcal{R}_q^{m-1} \rightarrow \{0, 1\}^*$ be a deterministic auxiliary-information function. The GA-MLWE $_{\mathcal{F}}$ problem asks to distinguish

the two distributions

$$(\mathbf{A}_{\text{gen}}, \mathcal{F}(\mathbf{s}_{\text{tail}}), \mathbf{A}_{\text{gen}}\mathbf{s}_{\text{gen}} + \mathbf{e}_{\text{gen}} \bmod q) \quad \text{and} \quad (\mathbf{A}_{\text{gen}}, \mathcal{F}(\mathbf{s}_{\text{tail}}), \mathbf{u}),$$

where $\mathbf{A}_{\text{gen}} \leftarrow \mathcal{R}_q^{k \times (l-1)}$ and $\mathbf{u} \leftarrow \mathcal{R}_q^k$ are chosen uniformly at random. The **GA-MLWE $_{\mathcal{F}}$ assumption** asserts that every probabilistic polynomial-time adversary has at most a negligible advantage in solving this distinguishing problem.

Instantiations. We instantiate $\mathcal{F}$ with three concrete functions that capture the auxiliary information appearing in our three constructions.

- **GA-MLWE $_{\text{diag}}$:** $\mathcal{F}_{\text{diag}}(\mathbf{s}_{\text{tail}}) = \{\mathcal{G}_{G_w}(s_i)\}_{i=2}^m$, where the $\mathcal{G}_{G_w}(s_i)$ are the diagonal Gram blocks defined in Construction 1 (Section 3.3.2), and $G_w \subset \text{Gal} \setminus \{\text{id}\}$ is a uniformly random subset of non-identity Galois automorphisms explicitly chosen not to form a subgroup.
- **GA-MLWE $_{\text{full}}$:** $\mathcal{F}_{\text{full}}(\mathbf{s}_{\text{tail}}) = \mathbf{B}\mathbf{B}^*$, where $\mathbf{B} \in \mathcal{R}_q^{d \times d}$ ($d = m - 1 = k + l - 1$) is a unimodular matrix output by **GenUnimod** whose first column equals $\mathbf{s}_{\text{tail}}$. For a ring element $a = \sum_{i=0}^{n-1} a_i x^i \in \mathcal{R}_q$, we write $a^* = a(x^{-1}) \bmod (x^n + 1)$; this is the *self-adjoint involution* induced by $x \mapsto x^{-1}$. For a matrix $\mathbf{M}$ over $\mathcal{R}_q$, $\mathbf{M}^*$ denotes the conjugate-transpose obtained by applying the involution $*$ to each entry and then transposing. (See Construction 2, Section 3.3.3.)
- **GA-MLWE $_{\text{short}}$:** $\mathcal{F}_{\text{short}}(\mathbf{s}_{\text{tail}}) = \mathbf{B}_{\text{short}}\mathbf{B}_{\text{short}}^*$, where $\mathbf{B}_{\text{short}}$ consists of the first d rows of a larger unimodular matrix (see Construction 3, Section 3.3.4).

Plausibility and relations among the assumptions. All three variants are plausible under the current state of algebraic cryptanalysis. Since $\mathcal{F}_{\text{full}}$ leaks strictly more information than $\mathcal{F}_{\text{diag}}$, the assumption **GA-MLWE $_{\text{full}}$** is formally stronger than **GA-MLWE $_{\text{diag}}$** ; similarly, **GA-MLWE $_{\text{full}}$** is stronger than **GA-MLWE $_{\text{short}}$** . These problems are inspired by the search module Lattice Isomorphism Problem (smLIP) introduced in Hawk [21], adapted here to a setting where the adversary is given only partial Gram information (diagonal blocks, a full Gram matrix, or a submatrix thereof). The concrete hardness of the GA-MLWE instances is analysed in Section 4.6.

3.3 Core Strategy: Convolution of Constant Counts Sampling

As introduced before, the scheme follows the G+G structure over the cyclotomic ring $\mathcal{R}_q = \mathbb{Z}_q[x]/(x^n + 1)$ with $n = 2^\kappa \geq 2$ and $q \equiv 1 \pmod{2n}$ a prime. The public key is a matrix $\mathbf{A} \in \mathcal{R}_{2q}^{k \times m}$ and the secret key is a vector $\mathbf{s} = (s_1, \dots, s_m)^\top \in \mathcal{R}_{2q}^m$ satisfying $\mathbf{A}\mathbf{s} = q\mathbf{j} \bmod 2q$, where $\mathbf{j} = (1, 0, \dots, 0)^\top$. In our instantiation the secret takes the form $\mathbf{s} = (1, \mathbf{s}_{\text{gen}}, \mathbf{e}_{\text{gen}})^\top$ with the first component fixed to 1 and the remaining components having small coefficients (e.g. drawn from $\{\pm 1\}$ or a binomial distribution). Signing proceeds by choosing

a masking vector $\mathbf{y}$, computing $\mathbf{w} = \mathbf{A}\mathbf{y} \bmod 2q$, deriving a challenge ring element $c \in \mathcal{R}_q$ with coefficients in $\{0, 1\}$, and then forming $\mathbf{z} = \mathbf{y} + \mathbf{s}c + \mathbf{h}$ where the auxiliary vector $\mathbf{h}$ satisfies $\mathbf{A}\mathbf{h} = \mathbf{0} \bmod 2q$ and is chosen to center the distribution of $\mathbf{z}$, e.g. in $\text{G}+\text{G}$ $\mathbf{h} = 2\mathbf{s}x$ where x follows a Gaussian distribution of center $-c/2$. Then the goal is to ensure that the distribution of $\mathbf{z}$ is (statistically or computationally) independent of $\mathbf{s}$.

The *3C Sampling* (Convolution of Constant Count Sampling) paradigm realises this goal by constructing $\mathbf{z}$ as $\mathbf{z} = \tilde{\mathbf{B}}\mathbf{x}$, where $\tilde{\mathbf{B}}$ is a *deterministic* integer matrix derived from the secret key and $\mathbf{x}$ is a vector of *independent* discrete Gaussian noise polynomials, with the number of polynomials kept *constant* and independent of the security parameter. Crucially, all noise polynomials are sampled from the canonical spherical discrete Gaussian; no floating-point sampling from a varying, non-spherical covariance matrix is ever required. The three constructions that follow illustrate how $\tilde{\mathbf{B}}$ can be instantiated, progressively trading explicit algebraic structure for compactness and practical efficiency. Before we introduce the constructions, we start by introducing a lightweight rejection sampling process for the first component of $\mathbf{s}$ which is always $1 \in \mathcal{R}_q$.

3.3.1 Lightweight Rejection Sampling for the Fixed Component

Because the first secret component is the public constant $s_1 = 1$, its output $z_1 = y_1 + 2x + c$ must be handled separately to avoid ratio attack [39], where x is an integer-valued noise term (and $2x$ is later embedded into the auxiliary vector). We present a lightweight iterative rejection sampling method [24] that, given a masking coefficient y_1 and a challenge bit c , produces a pair $(z_1, 2x)$ such that z_1 follows a truncated Gaussian, the integer $v = 2x + c$ follows a discrete Gaussian over the integers of parity c , and z_1 is independent of x .

Let σ_y be the masking Gaussian parameter and $\sigma_{tar} = 2\sigma$ the target parameter. Fix a bound B_1 and a small integer R (e.g. $R = \lceil 5\sigma_{tar} \rceil$). For $c \in \{0, 1\}$ define $\mathcal{V}_c = \{v \in \mathbb{Z} : |v| \leq R, v \equiv c \pmod{2}\}$ and the discrete Gaussian distribution $\chi_{\sigma_{tar}, R}^{(c)}$ over $\mathcal{V}_c$ with probability mass function $p_c(v) = \rho_{\sigma_{tar}}(v)/S_c$ where $\rho_{\sigma_{tar}}(v) = \exp(-\pi v^2/\sigma_{tar}^2)$ and $S_c = \sum_{u \in \mathcal{V}_c} \rho_{\sigma_{tar}}(u)$.

Theorem 3.1 (Properties of Rejection Sampling). *For inputs y drawn from D_{σ_y} conditioned on $|y| \leq B_1 + R$, Algorithm 3 outputs a pair $(z, 2x)$ such that:*

1. z follows D_{σ_y} conditioned on $|z| \leq B_1$;
2. $v = 2x + c$ follows $\chi_{\sigma_{tar}, R}^{(c)}$ (a discrete Gaussian over $\mathcal{V}_c$);
3. x is independent of z .

Algorithm 3 Rejection_Sampling

Input: $y \in \mathbb{Z}$ with $|y| \leq B_1 + R$, challenge bit $c \in \{0, 1\}$, random tape rt ;

Output: $(z, 2x) \in \mathbb{Z}^2$ or $\perp$;

```
1:  $r \leftarrow U_{(0,1)}(rt)$  {Uniform random number in  $(0, 1)$ }
2: Set  $\mathcal{V} = \mathcal{V}_c$ ,  $p = p_c$ , and let  $M$  be the constant defined in Theorem 3.1
3:  $cum \leftarrow 0$ 
4: for  $v \in \mathcal{V}$  in a fixed order do
5:   if  $|y + v| \leq B_1$  then
6:      $p_{acc} \leftarrow p(v) \cdot \rho_{\sigma_y}(y + v) / (M \cdot \rho_{\sigma_y}(y))$ 
7:   else
8:      $p_{acc} \leftarrow 0$ 
9:   end if
10:   $cum \leftarrow cum + p_{acc}$ 
11:  if  $r < cum$  then
12:     $z \leftarrow y + v$ ,  $2x \leftarrow v - c$ 
13:    return  $(z, 2x)$ 
14:  end if
15: end for
16: return  $\perp$ 
```

Algorithm 4 Sample_Z

Input: $\mathbf{c} \in \{0, 1\}^n$, $\mathbf{y} \in \mathbb{Z}^n$ with $|y_i| \leq B_1 + R$, random tape rt ;

Output: $(z_i, 2x_i)_{i=1}^n$ or $\perp$;

```
1: for  $i \in [n]$  do
2:    $(z_i, 2x_i) \leftarrow \text{Rejection\_Sampling}(y_i, c_i, rt)$ 
3:   if  $(z_i, 2x_i) = \perp$  or  $|z_i| > B_1$  then
4:     return  $\perp$ 
5:   end if
6: end for
7: return  $(z_i, 2x_i)_{i=1}^n$ 
```

The acceptance probability is $\Pr_{acc} = \frac{1}{M} \cdot \frac{\sum_{|z| \leq B_1} \rho_{\sigma_y}(z)}{\sum_{|y| \leq B_1 + R} \rho_{\sigma_y}(y)}$. With the optimal choice of M at the boundary $y = B_1 + R_c$ (where R_c is the largest element of $\mathcal{V}_c$), one can set $M = \frac{\rho_{\sigma_{tar}}(R_c)}{\sigma_y^2} \exp(\frac{2\pi B_1 R_c + \pi R_c^2}{\sigma_y^2})$.

The proof follows by direct calculation of the joint distribution; see the Appendix A for details. Note that the output $v = 2x + c$ is exactly the coefficient we will later embed into $2X_0$ (i.e. the coefficient of $2X_0$ is v), so $2X_0$ has coefficients in $\mathcal{V}_c$ and follows the prescribed parity-conditioned Gaussian.

3.3.2 Construction 1: Automorphism Isotropic Scheme

This construction directly instantiates the 3C principle by introducing explicit *cancellation* columns and *automorphism* columns, showing how to achieve *statistical* zero-knowledge when all secret components s_i have coefficients in $\{\pm 1\}$ (the case $s_1 = 1$ can be cancelled by rejection sampling described above). Its purpose is pedagogical: it makes the covariance cancellation mechanism completely transparent, at the cost of a prohibitively large number of noise columns.

Setup. Let $\text{Gal} = \{\tau_j : x \mapsto x^j \mid j \in (\mathbb{Z}_{2n})^\times\}$ be the Galois group of $\mathcal{R}_q$ over $\mathbb{Z}_q$. The secret $\mathbf{s} = (s_2, \dots, s_m)^\top$ has each coefficient of every s_i chosen uniformly from $\{\pm 1\}$.

Cancellation part. For each unordered pair (i, j) with $i < j$, we add a column that contains $-2s_i$ in row i and $2s_j$ in row j . These $(m-2)(m-1)/2$ columns are multiplied by independent Gaussian polynomials $E_{i,j} \sim D_{\mathbb{Z}, \sigma}^n$. They contribute $-4\sigma^2 s_i s_j^* / (2\pi)$ to the (i, j) off-diagonal block of the covariance matrix and $4(m-2)\sigma^2 s_i s_i^* / (2\pi)$ to the i -th diagonal block.

Automorphism part. For each index i and each non-identity $\tau \in \text{Gal}$ we add $4(m-1)$ columns, each containing the element $\tau(s_i)$. By the four-square theorem, these can be compressed to four independent Gaussian polynomials per (i, τ) , using coefficients a, b, c, d with $a^2 + b^2 + c^2 + d^2 = 4(m-1)$. Summing over all $\tau \neq \text{id}$ gives a contribution of $4(m-1)\sigma^2 \sum_{\tau \neq \text{id}} \tau(s_i s_i^*) / (2\pi)$ to the i -th diagonal block.

Combined output. Together with a shared base noise $2s_i X_0$ (where X_0 is a fresh Gaussian) and individual noise e'_i (each coefficient drawn from $D_{\mathbb{Z}, \sigma}$), one obtains

$$z_i = 2s_i \left(\sum_{j < i} E_{j,i} - \sum_{j > i} E_{i,j} \right) + \sum_{\tau \neq \text{id}} \tau(s_i) (aU_{i,\tau}^{(1)} + \dots + dU_{i,\tau}^{(4)}) + 2s_i X_0 + e'_i.$$

A direct computation shows that the covariance matrix of the coefficient vector of $\mathbf{z}$ is

$$\Sigma_{ii} = \frac{4(m-1)\sigma^2}{2\pi} \sum_{\tau \in \text{Gal}} \tau(s_i s_i^*) = \frac{4(m-1)\sigma^2 \text{Tr}(s_i s_i^*)}{2\pi} \mathbf{Id}_n, \quad \Sigma_{ij} = 0 \ (i \neq j).$$

For $\{\pm 1\}$ coefficients we have $\text{Tr}(s_i s_i^*) = n^2$ because each coefficient of $s_i s_i^*$ is a sum of n terms of the form $(\pm 1)^2$, yielding n^2 after summing the (constant) trace, hence $\Sigma = \frac{4(m-1)n^2\sigma^2}{2\pi} \mathbf{Id}_{(m-1)n}$ plus the individual e'_i terms, yielding a spherical discrete Gaussian after appropriate rounding. The statistical hiding property then follows from the smoothing parameter of the kernel lattice; a tight bound requires $\sigma \geq 2\sqrt{(m-1)n} \cdot \sqrt{\ln(2(N - (m-1)n)/\varepsilon)/\pi}$ where $N = \binom{m-1}{2} + 4(m-1)(n-1) + m$.

To reduce the number of automorphism columns, we replace the full Galois group by a random subset $G_w \subset \text{Gal} \setminus \{\text{id}\}$ of size w (e.g. $w = \lfloor \sqrt{n} \rfloor$), chosen such that $G_w \cup \{\text{id}\}$ does not form a subgroup. The output is still given by a deterministic integer matrix acting on independent Gaussians, but now the covariance is only block-diagonal, not spherical.

Construction 3.1 (Isotropic Construction with Random Subset). *Given $\mathbf{s}$ as above (all coefficients $\{\pm 1\}$; again $s_1 = 1$ will be handled analogously), define for each $i = 2, \dots, m$:*

$$z_i = 2s_i \left(\sum_{j < i} E_{j,i} - \sum_{j > i} E_{i,j} \right) + \sum_{\tau \in G_w} \tau(s_i) (aU_{i,\tau}^{(1)} + \dots + dU_{i,\tau}^{(4)}) + 2s_i X_0 + e'_i,$$

where all noise polynomials are independent with coefficients from $D_{\mathbb{Z},\sigma}$, and a, b, c, d are integers with $a^2 + b^2 + c^2 + d^2 = 4(m-1)$.

Flattening to coefficient vectors, the covariance matrix becomes block-diagonal with the i -th block

$$\Sigma_{ii} = \frac{4(m-1)\sigma^2}{2\pi} \sum_{\tau \in G_w \cup \{\text{id}\}} \tau(s_i s_i^*) + \frac{\sigma^2}{2\pi} \mathbf{Id}_n.$$

Define the *Gram block* of s_i (under G_w) as

$$\mathcal{G}_{G_w}(s_i) = 4(m-1) \sum_{\tau \in G_w \cup \{\text{id}\}} \mathbf{M}_{\tau}(s_i s_i^*) + \mathbf{Id}_n \in \mathbb{Z}^{n \times n},$$

where $\mathbf{M}_{\tau}(a)$ is the circulant matrix of multiplication by $\tau(a)$. Then $\Sigma_{ii} = \frac{\sigma^2}{2\pi} \mathcal{G}_{G_w}(s_i)$. Because G_w does not contain all automorphisms, the Gram block is not a scalar matrix; nevertheless, recovering s_i from $\mathcal{G}_{G_w}(s_i)$ appears to be hard under $\text{GA-MLWE}_{\text{diag}}$ assumption defined in Section 3.2, and the proof of this construction can be found in Appendix B.

Plausibility. For $w = 0$ the Gram block reveals $s_i s_i^*$ and the secret can be recovered by factoring the polynomial [25, 28]. For small w the sum destroys the product structure, and no efficient attack is known. With the recommended $w = \lfloor \sqrt{n} \rfloor$, the algebraic system is highly over-determined and non-linear in the $\{\pm 1\}$ coefficients, providing a comfortable margin. The assumption can be seen as a module-LIP variant where only diagonal Gram blocks are given, and it inherits confidence from the cryptanalysis of Hawk [21] and related NTRU-based designs.

3.3.3 Construction 2: Unimodular Matrix Scheme

The previous constructions realise 3C Sampling via an explicit assembly of columns. We now present the second construction, which encapsulates all the

required correlations into a short unimodular matrix $\mathbf{B} \in \mathcal{R}^{d \times d}$ (with $d = m - 1 = k + l - 1$) and handles the fixed component $s_1 = 1$ via the rejection sampler of Section 3.3.1. We first describe the required building block: a generator of short unimodular matrices.

Short Unimodular Matrices via Module-NTRU Trapdoors. We require a generator $\text{GenUnimod}(1^\lambda, d)$ that outputs a matrix $\mathbf{B} \in \mathcal{R}^{d \times d}$ with $\det \mathbf{B} = 1$ and $\|\mathbf{B}_{[1:d-1, :]}\|_\infty \leq B_{\max}$ (the first $d - 1$ rows are short). For $d = 2$ one can use the classical NTRU/Hawk construction; for $d \geq 3$ the module-NTRU trapdoor technique [15] provides a Gaussian sampling method. In the following we treat GenUnimod as a black-box.

Construction 3.2 (The Scheme Π_{th} (with Gram matrix)).

- $\text{KeyGen}(1^\lambda)$:
 1. Choose uniform $\mathbf{A}_{\text{gen}} \leftarrow \mathcal{R}_q^{k \times (l-1)}$.
 2. Generate $(\mathbf{B}, \mathbf{s}_{\text{tail}}) \leftarrow \text{GenUnimod}(1^\lambda, d)$ and parse the first column of $\mathbf{B}$ as $\mathbf{s}_{\text{tail}} = (\mathbf{s}_{\text{gen}}, \mathbf{e}_{\text{gen}})^\top$ with $\mathbf{s}_{\text{gen}} \in \mathcal{R}_{2q}^{l-1}$, $\mathbf{e}_{\text{gen}} \in \mathcal{R}_{2q}^k$.
 3. Compute $\mathbf{b}_{\text{gen}} = \mathbf{A}_{\text{gen}} \mathbf{s}_{\text{gen}} + \mathbf{e}_{\text{gen}} \pmod{q}$.
 4. Form $\mathbf{A} = (-2\mathbf{b}_{\text{gen}} + q\mathbf{j} \mid 2\mathbf{A}_{\text{gen}} \mid 2\mathbf{Id}_k) \in \mathcal{R}_{2q}^{k \times m}$ and set $\mathbf{s} = (1, \mathbf{s}_{\text{gen}}, \mathbf{e}_{\text{gen}})^\top$. Note that $\mathbf{A}\mathbf{s} = q\mathbf{j} \pmod{2q}$.
 5. Compute $\mathbf{G} = \mathbf{B}\mathbf{B}^* \in \mathcal{R}_{2q}^{d \times d}$ (in the coefficient embedding an $nd \times nd$ positive definite matrix).
 6. Output $\text{pk} = (\mathbf{A}, \mathbf{G})$ and $\text{sk} = (\mathbf{B}, \mathbf{s})$.
- $\text{Sign}(\text{sk}, \mu)$:
 1. Sample $\mathbf{X}' \in \mathcal{R}_q^{d-1}$ with each coefficient from $D_{\mathbb{Z}, \sigma}$.
 2. Let $\mathbf{B}_{\text{re}}$ be $\mathbf{B}$ without its first column. Sample $y_1 \in \mathcal{R}_q$ with each coefficient from D_{σ_y} conditioned on $|y_{1j}| \leq B_1 + R$, and sample an independent noise vector $\mathbf{e}_{\text{bottom}} \in \mathcal{R}_q^d$ with each coefficient drawn from $D_{\mathbb{Z}, 2\sigma}$. Then set $\mathbf{y}_{\text{bottom}} = 2\mathbf{B}_{\text{re}}\mathbf{X}' + \mathbf{e}_{\text{bottom}}$ and $\mathbf{y} = \begin{pmatrix} y_1 \\ \mathbf{y}_{\text{bottom}} \end{pmatrix}$.
 3. $\mathbf{w} = \mathbf{A}\mathbf{y} \pmod{2q}$.
 4. $c = \text{Hash}(\mathbf{w}, \mu) \in \mathcal{R}_q$ with coefficients in $\{0, 1\}$.
 5. For each coefficient c_j of c and y_{1j} of y_1 , run $\text{Rejection_Sampling}(y_{1j}, c_j)$ (Algorithm 3) to obtain $(z_{1j}, 2x_j)$.
 - + Set $v_j = 2x_j + c_j$; assemble the polynomial $V \in \mathcal{R}_q$ with coefficients $\{v_j\}$.
 - + Define $X_0 = V/2$ (so the j -th coefficient of $2X_0$ is v_j). By construction the coefficients of $2X_0$ follow $\chi_{2\sigma, R}^{(c)}$.
 - + Assemble $z_1 \in \mathcal{R}_q$ from $\{z_{1j}\}$.
 6. Compute $\mathbf{z} = \mathbf{y} + 2\mathbf{s}X_0$. Observe that z_1 is generated independently by the rejection sampler, while the bottom part satisfies $\mathbf{z}_{\text{bottom}} = 2\mathbf{B}_{\text{re}}\mathbf{X}' + \mathbf{e}_{\text{bottom}} + 2\mathbf{s}_{\text{tail}}X_0 = 2\mathbf{B} \begin{pmatrix} X_0 \\ \mathbf{X}' \end{pmatrix} + \mathbf{e}_{\text{bottom}} \pmod{2q}$. Consequently, z_1 and $\mathbf{z}_{\text{bottom}}$ are independent.

7. If $\|\mathbf{z}\|_\infty > B$, restart from step 1.
 8. Output $(\mathbf{z}, c)$.
- **Verify**($\mathbf{pk}, \mu, (\mathbf{z}, c)$):
1. $\mathbf{w} = \mathbf{A}\mathbf{z} - q\mathbf{j}c \bmod 2q$.
 2. Check $c = \text{Hash}(\mathbf{w}, \mu)$ and $\|\mathbf{z}\|_\infty \leq B$.

Correctness and distribution. Using $\mathbf{A}\mathbf{s} = q\mathbf{j}$ and $2X_0 \equiv c \pmod{2}$, one verifies

$$\mathbf{A}\mathbf{z} = \mathbf{A}\mathbf{y} + 2\mathbf{A}\mathbf{s}X_0 = \mathbf{w} + 2q\mathbf{j}X_0 \equiv \mathbf{w} + q\mathbf{j}c \pmod{2q}$$

so verification succeeds. Before truncation, the bottom part has the form $\mathbf{z}_{\text{bottom}} = \mathbf{e}_{\text{bottom}} + 2\mathbf{B}(X_0, \mathbf{X}')^\top$. Since $\mathbf{e}_{\text{bottom}}$ is independent of $(X_0, \mathbf{X}')$ and all components are drawn from $D_{\mathbb{Z}, \sigma}$, the covariance matrix of $\mathbf{z}_{\text{bottom}}$ (in the coefficient embedding) is

$$\Sigma_{\text{bottom}} = \frac{4\sigma^2}{2\pi} \mathbf{Id}_{nd} + \frac{4\sigma^2}{2\pi} \mathbf{G},$$

where the first term is the contribution of $\mathbf{e}_{\text{bottom}}$ and $\mathbf{G} = \mathbf{B}\mathbf{B}^*$. We use the unified Gaussian parameterization $\rho_\sigma(x) = \exp(-\pi x^2/\sigma^2)$ so that the variance equals $\sigma^2/(2\pi)$. The first component z_1 follows a truncated discrete Gaussian with variance $\sigma_y^2/(2\pi)$ (conditioned on $|z_1| \leq B_1$) and is independent of $\mathbf{z}_{\text{bottom}}$. Hence the entire signature $\mathbf{z}$ is distributed as a product of independent Gaussians, with a block-diagonal covariance fully determined by public parameters and $\mathbf{G}$. A simulator that knows $\mathbf{G}$ can reproduce this distribution, guaranteeing statistical (or computational) hiding under the $\text{GA-MLWE}_{\text{full}}$ assumption.

Security. Under the $\text{GA-MLWE}_{\text{full}}$ assumption, the simulator receives the public Gram matrix $\mathbf{G} = \mathbf{B}\mathbf{B}^*$ together with the public key $\mathbf{A}$. It generates a signature on a message μ as follows:

1. Sample the first component $z_1 \in \mathbb{Z}^n$ directly from the truncated discrete Gaussian $D_{\sigma_y}^n$ conditioned on $\|z_1\|_\infty \leq B_1$, exactly as in the real signing algorithm.
2. For the bottom part $\mathbf{z}_{\text{bottom}}$, produce two independent components $\mathbf{U}_{\text{sim}}$ and $\mathbf{V}_{\text{sim}}$:
 - **Parity-masking noise** $\mathbf{V}_{\text{sim}}$. In the real scheme, the masking noise is sampled from $D_{\mathbb{Z}^{nd}, 2\sigma}$ precisely to ensure that its parity distribution is statistically uniform. The simulator therefore samples $\mathbf{V}_{\text{sim}} \leftarrow D_{\mathbb{Z}^{nd}, 2\sigma}$, which is a perfect simulation of the true noise (zero statistical distance). The choice of parameter 2σ guarantees

$$2\sigma \geq 2\eta_\varepsilon(\mathbb{Z}^{nd}) = \eta_\varepsilon(2\mathbb{Z}^{nd}),$$

so the distribution of $\mathbf{V}_{\text{sim}} \bmod 2$ is 2ε -close to uniform over all parity patterns. Consequently, for *any* fixed shift $\mathbf{t} \in \mathbb{Z}^{nd}$, the sum $\mathbf{t} + \mathbf{V}_{\text{sim}}$ taken modulo 2 is also 2ε -close to uniform. This completely masks the

parity information that the structured component $\mathbf{U}$ might carry, in particular the secret-dependent term ($\mathbf{s}_{\text{tail}}X_0 \bmod 2 = \mathbf{s}_{\text{tail}}c \bmod 2$) that determines the true parity shift.

- **Structured component $\mathbf{U}_{\text{sim}}$.** Recall that $\mathbf{B} \in \mathbb{Z}^{nd \times nd}$ is a unimodular matrix ($\det(\mathbf{B}) = 1$) whose first column is the secret $\mathbf{s}_{\text{tail}}$. In the real scheme, $\mathbf{U} = 2\mathbf{B}\mathbf{X}_{\text{full}}$ with $\mathbf{X}_{\text{full}} \sim D_{\mathbb{Z}^{nd}, \sigma}$. Because $\mathbf{B}$ is invertible over $\mathbb{Z}$, the kernel of $2\mathbf{B}$ is trivial, and the linear transformation lemma (Lemma 2.3) applies under the condition $\sigma \geq \eta_\varepsilon(\mathbb{Z}^{nd})$. The lemma guarantees that the distribution of $\mathbf{U}$ is within $4\varepsilon + O(\varepsilon^2)$ of the rounded continuous Gaussian $\lfloor \mathcal{N}(\mathbf{0}, \frac{4\sigma^2}{2\pi}\mathbf{G}) \rfloor$. The simulator therefore samples $\mathbf{U}_{\text{sim}}$ from this continuous Gaussian and rounds each coordinate to the nearest integer, achieving a statistical defect of at most $4\varepsilon + O(\varepsilon^2)$ relative to the real $\mathbf{U}$.

3. Assemble $\mathbf{z} = (z_1, \mathbf{U}_{\text{sim}} + \mathbf{V}_{\text{sim}})$. If $\|\mathbf{z}\|_\infty > B$, restart.
4. Choose a uniformly random challenge $c \in \mathcal{C}$ and compute $\mathbf{w} = \mathbf{A}\mathbf{z} - q\mathbf{j}c \bmod 2q$. Program the random oracle so that $\text{Hash}(\mathbf{w}, \mu) = c$.
5. Output $(\mathbf{z}, c)$.

Statistical closeness. The component $\mathbf{V}_{\text{sim}}$ is perfectly simulated (distance 0) and its parameter 2σ independently satisfies the parity-masking condition $2\sigma \geq \eta_\varepsilon(2\mathbb{Z}^{nd})$. The component $\mathbf{U}_{\text{sim}}$ deviates from the real $\mathbf{U}$ by at most $4\varepsilon + O(\varepsilon^2)$ under the separate condition $\sigma \geq \eta_\varepsilon(\mathbb{Z}^{nd})$, which controls the rounding error of the continuous Gaussian approximation. The two conditions are satisfied simultaneously without conflating their parameters: the structured noise uses standard deviation σ , while the masking noise uses 2σ . By the additivity of statistical distance for independent pairs, the overall max-log distance between the simulated and real $\mathbf{z}_{\text{bottom}}$ is $4\varepsilon + O(\varepsilon^2)$. Together with the perfect simulation of z_1 , this shows that the signature distribution produced by the simulator is statistically close to the real one. The simulator then programs the random oracle to match the computed commitment $\mathbf{w} = \mathbf{A}\mathbf{z} - q\mathbf{j}c \bmod 2q$, completing the EUF-CMA reduction.

Practical Variant Π_{real} (without Gram matrix). For deployment we simply drop $\mathbf{G}$ from the public key:

$\text{KeyGen}_{\text{real}}(1^\lambda) : \text{run } \text{KeyGen}_{\text{th}}(1^\lambda) \text{ and output } \text{pk} = (\mathbf{A}), \text{ sk} = (\mathbf{B}, \mathbf{s}).$

Signing and verification are unchanged. Any EUF-CMA adversary against Π_{real} can be transformed into an adversary against Π_{th} by forwarding queries and ignoring $\mathbf{G}$; the reduction is perfect. Hence both variants enjoy the same provable security under GA-MLWE_{full}, while Π_{real} achieves a smaller public key with no computational overhead, the proof of this construction can be found in Appendix C.

3.3.4 Construction 3: Extended Unimodular Matrix Scheme

In Construction 2, the signature length is tied to $d = k + l - 1$. To decouple the security requirement on the module dimensions from the compactness of the secret, we embed the short $d \times d$ unimodular matrix into a larger $D \times D$ unimodular matrix $\mathbf{B}_{\text{ext}}$, with $D \geq d + 1$ chosen large enough to guarantee resistance against algebraic distinguishers (see Section 4.6). This completely removes the contribution of the (potentially long) last row of $\mathbf{B}_{\text{ext}}$ to the signature norm.

Consider a generator **GenUnimod** which outputs a unimodular matrix $\mathbf{B}_{\text{ext}} \in \mathcal{R}_q^{D \times D}$ with $\mathbf{B}_{\text{ext}} \cdot \mathbf{e}_1 = \mathbf{s}'_{\text{tail}}$ and the following properties:

- The first $D - 1$ rows of $\mathbf{B}_{\text{ext}}$ have infinity norm $\leq B_{\text{max}} = O(\sqrt{n})$ (short rows).
- The last row may have larger coefficients; its norm is polynomially bounded (e.g., via Babai reduction to the orthogonal lattice of the first $D - 1$ rows). For our construction it is *not* required to be short.

Such a generator can be realized using module-NTRU trapdoor techniques [15], where the first $D - 1$ rows are sampled as random short vectors and the last row is determined by solving a linear equation to make the determinant 1, followed by Babai reduction if a bound is desired.

Construction 3.3 (Π_{direct}). *Let $d = l - 1 + k$ (the original secret length). We denote by $\mathbf{B}_{\text{short}} \in \mathcal{R}_q^{d \times D}$ the matrix consisting of the first d rows of $\mathbf{B}_{\text{ext}}$, and by $\mathbf{B}_{\text{re}} \in \mathcal{R}_q^{D \times (D-1)}$ the matrix obtained by deleting the first column of $\mathbf{B}_{\text{ext}}$.*

- **KeyGen**(1^λ):
 1. Sample $\mathbf{A}_{\text{gen}} \leftarrow \mathcal{R}_q^{k \times (l-1)}$.
 2. Generate $(\mathbf{B}_{\text{ext}}, \mathbf{s}'_{\text{tail}}) \leftarrow \text{GenUnimod}(1^\lambda, D)$ where $\mathbf{B}_{\text{ext}} \in \mathcal{R}_{2q}^{D \times D}$ is unimodular. Parse the first column as $\mathbf{s}'_{\text{tail}} = (\mathbf{s}_{\text{tail}}, \mathbf{v})^\top$ with $\mathbf{s}_{\text{tail}} = (\mathbf{s}_{\text{gen}}, \mathbf{e}_{\text{gen}})^\top \in \mathcal{R}_{2q}^d$ and $\mathbf{v} \in \mathcal{R}_{2q}^{D-d}$ (the latter is discarded).
 3. Compute $\mathbf{b}_{\text{gen}} = \mathbf{A}_{\text{gen}} \mathbf{s}_{\text{gen}} + \mathbf{e}_{\text{gen}} \pmod q$ and form $\mathbf{A} = (-2\mathbf{b}_{\text{gen}} + q\mathbf{j} \mid 2\mathbf{A}_{\text{gen}} \mid 2\mathbf{Id}_k) \in \mathcal{R}_{2q}^{k \times m}$ where $m = 1 + d$.
 4. Append $D - d$ zero columns to obtain $\mathbf{A}' = [\mathbf{A} \mid \mathbf{0}_{k \times (D-d)}] \in \mathcal{R}_{2q}^{k \times (D+1)}$. This zero columns align with the component that will later be discarded.
 5. Set $\mathbf{s} = (1, \mathbf{s}_{\text{tail}})^\top$ and $\mathbf{s}' = (1, \mathbf{s}_{\text{tail}}, \mathbf{v})^\top$; then $\mathbf{A}' \mathbf{s}' = \mathbf{A} \mathbf{s} = q\mathbf{j} \pmod{2q}$.
 6. Compute the public Gram matrix $\mathbf{G}_0 = \mathbf{B}_{\text{short}} \mathbf{B}_{\text{short}}^* \in \mathcal{R}_{2q}^{d \times d}$.
 7. Output $\text{pk} = (\mathbf{A}, \mathbf{G}_0)$ and $\text{sk} = (\mathbf{B}_{\text{ext}}, \mathbf{s})$. (Note: $\mathbf{A}$ alone suffices for verification because $\mathbf{A}' \mathbf{z}' = \mathbf{A} \mathbf{z}$ after discarding the components that correspond to the zero columns.)
- **Sign**(sk, μ):
 1. Sample $\mathbf{X}' \in \mathcal{R}_q^{D-1}$ with coefficients from $D_{\mathbb{Z}, \sigma}$. (Note: $\mathbf{X}'$ has dimension $D - 1$ because $\mathbf{B}_{\text{re}}$ has $D - 1$ columns.)

2. Sample $y_1 \in \mathcal{R}_q$ with each coefficient from D_{σ_y} conditioned on $|y_{1j}| \leq B_1 + R$. Sample $\mathbf{e}_{\text{bottom}} \in \mathcal{R}_{2q}^d$ and $\mathbf{e}_{\text{last}} \in \mathcal{R}_{2q}^{D-d}$ with independent $D_{\mathbb{Z}, 2\sigma}$ coefficients. Set $\mathbf{e}'_{\text{bottom}} = (\mathbf{e}_{\text{bottom}}, \mathbf{e}_{\text{last}})^\top \in \mathcal{R}_{2q}^D$. Compute $\mathbf{y}'_{\text{bottom}} = 2\mathbf{B}_{\text{re}}\mathbf{X}' + \mathbf{e}'_{\text{bottom}}$ and $\mathbf{y}' = \begin{pmatrix} y_1 \\ \mathbf{y}'_{\text{bottom}} \end{pmatrix} \in \mathcal{R}_{2q}^{D+1}$. Let $\mathbf{y}$ be $\mathbf{y}'$ with its last $D - d$ elements removed.
 3. $\mathbf{w} = \mathbf{A}'\mathbf{y}' = \mathbf{A}\mathbf{y} \bmod 2q$.
 4. $c = \text{Hash}(\mathbf{w}, \mu)$; run `Rejection_Sampling` to obtain z_1 and X_0 as in Construction 2.
 5. $\mathbf{z}' = \mathbf{y}' + 2\mathbf{s}'X_0 \in \mathcal{R}_{2q}^{D+1}$.
 6. **Discard the last $D - d$ component:** $\mathbf{z} = \mathbf{z}'_{[1:1+d]}$. The discarded components involve last rows (including the last row with large coefficients) of $\mathbf{B}_{\text{ext}}$ and the noise $\mathbf{e}_{\text{last}}$; it is never revealed. The retained vector is $\mathbf{z} = (z_1, \mathbf{z}_{\text{bottom}})^\top$ with $\mathbf{z}_{\text{bottom}} \in \mathcal{R}_{2q}^d$.
 7. If $\|\mathbf{z}\|_\infty > B$, restart; else output $(\mathbf{z}, c)$.
- `Verify(pk,  $\mu$ ,  $(\mathbf{z}, c)$ ):`
1. $\mathbf{w} = \mathbf{A}\mathbf{z} - q\mathbf{j}c \bmod 2q$.
 2. Check $c = \text{Hash}(\mathbf{w}, \mu)$ and $\|\mathbf{z}\|_\infty \leq B$.

Correctness and distribution. Because the last $D - d$ columns of $\mathbf{A}'$ are zero and $\mathbf{z}$ discards the corresponding components, $\mathbf{A}'\mathbf{z}' = \mathbf{A}\mathbf{z}$. Verification therefore succeeds.

We now derive the form of $\mathbf{z}_{\text{bottom}}$ explicitly. Write $\mathbf{B}_{\text{ext}} = [\mathbf{s}'_{\text{tail}} \mid \mathbf{B}_{\text{re}}]$. Then

$$\mathbf{z}'_{\text{bottom}} = 2\mathbf{B}_{\text{re}}\mathbf{X}' + \mathbf{e}'_{\text{bottom}} + 2\mathbf{s}'_{\text{tail}}X_0 = 2\mathbf{B}_{\text{ext}} \begin{pmatrix} X_0 \\ \mathbf{X}' \end{pmatrix} + \mathbf{e}'_{\text{bottom}}.$$

Discarding the last $D - d$ components of $\mathbf{z}'_{\text{bottom}}$ (corresponding to the last $D - d$ rows of $\mathbf{B}_{\text{ext}}$ and the noise $\mathbf{e}_{\text{last}}$) gives the retained part

$$\mathbf{z}_{\text{bottom}} = 2\mathbf{B}_{\text{short}} \begin{pmatrix} X_0 \\ \mathbf{X}' \end{pmatrix} + \mathbf{e}_{\text{bottom}},$$

where all components of $\mathbf{e}_{\text{bottom}}$ and the vector $(X_0, \mathbf{X}')$ are independent with coefficients from $D_{\mathbb{Z}, 2\sigma}$ and $D_{\mathbb{Z}, \sigma}$. Hence the covariance matrix of $\mathbf{z}_{\text{bottom}}$ (in the coefficient embedding) is

$$\Sigma_{\text{bottom}} = \frac{4\sigma^2}{2\pi} \mathbf{Id}_{nd} + \frac{4\sigma^2}{2\pi} \mathbf{G}_0, \quad \mathbf{G}_0 = \mathbf{B}_{\text{short}} \mathbf{B}_{\text{short}}^*.$$

Security. Π_{direct} is EUF-CMA secure in the random oracle model under the GA-MLWE_{short} assumption, where the auxiliary information is $\mathcal{F}_{\text{short}}(\mathbf{s}) = \mathbf{G}_0$. Because $\mathbf{G}_0$ is a submatrix of the full Gram matrix, this assumption is *weaker* (hence more conservative) than GA-MLWE_{full}. The proof follows that of Construction 2, with the only change that the simulator uses the submatrix $\mathbf{G}_0 =$

$\mathbf{B}_{\text{short}}\mathbf{B}_{\text{short}}^*$ instead of the full Gram matrix $\mathbf{G}$. Concretely, the simulator decomposes the bottom signature vector as $\mathbf{z}_{\text{bottom}} = \mathbf{U} + \mathbf{V}$ with $\mathbf{V} \sim D_{\mathbb{Z}^{nd}, 2\sigma}$ and $\mathbf{U} = 2\mathbf{B}_{\text{short}}(\mathbf{X}_0, \mathbf{X}')^\top$. It then:

1. samples $\mathbf{V}_{\text{sim}}$ perfectly from the public discrete Gaussian $D_{\mathbb{Z}^{nd}, 2\sigma}$;
2. samples $\mathbf{U}_{\text{sim}}$ by extracting the first nd coordinates of the (continuous approximation of the) extended vector $2\mathbf{B}_{\text{ext}}\mathbf{X}_{\text{ext}}$ with $\mathbf{X}_{\text{ext}} \sim D_{\mathbb{Z}^{nD}, \sigma}$. Thanks to the unimodularity of $\mathbf{B}_{\text{ext}}$, the marginal distribution of these coordinates is well-approximated by $\mathcal{N}(\mathbf{0}, \frac{4\sigma^2}{2\pi}\mathbf{G}_0)$; rounding this continuous Gaussian gives a distribution within $4\varepsilon + O(\varepsilon^2)$ of the true $\mathbf{U}$.
3. sets $\mathbf{z}_{\text{bottom}} = \mathbf{U}_{\text{sim}} + \mathbf{V}_{\text{sim}}$.

No coordinate-dropping is needed on the simulator's side, because the marginalisation over the discarded components is already encoded in $\mathbf{G}_0$. The overall statistical distance from the real distribution is bounded by $4\varepsilon + O(\varepsilon^2)$ under the smoothing condition $\sigma \geq \eta_\varepsilon(\mathbb{Z}^{nD})$ (which is only logarithmically larger than $\eta_\varepsilon(\mathbb{Z}^{nd})$ and is satisfied by our parameter choice).

Practical variant. As before, $\mathbf{G}_0$ can be omitted from the public key, yielding a minimal public key $\mathbf{A}$ with the same EUF-CMA security. A detailed proof can be found in Section 4.

3.3.5 The 3C Sampling Paradigm Revisited

All three constructions adhere to a common blueprint: the final output $\mathbf{z}$ is obtained by applying a *deterministic* integer matrix (derived from the secret key) to a vector of independent discrete Gaussian noises, whose number is *constant* and independent of the security parameter. This is the essence of *Convolution of Constant Count Sampling* (3C Sampling): no floating-point sampling from a varying covariance matrix is required, and the signature distribution can be simulated from public parameters alone.

- **Construction 1 (Automorphism Isotropic).** The matrix is built from explicit cancellation columns and automorphism columns. When the full Galois group is used, the output is statistically indistinguishable from a spherical Gaussian, achieving *statistical* hiding at the cost of a large number of noise columns. Replacing the full Galois group by a random subset G_w of size w eliminates the spherical property but drastically reduces the matrix width; security then relies on the $\text{GA-MLWE}_{\text{diag}}$ assumption, where the adversary is given only the diagonal Gram blocks of the secret components.
- **Construction 2 (Unimodular Matrix).** The deterministic matrix is encapsulated into a short unimodular $\mathbf{B} \in \mathcal{R}_{2q}^{d \times d}$, together with the identity matrix for the additional noise $\mathbf{e}_{\text{bottom}}$. The fixed component $s_1 = 1$ is handled by the lightweight rejection sampler of Section 3.3.1. The theoretical

variant Π_{th} includes the full Gram matrix $\mathbf{G} = \mathbf{B}\mathbf{B}^*$ in the public key, enabling simulation via rounded continuous Gaussians under the $\text{GA-MLWE}_{\text{full}}$ assumption. The practical variant Π_{real} omits $\mathbf{G}$ entirely, retaining the same security guarantee because the signing algorithm never uses it. Unimodularity ensures that only $\sigma \geq \eta_\varepsilon(\mathbb{Z}^{nd})$ is required for statistical hiding, yielding exceptionally compact signatures.

- **Construction 3 (Extended Unimodular Matrix).** Building on Construction 2, a larger $D \times D$ unimodular matrix $\mathbf{B}$ is generated, and the last $D - d$ components of the signature are simply discarded. The public key contains only the Gram matrix $\mathbf{G}_0 = \mathbf{B}_{\text{short}}\mathbf{B}_{\text{short}}^*$ of the first d (short) rows. This completely eliminates any penalty from a potentially longer last row, yielding an even more compact signature. Security relies on $\text{GA-MLWE}_{\text{short}}$, which is *weaker* (hence more conservative) than the full Gram-aided assumption.

The evolution illustrates how the same core idea leads to a flexible design space with trade-offs between information-theoretic security, compactness, and the strength of the underlying algebraic assumptions. All three constructions fit into the unified GA-MLWE framework of Section 3.2, with progressively weaker auxiliary information and progressively better practical efficiency. Among them, the practical variant of Construction 3 achieves the smallest public key and signature size, making it the recommended instantiation of the Rhyme signature scheme.

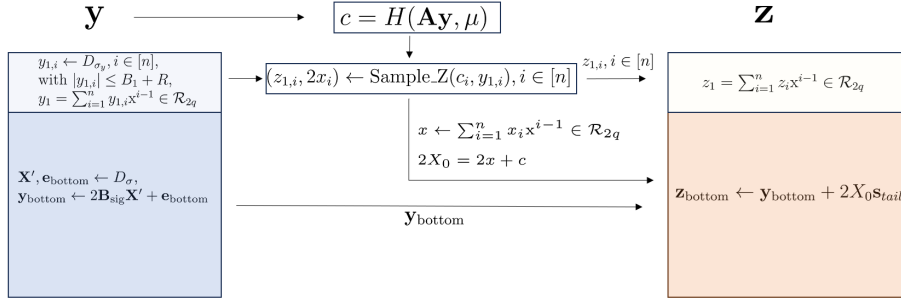


Figure 1. 3C Sampling Method.

3.4 Signature Algorithms

Relation to Construction 3. The algorithms below instantiate the practical variant of Construction 3 (Section 3.3.4), omitting the Gram matrix entirely.

In that construction, a signature is formed from an extended vector and the last components are discarded; this allows using a unimodular matrix $\mathbf{B}_{\text{ext}}$ of dimension D . To avoid explicitly computing and then discarding components, our key generation precomputes a matrix $\mathbf{B}_{\text{sig}} \in \mathcal{R}_q^{(k+l-1) \times (D-1)}$ by deleting the first column and the last $D - (k + l - 1)$ rows of $\mathbf{B}_{\text{ext}}$. Signing then directly uses $\mathbf{B}_{\text{sig}}$ to produce a compact output of size $k+l$ without ever handling a superfluous coordinate. The public key consists solely of $(\text{seed}_A, \mathbf{b}_{\text{gen}})$ and contains no Gram information, realising the most bandwidth-efficient instantiation of Rhyme.

Key generation. Our key generation method follows a classical framework of bimodal distributions where we need to ensure $\mathbf{A}\mathbf{s} \bmod 2q = \mathbf{q}\mathbf{j}$ as in HAETAE and G+G. The difference is that $\mathbf{s}_{\text{gen}}, \mathbf{e}_{\text{gen}}$ are the first $k+l-1$ components of the first column of a $D \times D$ unimodular matrix output by `GenUnimod`. Besides, since the last $D - (k + l - 1)$ rows of $\mathbf{B}_{\text{ext}}$ will be discarded and never used in signing or verification, we can run a simplified `GenUnimod` procedure that only guarantees the existence of a unimodular completion without explicitly computing the discarded rows, which greatly improves the efficiency of key generation.

To build a key pair $(\mathbf{A}, \mathbf{s})$ satisfying our requirements, we start by generating a unimodular matrix of dimension D by using simplified `GenUnimod`, then extracting the first $k + l - 1$ elements of the first column of the matrix as $\mathbf{s}_{\text{gen}}, \mathbf{e}_{\text{gen}}$. Then we generate an MLWE sample $\mathbf{b}_{\text{gen}} = \mathbf{A}_{\text{gen}}\mathbf{s}_{\text{gen}} + \mathbf{e}_{\text{gen}} \bmod q$, where $\mathbf{A}_{\text{gen}} \leftarrow U(\mathcal{R}_q^{k \times (l-1)})$.

Then we define $\mathbf{A} = (-2\mathbf{b}_{\text{gen}} + \mathbf{q}\mathbf{j} \mid 2\mathbf{A}_{\text{gen}} \mid 2\mathbf{Id}_k)$. And $\mathbf{s} = (1, \mathbf{s}_{\text{gen}}, \mathbf{e}_{\text{gen}})$ which ensures that $\mathbf{A}\mathbf{s} = \mathbf{q}\mathbf{j} \bmod 2q$. In practice, the verification key is then a combination of $\mathbf{b}_{\text{gen}}$ and the seed that allows generating $\mathbf{A}_{\text{gen}}$.

Algorithm 5 Key Generation

Input: 1^λ
Output: sk, pk

- 1: $\text{seed} \leftarrow \{0, 1\}^{\rho_0}$
- 2: $(\text{seed}_A, \text{seed}_{sk}, K) \leftarrow \text{H}_{\text{gen}}(\text{seed})$
- 3: $\mathbf{A}_{\text{gen}} \leftarrow \text{expandA}(\text{seed}_A) \in \mathcal{R}_q^{k \times (l-1)}$
- 4: $(\mathbf{B}_{\text{ext}}, \mathbf{s}'_{\text{tail}}) \leftarrow \text{GenUnimod}(1^\lambda, D)$
- 5: $(\mathbf{s}_{\text{gen}}, \mathbf{e}_{\text{gen}}, \mathbf{v}) \leftarrow \mathbf{s}'_{\text{tail}}$, where $\mathbf{s}_{\text{gen}} \in \mathcal{R}_q^{l-1}$, $\mathbf{e}_{\text{gen}} \in \mathcal{R}_q^k$, $\mathbf{v} \in \mathcal{R}_q^{D-(k+l-1)}$ // Parse first column: last $D - (k + l - 1)$ elements are discarded as $\mathbf{v}$
- 6: $\mathbf{b}_{\text{gen}} \leftarrow \mathbf{A}_{\text{gen}}\mathbf{s}_{\text{gen}} + \mathbf{e}_{\text{gen}} \bmod q \in \mathcal{R}_q^k$
- 7: $\mathbf{A} \leftarrow (-2\mathbf{b}_{\text{gen}} + \mathbf{q}\mathbf{j} \mid 2\mathbf{A}_{\text{gen}} \mid 2\mathbf{I}_k) \bmod 2q \in \mathcal{R}_{2q}^{k \times (k+l)}$
- 8: $\mathbf{s} \leftarrow (1, \mathbf{s}_{\text{gen}}, \mathbf{e}_{\text{gen}}) = (1, \mathbf{s}_{\text{tail}}) \in \mathcal{R}_{2q}^{k+l}$
- 9: $\mathbf{B}_{\text{sig}} \leftarrow \mathbf{B}_{\text{ext}}[1:k+l-1, 2:D] \in \mathcal{R}_{2q}^{(k+l-1) \times (D-1)}$ {remove the last $D - (k + l - 1)$ rows and first column}
- 10: **return** $\text{sk} = (\mathbf{s}, \mathbf{B}_{\text{sig}}, K)$, $\text{pk} = (\text{seed}_A, \mathbf{b}_{\text{gen}})$

Sign process. The signing algorithm produces a signature $\sigma = (\mathbf{z}_{\text{comp}}, c)$ on a message M using the secret key $\text{sk} = (\mathbf{s}, \mathbf{B}_{\text{sig}}, K)$ and the public key $\text{pk} = (\text{seed}_A, \mathbf{b}_{\text{gen}})$. First, the public matrix is reconstructed: $\mathbf{A}_{\text{gen}}$ is expanded from seed_A and $\mathbf{A} \in \mathcal{R}_{2q}^{k \times (k+l)}$ is formed as in key generation. A random seed seed_{sig} is derived from K, M and a deterministic nonce.

The core of signing is a rejection loop. In each iteration, three noise terms are sampled using fresh randomness derived from seed_{sig} and a counter: a polynomial $y_1 \in \mathcal{R}_q$ (with coefficients truncated to a fixed range), a vector $\mathbf{X}' \in \mathcal{R}_q^{D-1}$, and an independent Gaussian vector $\mathbf{e}_{\text{bottom}} \in \mathcal{R}_q^{k+l-1}$. The matrix $\mathbf{B}_{\text{sig}} \in \mathcal{R}_{2q}^{(k+l-1) \times (D-1)}$ is obtained from the secret unimodular matrix $\mathbf{B}_{\text{ext}}$ by deleting its first column and its last $D - (k + l - 1)$ rows. The bottom part of the masking vector is computed as $\mathbf{y}_{\text{bottom}} = 2\mathbf{B}_{\text{sig}}\mathbf{X}' + \mathbf{e}_{\text{bottom}} \bmod 2q$. Together with y_1 , this yields the full masking vector $\mathbf{y} = (y_1, \mathbf{y}_{\text{bottom}})^\top$.

The hash $\mathbf{w} = \mathbf{A}\mathbf{y} \bmod 2q$ is computed, and a binary challenge $c \in \mathcal{R}_q$ is sampled from the distribution $\text{SampleInBall}(\mathbf{w}, \mu)$ where μ is a hash binding the message and public key. The coefficients of y_1 and the corresponding bits of c are fed to the lightweight rejection sampler Sample.Z (Algorithm 3), which either returns a pair $(z_1, 2x)$ or fails. On failure, the loop retries with an increased counter.

If the rejection sampler succeeds, set $2X_0 = 2x + c \in \mathcal{R}_q$ (so that the coefficients of $2X_0$ obtain the correct parity). The remaining part of the signature is then $\mathbf{z}_{\text{bottom}} = \mathbf{y}_{\text{bottom}} + 2\mathbf{s}_{\text{tail}}X_0 \in \mathcal{R}_{2q}^{k+l-1}$, where $\mathbf{s}_{\text{tail}}$ is the tail of the secret vector $\mathbf{s} = (1, \mathbf{s}_{\text{tail}})^\top$. The complete output vector is $\mathbf{z} = (z_1, \mathbf{z}_{\text{bottom}})$, which coincides with the first $k + l$ components of the extended signature; the discarded component does not need to be computed.

If the signature's norm satisfies the preset conditions, the loop terminates; otherwise the counter is incremented and the whole iteration restarts. Finally, the signature components are compressed using an asymmetric numeral system (ANS) encoder, yielding the compact representation $\mathbf{z}_{\text{comp}}$. The output signature is $(\mathbf{z}_{\text{comp}}, c)$.

Verify process. Our verify process is similar to HAETA and G+G without using hints. However, the computation of $\mathbf{A}\mathbf{z}$ can be simplified in implementation. Since the last k columns of $\mathbf{A}$ are $2\mathbf{Id}_k$, we can write $\mathbf{A} = (\mathbf{A}_0 \mid 2\mathbf{Id}_k)$ and partition $\mathbf{z} = (\mathbf{z}_l, \mathbf{z}_k)^\top$ with $\mathbf{z}_l \in \mathcal{R}_{2q}^l$ and $\mathbf{z}_k \in \mathcal{R}_{2q}^k$. Then we can compute $\mathbf{A}\mathbf{z} = \mathbf{A}_0\mathbf{z}_l + \mathbf{z}_k + \mathbf{z}_k$ to decrease the number of multiplications for better efficiency.

Specification. We give the description of our signature scheme with the following building blocks:

- Hash function H_{gen} for generating the seeds and hashing the messages,
- Extendable output function expandA for deriving $\mathbf{A}_{\text{gen}}$ from seed_A ,
- Extendable output function expand for deriving a uniformly random bit string

Algorithm 6 SIG-Sign

Input: sk, pk, M **Output:** σ

```
1:  $\mathbf{A}_{\text{gen}} \leftarrow \text{expandA}(\text{seed}_A) \in \mathcal{R}_q^{k \times (l-1)}$ 
2:  $\mathbf{A} \leftarrow (-2\mathbf{b}_{\text{gen}} + q\mathbf{j} \mid 2\mathbf{A}_{\text{gen}} \mid 2\mathbf{Id}_k) \bmod 2q \in \mathcal{R}_{2q}^{k \times (k+l)}$ 
3:  $\mu \leftarrow H_{\text{gen}}(\text{seed}_A, \mathbf{b}_{\text{gen}}, M)$ 
4:  $\text{rnd} \leftarrow 0^{\rho_0}$  {zeroes (from RBG) for the deterministic (random) version}
5:  $\text{seed}_{\text{sig}} \leftarrow H_{\text{gen}}(K, \mu, \text{rnd})$ 
6:  $\text{count} \leftarrow 0, (z_1, 2x) \leftarrow \perp$ 
7: while  $(z_1, 2x) = \perp$  do
8:    $(\text{rt}_1, \text{rt}_2, \text{rt}_3) \leftarrow \text{expand}(\text{seed}_{\text{sig}}, \text{count})$ 
9:    $y_1 \leftarrow \text{Sample\_Y}_1(\text{rt}_1) \in \mathbb{Z}^n$ 
10:   $\mathbf{X}' \leftarrow \text{Sample\_Gaussian}(\sigma, \text{rt}_2) \in \mathcal{R}_{2q}^{D-1}$ 
11:   $\mathbf{e}_{\text{bottom}} \leftarrow \text{Sample\_Gaussian}(2\sigma, \text{rt}_3) \in \mathcal{R}_{2q}^{k+l-1}$ 
12:   $\mathbf{y}_{\text{bottom}} \leftarrow 2\mathbf{B}_{\text{sig}}\mathbf{X}' + \mathbf{e}_{\text{bottom}} \in \mathcal{R}_{2q}^{k+l-1}$ 
13:   $\mathbf{w} \leftarrow \mathbf{A} \begin{pmatrix} y_1 \\ \mathbf{y}_{\text{bottom}} \end{pmatrix} \bmod 2q$ 
14:   $c \leftarrow \text{SampleInBall}(\mathbf{w}, \mu) \in \mathcal{R}_2$ 
15:   $(z_1, 2x) \leftarrow \text{Sample\_Z}(c, y_1, \text{rt}_2)$ 
16:  if  $(z_1, 2x) = \perp$  then
17:     $\text{count} \leftarrow \text{count} + 1$ 
18:  else
19:     $2X_0 \leftarrow 2x + c \in \mathcal{R}_{2q}$ 
20:     $\mathbf{z}_{\text{bottom}} \leftarrow \mathbf{y}_{\text{bottom}} + 2\mathbf{s}_{\text{tail}}X_0 \in \mathcal{R}_{2q}^{k+l-1}$ 
21:     $\mathbf{z} \leftarrow (z_1, \mathbf{z}_{\text{bottom}})$ 
22:    if  $\|z_1\|_{\infty} > B_1$  or  $\|\mathbf{z}_{\text{bottom}}\|_{\infty} > B_2$  or  $\|\mathbf{z}\|_2 > B_{L2}$  then
23:       $\text{count} \leftarrow \text{count} + 1, (z_1, 2x) \leftarrow \perp$ 
24:    end if
25:  end if
26: end while
27:  $\mathbf{z}_{\text{comp}} \leftarrow \text{rANSEncode}(\mathbf{z})$ 
28: return  $(\mathbf{z}_{\text{comp}}, c)$ 
```

from seed ,

The above building blocks can be implemented with symmetric primitives. Specifically, we use SHAKE256/SM3 for the hash functions and extendable output functions.

4 Security Analysis

The security proof of the scheme follows the idea of G+G with a few modifications (especially about HVZK). More specifically, we follow the well-established “Lossy Identification + Fiat-Shamir” framework [16, 27], by first designing an

Algorithm 7 SIG_Verify

Input: $pk, M, \mathbf{z}', c$;**Output:** $\{0, 1\}$;

```
1:  $\mathbf{z} \leftarrow \text{rANSDecode}(\mathbf{z}')$ 
2:  $\mathbf{A}_{\text{gen}} \leftarrow \text{expandA}(\text{seed}_A) \in \mathcal{R}_q^{k \times (l-1)}$ 
3:  $\mathbf{A} \leftarrow (-2\mathbf{b}_{\text{gen}} + q\mathbf{j}|2\mathbf{A}_{\text{gen}}|2\mathbf{Id}_k) \bmod 2q \in \mathcal{R}_{2q}^{k \times (k+l)}$ 
4:  $\mathbf{w}' \leftarrow \mathbf{A}\mathbf{z} - qc \cdot \mathbf{j} \bmod 2q$ 
5:  $\mu \leftarrow \text{H}_{\text{gen}}(\text{seed}_A, \mathbf{b}_{\text{gen}}, M)$ 
6: if  $c = \text{SampleInBall}(\mathbf{w}', \mu)$  and  $\|\mathbf{z}_1\|_\infty \leq B_1$  and  $\|\mathbf{z}_{\text{bottom}}\|_\infty \leq B_2$  and  $\|\mathbf{z}\|_2 \leq B_{L2}$  then
7:   return 1
8: else
9:   return 0
10: end if
```

identification protocol (Rhyme_ID) and then applying the Fiat-Shamir transform to obtain the signature scheme.

Existential Unforgeability under Chosen Message Attacks (EU-CMA) is regarded as a standard security notion for digital signature schemes. The adversary is given the verification key and has access to a signing oracle to call on (adaptively) chosen messages. The adversary wins if it forges a valid signature of a new, non-queried message. Strong Existential Unforgeability under Chosen Message Attacks (sEU-CMA) is a slightly stronger security notion than EU-CMA: the adversary wins if it forges a valid signature-message pair that it did not already see.

The security of our signature scheme follows that in HAETAE and G+G, where the reduction can be proved in two ways. The first way is performed in the classical Random Oracle Model (ROM) under the standard MLWE and MSIS assumptions.

And the second way is using the zero-knowledge property of the underlying identification scheme, Existential Unforgeability under No Message Attacks (EU-NMA) reduces to (s)EU-CMA security, and can be proved both in the ROM and the QROM, which is introduced in [16] and [6].

EU-NMA is related to a problem defined in HAETAE, denoted as Bimodal-SelfTargetMSIS. Note that in the ROM, MSIS reduces to BimodalSelfTargetMSIS, but the reduction is not tight and does not readily extend to quantum adversaries (it relies on the forking lemma). This said, this non-tightness and limitation to classical adversaries is not known to reflect any weakness. For setting parameters, we consider the hardness of MSIS and MLWE for relevant parameters. Intuitively, the MLWE assumption is used for security against key-recovery attacks, and the BimodalSelfTargetMSIS used for security against forgeries is identified to the MSIS assumption.

In this section, we first describe the Rhyme identification protocol established on $\mathbb{Z}$, denoted as `Rhyme_ID`. Then in the next subsection, we prove the required properties to compile it into a signature using the Fiat-Shamir heuristic.

4.1 Description of the Rhyme Identification Protocol

Let us first introduce the parameters of the scheme as well as some notations. Let $k, l, n > 0$ and $\mathbf{I} = (\mathbf{Id}_n, \mathbf{0}_{n \times (k-1)n})^\top \in \mathbb{Z}^{kn \times n}$ (i.e. the first n rows form the identity, the remaining $(k-1)n$ rows are zero). Let σ, σ_y and χ be distributions as previously described. The challenge space $\mathcal{C}$ consists of all vectors in $\{0, 1\}^n$ of Hamming weight τ ; we identify such a vector with a ring element $c \in \mathcal{R}_q$ having the same coefficient vector. The scheme is also parametrized by an odd modulus q and acceptance bounds B_1, B_2 .

The identification protocol operates on the *integer lattice* obtained from the ring setting by the coefficient embedding. Dimensions are multiplied by n : a matrix originally of size $a \times b$ over $\mathcal{R}_q$ becomes a $an \times bn$ integer matrix. Accordingly, the unimodular generator `GenUnimod` is called with the integer dimension Dn , and it outputs an integer unimodular matrix $\mathbf{B}_{\text{ext}} \in \mathbb{Z}^{Dn \times Dn}$ whose first column is $\mathbf{S}'_{\text{tail}} \in \mathbb{Z}^{Dn \times n}$.

The Rhyme identification protocol corresponding to Π_{direct} is described in Table 2. The instance generation algorithm samples a verification key $\mathbf{A} \in \mathbb{Z}_{2q}^{kn \times (k+l)n}$ and a signing key $\mathbf{S} \in \mathbb{Z}^{(k+l)n \times n}$ with small-magnitude coefficients such that $\mathbf{A} \cdot \mathbf{S} = q\mathbf{I} \bmod 2q$. In the first phase of the interaction, the prover samples $y_i \leftarrow D_{\sigma_y}$ for $i \in [n]$ (within the fixed range $B_1 + R$), a vector $\mathbf{X}' \in \mathbb{Z}^{(D-1)n}$ and an independent Gaussian vector $\mathbf{e}_{\text{bottom}} \in \mathbb{Z}^{(k+l-1)n}$ with each component following D_σ and $D_{2\sigma}$.

Let $\mathbf{B}_{\text{sig}} \in \mathbb{Z}^{(k+l-1)n \times (D-1)n}$ be the matrix obtained from $\mathbf{B}_{\text{ext}}$ by deleting its first column and its last $D - (k + l - 1)$ rows. The prover computes

$$\mathbf{y}_{\text{bottom}} = 2\mathbf{B}_{\text{sig}}\mathbf{X}' + \mathbf{e}_{\text{bottom}}$$

and assembles the full masking vector

$$\mathbf{y} = (y_1, \dots, y_n, \mathbf{y}_{\text{bottom}}^\top)^\top \in \mathbb{Z}^{(k+l)n}.$$

The prover then sends the commitment $\mathbf{w} = \mathbf{A}\mathbf{y} \bmod 2q$ to the verifier. The protocol is public-coin: the verifier simply samples $\mathbf{c} \leftarrow U(\mathcal{C})$ and sends it to the prover. After receiving $\mathbf{c}$, the prover generates a carefully designed vector $\mathbf{x}$ using the lightweight rejection sampler; the distributions of $\mathbf{y}$ and $\mathbf{x}$ are arranged so that the final response $\mathbf{z}$ follows a publicly sampleable symmetric distribution.

By integrating this identification protocol with the Fiat-Shamir paradigm, we derive a lattice-based signature scheme denoted as `FS[Rhyme_ID, H]`, as formalized in the following theorem. The correctness and security of the scheme directly follow from the properties of the underlying identification protocol.

Igen(1^λ)

1. $\mathbf{A}_{\text{gen}} \leftarrow U(\mathbb{Z}_q^{kn \times (l-1)n})$
2. $(\mathbf{B}_{\text{ext}}, \mathbf{S}'_{\text{tail}}) \leftarrow \text{GenUnimod}(Dn)$
3. $(\mathbf{S}_{\text{gen}}, \mathbf{E}_{\text{gen}}, \mathbf{V}) \leftarrow \mathbf{S}'_{\text{tail}}$, where $\mathbf{S}_{\text{gen}} \in \mathbb{Z}_{2q}^{(l-1)n \times n}$, $\mathbf{E}_{\text{gen}} \in \mathbb{Z}_{2q}^{kn \times n}$, $\mathbf{V} \in \mathbb{Z}_{2q}^{(D-(k+l-1))n \times n}$
4. $\mathbf{B}_{\text{gen}} \leftarrow \mathbf{A}_{\text{gen}} \mathbf{S}_{\text{gen}} + \mathbf{E}_{\text{gen}} \bmod q \in \mathbb{Z}_q^{kn \times n}$
5. $\mathbf{A} \leftarrow (-2\mathbf{B}_{\text{gen}} + q\mathbf{I} \mid 2\mathbf{A}_{\text{gen}} \mid 2\mathbf{Id}_{kn}) \bmod 2q \in \mathbb{Z}_{2q}^{kn \times (k+l)n}$
6. $\mathbf{S} \leftarrow (\mathbf{Id}_n, \mathbf{S}_{\text{gen}}, \mathbf{E}_{\text{gen}})^\top = (\mathbf{Id}_n, \mathbf{S}_{\text{tail}})^\top \in \mathbb{Z}_{2q}^{(k+l)n \times n}$
7. $\mathbf{B}_{\text{short}} \leftarrow \mathbf{B}_{\text{ext}}[1:(k+l-1)n, 1:Dn]$
8. $\mathbf{G}_0 \leftarrow \mathbf{B}_{\text{short}} \mathbf{B}_{\text{short}}^\top \in \mathbb{Z}_{2q}^{(k+l-1)n \times (k+l-1)n}$
9. $vk \leftarrow (\mathbf{A}, \mathbf{G}_0)$, $sk \leftarrow (\mathbf{B}_{\text{ext}}, \mathbf{S})$
10. **return** (vk, sk)

Prover $\mathcal{P}(\mathbf{A}, \mathbf{S})$	Verifier $\mathcal{V}(\mathbf{A})$
$y_i \leftarrow D_{\sigma_y}$ for $i \in [n]$ (bounded by $B_1 + R$) $\mathbf{B}_{\text{sig}} \leftarrow \mathbf{B}_{\text{ext}}[1:(k+l-1)n, n+1:n]$ $\mathbf{X}' \leftarrow D_\sigma, \mathbf{E}_{\text{bottom}} \leftarrow D_{2\sigma}$ (each coefficient) $\mathbf{Y}_{\text{bottom}} \leftarrow 2\mathbf{B}_{\text{sig}}\mathbf{X}' + \mathbf{E}_{\text{bottom}}$ $\mathbf{Y} = (y_1, \dots, y_n, \mathbf{Y}_{\text{bottom}}^\top)^\top$ $\mathbf{W} \leftarrow \mathbf{A}\mathbf{Y} \bmod 2q$	$\mathbf{c} \leftarrow U(\mathcal{C}) \in \{0, 1\}^n$
$(z_i, 2x_i) \leftarrow \text{Sample_Z}(c_i, y_i)$, $i \in [n]$ $\mathbf{X}_0 \in \mathbb{Z}^n$ is defined by $2\mathbf{X}_0 = 2\mathbf{x} + \mathbf{c}$ $\mathbf{Z}_{\text{bottom}} \leftarrow \mathbf{Y}_{\text{bottom}} + 2\mathbf{S}_{\text{tail}}\mathbf{X}_0 \bmod 2q$ $\mathbf{Z} = (z_1, \dots, z_n, \mathbf{Z}_{\text{bottom}}^\top)^\top$	$\xrightarrow{\mathbf{w}} \quad \xleftarrow{\mathbf{c}} \quad \xrightarrow{\mathbf{z}}$
	Accept if $\mathbf{AZ} = \mathbf{W} + q\mathbf{Ic} \bmod 2q$ and $\ z_i\ _\infty \leq B_1$, $i \in [n]$ and $\ \mathbf{Z}_{\text{bottom}}\ _\infty \leq B_2$

Table 2. The Rhyme Identification Protocol

4.2 Completeness and Commitment Recoverability

We progressively refine the security proofs of our cryptographic components required in the security reduction, which is tight as described in [27]. First, we should ensure our signature protocol is complete and commitment-recoverable. As a corollary, we obtain that the resulting Fiat-Shamir signature scheme $\text{FS}[\text{Rhyme_ID}, \text{H}]$ is correct.

Theorem 4.1 (Completeness and Commitment-Recoverability). *Let $k, l, n > 0$, $\varepsilon \in (0, 1/4]$, and $\sigma > \eta_\varepsilon(\mathbb{Z}^{(k+l)n})$ in the range of lgen . Let B_1, B_2 be acceptance bounds and let $q > 2B$ be an odd modulus with $B = \max(B_1, B_2)$. Then the Rhyme_ID identification protocol is ε_c -complete and achieves commitment-recoverability, where the completeness error ε_c is bounded by*

$$\varepsilon_c = 1 - (1 - \Pr[\|\mathbf{z}_0\|_\infty > B_1])(1 - \Pr[\|\mathbf{z}_1\|_\infty > B_2])$$

with

$$\Pr[\|\mathbf{z}_0\|_\infty > B_1] = 1 - (\Pr_{\text{acc}})^n, \quad \Pr[\|\mathbf{z}_1\|_\infty > B_2] \leq \Pr_{\mathbf{z}'_1}[\|\mathbf{z}'_1\|_\infty > B_2] + 4\varepsilon,$$

where $\Pr_{\text{acc}}$ is defined in Theorem 3.1 and $\mathbf{z}'_1 \sim \mathcal{N}(\mathbf{0}, \Sigma)$ with $\Sigma = \frac{\sigma^2}{2\pi}(\mathbf{I}_{nd} + 4\mathbf{G}_0)$ ($d = k + l - 1$).

Proof. First, note that $\mathbf{AS} = q\mathbf{I} \bmod 2q$ holds for any key pair output by lgen . For any honestly generated transcript $(\mathbf{W}, \mathbf{c}, \mathbf{Z})$ we have

$$\mathbf{Z} = \mathbf{Y} + \mathbf{S}(\mathbf{c} + 2\mathbf{x})$$

(where $2\mathbf{X}_0 = \mathbf{c} + 2\mathbf{x}$ as defined in the protocol). Hence

$$\mathbf{AZ} = \mathbf{AY} + \mathbf{AS}(\mathbf{c} + 2\mathbf{x}) = \mathbf{W} + q\mathbf{I}(\mathbf{c} + 2\mathbf{x}) \equiv \mathbf{W} + q\mathbf{Ic} \pmod{2q},$$

because $2q \mid q \cdot 2\mathbf{x}$. This shows that every honestly generated transcript passes the verification equation, and that the commitment can be recovered as $\mathbf{W} = \mathbf{AZ} - q\mathbf{Ic} \bmod 2q$. Thus the protocol is commitment-recoverable.

It remains to bound the probability that an honest transcript is rejected because of norm violations. The rejection occurs if either $\|\mathbf{z}_0\|_\infty > B_1$ or $\|\mathbf{z}_1\|_\infty > B_2$. Since $\mathbf{z}_0$ and $\mathbf{z}_1$ are generated from independent randomness (the former depends only on y_1 and the rejection sampling tape; the latter depends on $\mathbf{X}'$, $\mathbf{e}_{\text{bottom}}$ and X_0), these two events are independent. Hence the completeness error is exactly

$$\varepsilon_c = 1 - (1 - \Pr[\|\mathbf{z}_0\|_\infty > B_1])(1 - \Pr[\|\mathbf{z}_1\|_\infty > B_2]).$$

We now bound each term separately.

- The vector $\mathbf{z}_0$ is obtained coordinate-wise by the rejection sampling procedure of Section 3.3.1. According to Theorem 3.1, each of the n coordinates is accepted independently with probability

$$\Pr_{\text{acc}} = \frac{1}{M} \frac{\sum_{|z| \leq B_1} \rho_{\sigma_y}(z)}{\sum_{|y| \leq B_1 + R} \rho_{\sigma_y}(y)}.$$

Therefore

$$\Pr[\|\mathbf{z}_0\|_\infty > B_1] = 1 - (\Pr_{\text{acc}})^n.$$

- The vector $\mathbf{z}_1 \in \mathbb{Z}^{(k+l-1)n}$ is obtained by a linear transformation of independent discrete Gaussian noises. Let $d = k + l - 1$ and denote

$$\Sigma = \frac{4\sigma^2}{2\pi} \mathbf{I}_{nd} + \frac{4\sigma^2}{2\pi} \mathbf{G}_0 \in \mathbb{R}^{nd \times nd},$$

where $\mathbf{G}_0 = \mathbf{B}_{\text{short}} \mathbf{B}_{\text{short}}^\top$ is the public Gram sub-matrix. Since the theorem assumes $\sigma \geq \eta_\varepsilon(\mathbb{Z}^{nD})$. By Lemma 2.3, rounding samples from the continuous Gaussian $\mathcal{N}(\mathbf{0}, \Sigma)$ to the nearest integer yields a distribution $D_{\mathbf{z}'_1}$ that is within max-log distance $4\varepsilon + O(\varepsilon^2)$ of the discrete Gaussian distribution $D_{\mathbf{z}_1}$ of the real $\mathbf{z}_1$. Consequently,

$$|\Pr[\|\mathbf{z}_1\|_\infty > B_2] - \Pr_{\mathbf{z}'_1}[\|\mathbf{z}'_1\|_\infty > B_2]| \leq 4\varepsilon + O(\varepsilon^2).$$

Thus, for sufficiently small ε ,

$$\Pr[\|\mathbf{z}_1\|_\infty > B_2] \leq \Pr_{\mathbf{z}'_1}[\|\mathbf{z}'_1\|_\infty > B_2] + 4\varepsilon.$$

Substituting these bounds into the expression for ε_c gives the claimed estimate. For typical parameter choices both terms are negligible, so the protocol is overwhelmingly complete.

Corollary 1. Under the same hypotheses as in Theorem 4.1, the signature scheme $\text{FS}[\text{Rhyme_ID}, \text{H}]$ is ε_c -correct, where ε_c is the bound derived above.

4.3 Proof of EU-NMA

Then we need to prove that the **Rhyme_ID** scheme achieves EU-CMA security. We show that **Rhyme_ID** is a lossy identification scheme and lossy-special sound. In Corollaries 2 and 3, we obtain that the signature scheme $\text{FS}[\text{Rhyme_ID}, \text{H}]$ is EU-NMA-secure in the ROM, and in the QROM under some parameter constraints.

Theorem 4.2. Let $k, l, n > 0$ and let $B_1, B_2 > 0$ with $B = \max(B_1, B_2)$ and $q > 2B$ be an odd modulus. Then our **Rhyme_ID** identification protocol:

- satisfies key-indistinguishability under the $\text{GA-MLWE}_{\text{short}}$ assumption,
- is ε_{ls} -lossy sound for

$$\varepsilon_{\text{ls}} = \frac{1}{|\mathcal{C}|} + \frac{(4B+1)^{(k+l)n}}{q^{kn}},$$

- is also lossy-special-sound, under the $\text{SIS}_{kn, (k+l)n, q, 2B}$ assumption.

Proof. Since the key generation, signing and verification processes in the **Rhyme_ID** scheme are quite similar to those in the **G+G** scheme, the lossy key generation, along with the related proofs and analyses, also share similarities.

Key-indistinguishability. The lossy key generation algorithm samples $\mathbf{B}_{\text{gen}} \leftarrow U(\mathbb{Z}_q^{kn \times n})$ instead of computing $\mathbf{B}_{\text{gen}} = \mathbf{A}_{\text{gen}} \mathbf{S}_{\text{gen}} + \mathbf{E}_{\text{gen}} \bmod q$ as in **Igen**. The Gram sub-matrix $\mathbf{G}_0 = \mathbf{B}_{\text{short}} \mathbf{B}_{\text{short}}^\top$ is *not* altered; it is still computed from

the same secret $\mathbf{B}_{\text{ext}}$ as in the real key generation. Therefore, the joint distribution of the lossy verification key $(\mathbf{A}_{\text{gen}}, \mathbf{G}_0, \mathbf{B}_{\text{gen}})$ (where $\mathbf{B}_{\text{gen}}$ is uniform and $\mathbf{G}_0$ is derived from the real secret) is computationally indistinguishable from the real distribution under the $\text{GA-MLWE}_{\text{short}}$ assumption. Consequently, lossy verification keys are computationally indistinguishable from non-lossy ones.

ε_{ls} -lossy soundness. If a lossy verification key $\mathbf{A}$ satisfies the following condition: for every commitment $\mathbf{W}$ there exists at most one challenge $\mathbf{c}$ such that a response $\mathbf{Z}$ exists making $(\mathbf{W}, \mathbf{c}, \mathbf{Z})$ pass verification, then, because the challenge is sampled uniformly and independently of $\mathbf{W}$, the probability that the prover correctly guesses a valid $\mathbf{c}$ is at most $1/|\mathcal{C}|$.

We now prove that the above property holds with overwhelming probability over the choice of the lossy key $\mathbf{A}$. Suppose there exist $\mathbf{W}, \mathbf{c}_\alpha, \mathbf{c}_\beta, \mathbf{Z}_\alpha, \mathbf{Z}_\beta$ with $\|\mathbf{Z}_\alpha\|_\infty, \|\mathbf{Z}_\beta\|_\infty \leq B$ and $\mathbf{c}_\alpha \neq \mathbf{c}_\beta$ such that both $\mathbf{AZ}_\alpha = \mathbf{W} + q\mathbf{Ic}_\alpha \bmod 2q$ and $\mathbf{AZ}_\beta = \mathbf{W} + q\mathbf{Ic}_\beta \bmod 2q$ hold. Subtracting gives

$$\mathbf{A}(\mathbf{Z}_\alpha - \mathbf{Z}_\beta) = q\mathbf{I}(\mathbf{c}_\alpha - \mathbf{c}_\beta) \bmod 2q.$$

Recall that $\mathbf{A} = (-2\mathbf{B}_{\text{gen}} + q\mathbf{I} \mid 2\mathbf{A}_{\text{gen}} \mid 2\mathbf{Id}_{kn})$ with $\mathbf{A}_{\text{gen}}, \mathbf{B}_{\text{gen}}$ uniformly random. Let $\zeta = 2^{-1} \bmod q$ (which exists because q is odd). Multiplying the equation by ζ modulo q gives

$$(\mathbf{B}_{\text{gen}} \mid \mathbf{A}_{\text{gen}} \mid \mathbf{Id}_{kn})(\mathbf{Z}_\alpha - \mathbf{Z}_\beta) = \mathbf{0} \bmod q.$$

We now bound the probability that such a collision exists. For a fixed non-zero matrix $\mathbf{V} = \mathbf{Z}_\alpha - \mathbf{Z}_\beta$, the condition $(\mathbf{B}_{\text{gen}} \mid \mathbf{A}_{\text{gen}} \mid \mathbf{Id}_{kn})\mathbf{V} = \mathbf{0} \bmod q$ fails for all but a $1/q^{kn}$ fraction of the uniform choices of $\mathbf{A}_{\text{gen}}$ and $\mathbf{B}_{\text{gen}}$. There are at most $(4B+1)^{(k+l)n}$ possible non-zero difference vectors $\mathbf{V}$, because each coordinate of $\mathbf{Z}_\alpha, \mathbf{Z}_\beta$ lies in $[-B, B]$ and therefore their difference lies in $[-2B, 2B]$. A union bound shows that the probability over $\mathbf{A}$ that a commitment with two distinct valid challenges exists is at most $\frac{(4B+1)^{(k+l)n}}{q^{kn}}$. Hence the lossy identification scheme is ε_{ls} -lossy-sound with

$$\varepsilon_{\text{ls}} \leq \frac{1}{|\mathcal{C}|} + \frac{(4B+1)^{(k+l)n}}{q^{kn}}.$$

Lossy special soundness. Assume there exists a PPT adversary $\mathcal{A}$ which, on input a lossy verification key $\mathbf{A}$, outputs two valid transcripts $(\mathbf{W}, \mathbf{c}_\alpha, \mathbf{Z}_\alpha)$ and $(\mathbf{W}, \mathbf{c}_\beta, \mathbf{Z}_\beta)$ with $\mathbf{c}_\alpha \neq \mathbf{c}_\beta$. By definition these satisfy

$$\mathbf{AZ}_\alpha = \mathbf{W} + q\mathbf{Ic}_\alpha \bmod 2q, \quad \mathbf{AZ}_\beta = \mathbf{W} + q\mathbf{Ic}_\beta \bmod 2q.$$

Subtracting yields

$$\mathbf{A}(\mathbf{Z}_\alpha - \mathbf{Z}_\beta) = q\mathbf{I}(\mathbf{c}_\alpha - \mathbf{c}_\beta) \bmod 2q.$$

We first argue that $\mathbf{Z}_\alpha \neq \mathbf{Z}_\beta$. If $\mathbf{Z}_\alpha = \mathbf{Z}_\beta$, the left-hand side is $\mathbf{0}$, implying $q\mathbf{I}(\mathbf{c}_\alpha - \mathbf{c}_\beta) = \mathbf{0} \bmod 2q$. Since the top n rows of $\mathbf{I}$ form the identity matrix and $\mathbf{c}_\alpha \neq \mathbf{c}_\beta$, there exists some coordinate for which the corresponding row equals $\pm q \not\equiv 0 \bmod 2q$, a contradiction. Thus $\mathbf{V} = \mathbf{Z}_\alpha - \mathbf{Z}_\beta$ is non-zero. Moreover, reducing the equality modulo q gives $\mathbf{A}\mathbf{V} = \mathbf{0} \bmod q$, so $\mathbf{V}$ is a solution to the SIS instance defined by $\mathbf{A}$. Finally, $\|\mathbf{V}\|_\infty \leq 2B$, and because $\mathbf{V} \neq \mathbf{0} \bmod q$ it is a valid SIS solution. Hence we can construct an adversary $\mathcal{B}$ against $\text{SIS}_{kn, (k+l)n, q, 2B}$ such that

$$\text{Adv}(\mathcal{A}) \leq \text{Adv}^{\text{SIS}_{kn, (k+l)n, q, 2B}}(\mathcal{B}).$$

This completes the proof of the theorem. $\square$

Building on this foundation, we can derive the following corollaries.

Corollary 2. Under the same assumptions as in Theorem 4.2, the signature scheme $\text{FS}[\text{Rhyme_ID}, \text{H}]$ is EU-NMA-secure in the ROM. Namely, for any adversary $\mathcal{A}$ against the EU-NMA security of $\text{FS}[\text{Rhyme_ID}, \text{H}]$, there exists an adversary $\mathcal{B}$ against the $\text{SIS}_{kn, (k+l)n, q, 2B}$ assumption as well as an adversary $\mathcal{B}'$ against the $\text{GA-MLWE}_{\text{short}}$ assumption such that

$$\text{Adv}^{\text{EU-NMA}}(\mathcal{A}) \leq Q_H \cdot \left(\sqrt{\text{Adv}^{\text{SIS}_{kn, (k+l)n, q, 2B}}(\mathcal{B})} + \frac{2}{|\mathcal{C}|} \right) + \text{Adv}^{\text{GA-MLWE}_{\text{short}}}(\mathcal{B}').$$

Corollary 3. Under the same assumptions as in Theorem 4.2, and if ε_{ls} is negligible, the signature scheme $\text{FS}[\text{Rhyme_ID}, \text{H}]$ is EU-NMA-secure in the QROM. Namely, for any (possibly quantum) adversary $\mathcal{A}$ against the EU-NMA security of $\text{FS}[\text{Rhyme_ID}, \text{H}]$ making at most Q_H (possibly quantum) hash queries, there exists a quantum adversary $\mathcal{B}$ against the $\text{GA-MLWE}_{\text{short}}$ assumption such that

$$\text{Adv}^{\text{EU-NMA}}(\mathcal{A}) \leq \text{Adv}^{\text{GA-MLWE}_{\text{short}}}(\mathcal{B}) + 8(Q_H + 1)^2 \cdot \left(\frac{1}{|\mathcal{C}|} + \frac{(4B + 1)^{(k+l)n}}{q^{kn}} \right).$$

To conclude, we introduce an additional assumption of a similar flavor as the SelfTargetMSIS assumption [4], which allows to directly prove EU-NMA-security of $\text{FS}[\text{Rhyme_ID}, \text{H}]$ in the QROM as it is (up to LWE) the EU-NMA security game of the resulting signature. As for SelfTargetMSIS, this problem can be related in the ROM to SIS, using the special soundness property of the scheme.

Definition 4.1 (SelfTargetSIS). Let $k, l, n > 0$, $B > 0$ and $q > 2B$ be an odd modulus. The $\text{TargetSIS}_{kn, (k+l)n, B, q}$ states that given a matrix $\mathbf{A} := (-2\mathbf{B}_{\text{gen}} + q\mathbf{I} \mid 2\mathbf{A}_{\text{gen}} \mid 2\mathbf{Id}_{kn})$, where $\mathbf{A}_{\text{gen}} \leftarrow U\left(\mathbb{Z}_q^{kn \times (l-1)n}\right)$, and oracle access to a hash function H , it is computationally hard to find $\mathbf{c} \in \mathcal{C}$, $\mathbf{Z} \in \mathbb{Z}_{2q}^{(k+l)n}$ and $\mu \in \{0, 1\}^*$ such that $H(\mathbf{A}\mathbf{Z} - q\mathbf{I}\mathbf{c}, \mu) = \mathbf{c}$ and $\|\mathbf{Z}\|_\infty \leq B$.

By leveraging the previous theorems and conclusions, we have already established that our scheme achieves EU-NMA security. Next, we need to prove that the Rhyme_ID protocol is HVZK and has a large commitment min-entropy. From this, we can further deduce that our signature scheme $\text{FS}[\text{Rhyme_ID}, \text{H}]$ is EU-CMA secure.

4.4 Honest-Verifier Zero-Knowledge and Commitment Min-Entropy

To complete the security proof we show that Rhyme_ID satisfies statistical honest-verifier zero-knowledge (HVZK) and that its commitments have high min-entropy. Together with the lossy-soundness properties established in Theorem 4.2, this implies that the Fiat-Shamir signature $\text{FS}[\text{Rhyme_ID}, \text{H}]$ is EUF-CMA secure in the random oracle model (and, under the same conditions, in the QROM via the framework of [16]).

Theorem 4.3 (Statistical HVZK). *Let $k, l, n > 0$, $\varepsilon \in (0, 1/4]$, and let $\sigma \geq \eta_\varepsilon(\mathbb{Z}^{nD})$, where D is the extended dimension of the unimodular matrix in Construction 3 (so $nD \geq nd$ for $d = k + l - 1$). Then the Rhyme_ID identification protocol (Table 2) is ε_{zk} -statistically honest-verifier zero-knowledge with $\varepsilon_{\text{zk}} = 4\varepsilon + O(\varepsilon^2)$. Concretely, there exists a PPT simulator Sim that, on input the public key $(\mathbf{A}, \mathbf{G}_0)$, outputs transcripts $(\mathbf{W}, \mathbf{c}, \mathbf{Z})$ whose distribution is within max-log distance ε_{zk} of honestly generated transcripts.*

Proof. Recall the real protocol from Table 2. An honest transcript is obtained by sampling $\mathbf{y}$, computing $\mathbf{W} = \mathbf{A}\mathbf{y} \bmod 2q$, receiving a uniform challenge $\mathbf{c} \leftarrow \mathcal{C}$, and outputting $\mathbf{Z} = (z_1, \dots, z_n, \mathbf{Z}_{\text{bottom}}^\top)^\top$ where $z_1, \dots, z_n$ are generated by the rejection sampler Sample_Z and

$$\mathbf{Z}_{\text{bottom}} = \mathbf{Y}_{\text{bottom}} + 2\mathbf{S}_{\text{tail}}\mathbf{X}_0$$

with $\mathbf{Y}_{\text{bottom}} = 2\mathbf{B}_{\text{sig}}\mathbf{X}' + \mathbf{E}_{\text{bottom}}$. All noise components are independent, and their coefficients are drawn from discrete Gaussians with the following parameters:

- $y_1, \dots, y_n$ use parameter σ_y (conditioned on a finite range);
- $\mathbf{X}_0$ and $\mathbf{X}'$ have coefficients from $D_{\mathbb{Z}, \sigma}$;
- $\mathbf{E}_{\text{bottom}}$ (the parity-masking noise) has coefficients from $D_{\mathbb{Z}, 2\sigma}$.

The use of a larger width 2σ for $\mathbf{E}_{\text{bottom}}$ is essential to independently guarantee that its modulo-2 distribution is statistically close to uniform, which will be exploited to mask parity information from the secret.

We construct the simulator $\text{Sim}(\mathbf{A}, \mathbf{G}_0)$ as follows:

1. Sample $\mathbf{c} \leftarrow \mathcal{C}$ uniformly.

2. For $i = 1, \dots, n$, sample z_i independently from the truncated discrete Gaussian D_{σ_y} conditioned on $|z_i| \leq B_1$ (the same marginal distribution as in the real protocol).
3. Let $d = k + l - 1$. The bottom vector $\mathbf{Z}_{\text{bottom}}$ is simulated as a sum of two independent parts:
 - **Parity-masking noise** $\mathbf{V}_{\text{sim}}$. In the real protocol, $\mathbf{V} = \mathbf{E}_{\text{bottom}}$ follows $D_{\mathbb{Z}^{nd}, 2\sigma}$. The simulator samples $\mathbf{V}_{\text{sim}} \leftarrow D_{\mathbb{Z}^{nd}, 2\sigma}$ identically; this is a perfect simulation (zero statistical distance). The choice of parameter 2σ guarantees

$$2\sigma \geq 2\eta_\varepsilon(\mathbb{Z}^{nD}) \geq 2\eta_\varepsilon(\mathbb{Z}^{nd}) = \eta_\varepsilon(2\mathbb{Z}^{nd}),$$

where the first inequality holds by the theorem's hypothesis and the second because $nD \geq nd$. Hence the distribution of $\mathbf{V}_{\text{sim}} \bmod 2$ is 2ε -close to uniform over all parity patterns, and therefore any fixed shift added to $\mathbf{V}_{\text{sim}}$ also becomes 2ε -close to uniform modulo 2. This property will absorb the secret-dependent parity shift present in the structured component.

- **Structured component** $\mathbf{U}_{\text{sim}}$. The real $\mathbf{U} = 2\mathbf{B}_{\text{sig}}(\mathbf{X}_0, \mathbf{X}')^\top$ is the first nd coordinates of the extended vector $\mathbf{Y}_{\text{ext}} = 2\mathbf{B}_{\text{ext}}\mathbf{X}_{\text{ext}}$, where $\mathbf{X}_{\text{ext}} = (X_0, \mathbf{X}')^\top \sim D_{\mathbb{Z}^{nD}, \sigma}$ and $\mathbf{B}_{\text{ext}} \in \mathbb{Z}^{nD \times nD}$ is the complete unimodular matrix ($\det(\mathbf{B}_{\text{ext}}) = 1$) whose first column is $\mathbf{S}_{\text{tail}}$. Because $\mathbf{B}_{\text{ext}}$ is invertible over $\mathbb{Z}$, the kernel of $2\mathbf{B}_{\text{ext}}$ is trivial, and the linear transformation lemma (Lemma 2.3) applies under the condition $\sigma \geq \eta_\varepsilon(\mathbb{Z}^{nD})$. Consequently, $\mathbf{Y}_{\text{ext}}$ is within $4\varepsilon + O(\varepsilon^2)$ of the rounded continuous Gaussian $\lfloor \mathcal{N}(\mathbf{0}, \frac{4\sigma^2}{2\pi} \mathbf{G}_{\text{ext}}) \rfloor$, where $\mathbf{G}_{\text{ext}} = \mathbf{B}_{\text{ext}} \mathbf{B}_{\text{ext}}^*$. The marginal distribution of its first nd coordinates is therefore well-approximated by the rounded version of $\mathcal{N}(\mathbf{0}, \frac{4\sigma^2}{2\pi} \mathbf{G}_0)$, with $\mathbf{G}_0 = \mathbf{B}_{\text{short}} \mathbf{B}_{\text{short}}^*$ the public Gram matrix. The simulator samples $\mathbf{U}_{\text{sim}}$ from this marginal continuous Gaussian and rounds each coordinate to the nearest integer, which introduces a statistical defect of at most $4\varepsilon + O(\varepsilon^2)$ with respect to the true $\mathbf{U}$.

Finally set $\mathbf{Z}_{\text{bottom}} = \mathbf{U}_{\text{sim}} + \mathbf{V}_{\text{sim}}$.

4. Assemble $\mathbf{Z} = (z_1, \dots, z_n, \mathbf{Z}_{\text{bottom}}^\top)^\top$.
5. Compute $\mathbf{W} = \mathbf{AZ} - q\mathbf{Ic} \bmod 2q$.
6. If $\|\mathbf{Z}_{\text{bottom}}\|_\infty > B_2$, restart from step 1 (this only adds a negligible factor to the simulation cost).
7. Output $(\mathbf{W}, \mathbf{c}, \mathbf{Z})$.

We now argue that the simulated distribution is statistically close to the real one.

- The distribution of $\mathbf{c}$ is identical.

- The first n coordinates $z_1, \dots, z_n$ are generated by the same truncated discrete Gaussian as in the real protocol, so their marginal distributions coincide; moreover they are independent of the rest both in the real execution and in the simulation.
- For the bottom part, the simulated $\mathbf{V}_{\text{sim}}$ is perfect (distance 0). The simulated $\mathbf{U}_{\text{sim}}$ is within $4\varepsilon + O(\varepsilon^2)$ of the real $\mathbf{U}$, as guaranteed by the linear transformation lemma under the condition $\sigma \geq \eta_\varepsilon(\mathbb{Z}^{nD})$. Because $\mathbf{U}$ and $\mathbf{V}$ are independent, and $\mathbf{U}_{\text{sim}}, \mathbf{V}_{\text{sim}}$ are independent by construction, the overall distribution of $\mathbf{Z}_{\text{bottom}}$ is within $4\varepsilon + O(\varepsilon^2)$ of the true one. Notably, the parity-masking property of $\mathbf{V}_{\text{sim}}$ (its modulo-2 uniformity) ensures that the adversary cannot exploit any residual parity correlation in $\mathbf{U}_{\text{sim}}$; the 4ε bound already absorbs the contribution of the parity classes.
- The truncation step on $\|\mathbf{Z}_{\text{bottom}}\|_\infty$ is applied identically to both distributions and affects the statistical distance only negligibly for suitably chosen B_2 .

The simulator does not use the secret key, and the combined statistical distance is at most $4\varepsilon + O(\varepsilon^2)$. This proves ε_{zk} -HVZK. $\square$

Theorem 4.4 (Commitment Min-Entropy). *Let $k, l, n > 0$ and let $\mathbf{A}$ be generated as in `lgen`. Assume that the masking vector $\mathbf{y}$ contains at least $d = k + l - 1$ ring elements whose nd coefficients are drawn independently from $D_{\mathbb{Z}, \sigma}$ with $\sigma = \omega(\sqrt{\log n})$, and that the remaining n coefficients are bounded. Then, for any fixed public key $(\mathbf{A}, \mathbf{G}_0)$, the commitment $\mathbf{W} = \mathbf{A}\mathbf{y} \bmod 2q$ satisfies*

$$H_\infty(\mathbf{W} \mid \mathbf{A}) \geq kn.$$

Consequently, the probability that any (even unbounded) adversary correctly guesses the value of $\mathbf{W}$ is at most 2^{-kn} .

Proof. The matrix $\mathbf{A} \in \mathbb{Z}_{2q}^{kn \times (k+l)n}$ is of the form

$$\mathbf{A} = (-2\mathbf{B}_{\text{gen}} + q\mathbf{I} \mid 2\mathbf{A}_{\text{gen}} \mid 2\mathbf{Id}_{kn}).$$

Consider the rightmost kn columns of $\mathbf{A}$, which form the submatrix $2\mathbf{Id}_{kn}$. Since q is odd, $\gcd(2, 2q) = 2$, so 2 is not invertible modulo $2q$. However, the map $\mathbf{x} \mapsto 2\mathbf{x} \bmod 2q$ is still injective on $\mathbb{Z}_q^{kn}$, because $2\mathbf{x} \equiv 2\mathbf{y} \pmod{2q}$ implies $q \mid (\mathbf{x} - \mathbf{y})$ for each component, i.e. $\mathbf{x} \equiv \mathbf{y} \pmod{q}$. Viewing $\mathbb{Z}_q$ as a subset of representatives in $\{0, \dots, q-1\}$, this shows that the columns of $2\mathbf{Id}_{kn}$ are linearly independent over $\mathbb{Z}_{2q}$ when coefficients are restricted to $\mathbb{Z}_q$. More directly, the Smith normal form of $2\mathbf{Id}_{kn}$ over $\mathbb{Z}_{2q}$ has diagonal entries 2, so its column space has size q^{kn} . Consequently, the rightmost kn columns of $\mathbf{A}$ already span a subspace of $\mathbb{Z}_{2q}^{kn}$ of size q^{kn} . Since the target space $\mathbb{Z}_{2q}^{kn}$ has size $(2q)^{kn}$, the full matrix $\mathbf{A}$ is *not* necessarily surjective onto $\mathbb{Z}_{2q}^{kn}$ — its image has size exactly q^{kn} .

Nevertheless, the masking vector $\mathbf{y}$ contains nd independent discrete Gaussian coefficients with parameter $\sigma = \omega(\sqrt{\log n})$ in its last d ring elements. Let $\mathbf{y}_{\text{bottom}} \in \mathbb{Z}^{nd}$ denote the coefficient vector of these d components. By standard discrete Gaussian tail bounds, each coordinate of $\mathbf{y}_{\text{bottom}}$ has min-entropy $\omega(\log n)$, hence $H_\infty(\mathbf{y}_{\text{bottom}}) \geq nd \cdot \omega(\log n)$.

The commitment is $\mathbf{W} = \mathbf{A}\mathbf{y} \bmod 2q$. Write $\mathbf{A} = (\mathbf{A}_L \mid 2\mathbf{Id}_{kn})$ where $\mathbf{A}_L$ consists of the leftmost $(l-1)n$ columns, and correspondingly $\mathbf{y} = (\mathbf{y}_L, \mathbf{y}_R)^\top$ with $\mathbf{y}_R \in \mathbb{Z}^{kn}$ drawn from the tail of the discrete Gaussian. Then

$$\mathbf{W} = \mathbf{A}_L \mathbf{y}_L + 2\mathbf{y}_R \bmod 2q.$$

The term $2\mathbf{y}_R \bmod 2q$ only depends on $\mathbf{y}_R \bmod q$. Since $\mathbf{y}_R$ has min-entropy at least $\omega(\log n)$ per coordinate, and the map $\mathbf{y}_R \mapsto \mathbf{y}_R \bmod q$ reduces the entropy by at most $n \log_2 q$ bits (the modulus), we obtain

$$H_\infty(\mathbf{W} \mid \mathbf{A}) \geq H_\infty(2\mathbf{y}_R \bmod 2q \mid \mathbf{A}_L \mathbf{y}_L) - \text{negl}(\lambda) \geq kn \log_2 q - \text{negl}(\lambda) \geq kn.$$

A rigorous derivation follows from the leftover hash lemma with the universal hash family $\mathcal{H} = \{h_{\mathbf{A}}(\mathbf{y}) = \mathbf{A}\mathbf{y} \bmod 2q\}$: the conditional distribution of $\mathbf{W}$ given $\mathbf{A}$ is statistically close to uniform over the image of $\mathbf{A}$, which has size q^{kn} . Hence the min-entropy of $\mathbf{W}$ is at least $\log_2(q^{kn}) - \text{negl}(\lambda) \geq kn$ bits (since $q \geq 2$). Therefore $\Pr[\mathbf{W} = w] \leq 2^{-kn} + \text{negl}(\lambda)$ for any fixed w , and the negligible term can be dropped for sufficiently large n . $\square$

4.5 Proof of EU-CMA and sEU-CMA

We then obtain the following corollary.

Corollary 4. Under the same assumptions as in Theorem 4.3 and Theorem 4.4, the resulting signature scheme $\text{FS}[\text{Rhyme_ID}, \text{H}]$ is EU-CMA-secure in the QROM, provided it is EU-NMA-secure. Namely, for any (possibly quantum) adversary $\mathcal{A}$ against the EU-CMA security of $\text{FS}[\text{Rhyme_ID}, \text{H}]$ making at most Q_S classical sign queries and at most Q_H (possibly quantum) hash queries, there exists an adversary $\mathcal{B}$ against the EU-NMA security of $\text{FS}[\text{Rhyme_ID}, \text{H}]$ such that

$$\begin{aligned} \text{Adv}^{\text{EU-CMA}}(\mathcal{A}) &\leq (1 + 4\varepsilon)^{Q_S} \text{Adv}^{\text{EU-NMA}}(\mathcal{B}) \\ &\quad + 3Q_S/2 \cdot \sqrt{(Q_H + Q_S + 1) \cdot 2^{-kn}}. \end{aligned}$$

(The factor $(1 + 4\varepsilon)^{Q_S}$ accounts for the max-log distance accumulated over Q_S simulated signing queries due to the HVZK simulation error; see Theorem 4.3.)

Finally, in order to show the strong unforgeability of the scheme, we prove that Rhyme_ID has Computational Unique Response (CUR).

Theorem 4.5. *Let $k, l, n > 0$, $B > 0$ and $q > 2B$ be an odd modulus. Then the Rhyme_ID identification protocol satisfies the CUR property under the GA-MLWE_{short} and SIS _{$kn, (k+l)n, q, 2B$} assumptions. Namely, for any (possibly quantum) adversary $\mathcal{A}$ against the CUR property, there exist an adversary $\mathcal{B}$ against GA-MLWE_{short} and an adversary $\mathcal{B}'$ against SIS _{$kn, (k+l)n, q, 2B$} such that*

$$\text{Adv}^{\text{CUR}}(\mathcal{A}) \leq \text{Adv}^{\text{GA-MLWE}_{\text{short}}}(\mathcal{B}) + \text{Adv}^{\text{SIS}_{kn, (k+l)n, q, 2B}}(\mathcal{B}').$$

Proof. We first replace the verification key with a lossy one as defined in the proof of Theorem 4.2. In the lossy key the matrix $\zeta \mathbf{A} \bmod q$ (with $\zeta = 2^{-1} \bmod q$) is uniformly random among matrices in Hermite Normal Form, and the Gram sub-matrix $\mathbf{G}_0$ is still the one derived from the real secret. By the GA-MLWE_{short} assumption, this change is computationally indistinguishable.

Now suppose an adversary outputs a CUR collision, i.e. two distinct valid responses $\mathbf{Z}_\alpha \neq \mathbf{Z}_\beta$ together with a commitment $\mathbf{W}$ and a challenge $\mathbf{c}$ such that both $(\mathbf{W}, \mathbf{c}, \mathbf{Z}_\alpha)$ and $(\mathbf{W}, \mathbf{c}, \mathbf{Z}_\beta)$ pass verification. Verification gives

$$\mathbf{A}\mathbf{Z}_\alpha - q\mathbf{I}\mathbf{c} = \mathbf{W} = \mathbf{A}\mathbf{Z}_\beta - q\mathbf{I}\mathbf{c} \bmod 2q,$$

hence $\mathbf{A}(\mathbf{Z}_\alpha - \mathbf{Z}_\beta) = \mathbf{0} \bmod 2q$, and therefore $\mathbf{A}(\mathbf{Z}_\alpha - \mathbf{Z}_\beta) = \mathbf{0} \bmod q$. Because the lossy matrix $\zeta \mathbf{A} \bmod q$ is uniformly random, $\mathbf{V} = \mathbf{Z}_\alpha - \mathbf{Z}_\beta$ is a non-zero vector with $\|\mathbf{V}\|_\infty \leq 2B$ satisfying $(\zeta \mathbf{A})\mathbf{V} = \mathbf{0} \bmod q$. This is exactly a solution to the SIS _{$kn, (k+l)n, q, 2B$} problem for the instance defined by $\zeta \mathbf{A}$. Thus any CUR adversary can be converted into a solver for either GA-MLWE_{short} (to switch to a lossy key) or SIS. $\square$

Corollary 5. Under the same assumptions as in Theorem 4.5, the signature scheme FS[Rhyme_ID, H] is sEU-CMA-secure in the QROM, provided it is EU-NMA-secure. Namely, for any (possibly quantum) adversary $\mathcal{A}$ against the sEU-CMA security making at most Q_S classical sign queries and at most Q_H (possibly quantum) hash queries, there exist an adversary $\mathcal{B}$ against the EU-NMA security of FS[Rhyme_ID, H], an adversary $\mathcal{B}'$ against GA-MLWE_{short}, and an adversary $\mathcal{B}''$ against SIS _{$kn, (k+l)n, q, 2B$} such that

$$\begin{aligned} \text{Adv}^{\text{sEU-CMA}}(\mathcal{A}) &\leq (1 + 4\varepsilon)^{Q_S} \text{Adv}^{\text{EU-NMA}}(\mathcal{B}) \\ &\quad + 3Q_S/2 \cdot \sqrt{(Q_H + Q_S + 1) \cdot 2^{-kn}} \\ &\quad + Q_S \cdot 2^{-kn} \\ &\quad + \text{Adv}^{\text{GA-MLWE}_{\text{short}}}(\mathcal{B}') + \text{Adv}^{\text{SIS}_{kn, (k+l)n, q, 2B}}(\mathcal{B}''). \end{aligned}$$

Building upon the above corollaries, which are derived from the properties of our identification protocol, we can state the final security theorem for the signature scheme FS[Rhyme_ID, H].

Theorem 4.6. *Let $k, l, n > 0$, $B > 0$ and $q > 2B$ be an odd modulus. Then the signature scheme $\text{FS}[\text{Rhyme_ID}, H]$ is:*

- *EU-CMA-secure in the ROM under the $\text{GA-MLWE}_{\text{short}}$ and $\text{SIS}_{kn, (k+l)n, q, 2B}$ assumptions. Namely, for any adversary $\mathcal{A}$ against EU-CMA making at most Q_S sign queries and Q_H hash queries, there exist an adversary $\mathcal{B}$ against $\text{SIS}_{kn, (k+l)n, q, 2B}$ and an adversary $\mathcal{B}'$ against $\text{GA-MLWE}_{\text{short}}$ such that*

$$\begin{aligned} \text{Adv}^{\text{EU-CMA}}(\mathcal{A}) \leq (1 + 4\varepsilon)^{Q_S} & \left[Q_H \left(\sqrt{\text{Adv}^{\text{SIS}_{kn, (k+l)n, q, 2B}}(\mathcal{B})} + \frac{2}{|\mathcal{C}|} \right) + \text{Adv}^{\text{GA-MLWE}_{\text{short}}}(\mathcal{B}') \right] \\ & + 3Q_S/2 \cdot \sqrt{(Q_H + Q_S + 1) \cdot 2^{-kn}}. \end{aligned}$$

- *EU-CMA-secure in the QROM under the $\text{GA-MLWE}_{\text{short}}$ assumption, provided ε_{ls} is negligible. Namely, for any quantum adversary against EU-CMA making at most Q_S classical sign queries and Q_H quantum hash queries, there exists an adversary $\mathcal{B}'$ against $\text{GA-MLWE}_{\text{short}}$ such that*

$$\begin{aligned} \text{Adv}^{\text{EU-CMA}}(\mathcal{A}) \leq (1 + 4\varepsilon)^{Q_S} & \left[\text{Adv}^{\text{GA-MLWE}_{\text{short}}}(\mathcal{B}') + 8(Q_H + 1)^2 \left(\frac{1}{|\mathcal{C}|} + \frac{(4B + 1)^{(k+l)n}}{q^{kn}} \right) \right] \\ & + 3Q_S/2 \cdot \sqrt{(Q_H + Q_S + 1) \cdot 2^{-kn}}. \end{aligned}$$

Moreover, the same bounds hold for sEU-CMA security after adding the extra term

$$+ Q_S \cdot 2^{-kn} + \text{Adv}^{\text{GA-MLWE}_{\text{short}}}(\mathcal{B}'') + \text{Adv}^{\text{SIS}_{kn, (k+l)n, q, 2B}}(\mathcal{B}''')$$

to the right-hand side, where $\mathcal{B}'', \mathcal{B}'''$ are the additional reductions required by Theorem 4.5.

4.6 Cost of Known Attacks

For the concrete security analysis, we list the best known lattice attacks and consider their costs for attacking our scheme. All the best known attacks rely on the Block–Korkine–Zolotarev (BKZ) lattice reduction algorithm [12, 26, 37]. The BKZ algorithm is a lattice basis reduction algorithm that repeatedly uses a Shortest Vector Problem (SVP) solver in small-dimensional projected sublattices. The dimension b of these projected sublattices is called the block-size. BKZ with block-size b hence relies on an SVP solver in dimension b . The block-size b governs both the runtime cost of BKZ and the quality of the reduced basis it produces. It provides a quality/time trade-off: If b gets larger, better quality will be guaranteed, but the time complexity for the SVP solver will exponentially increase. The time complexity of the b -BKZ algorithm is the same as the SVP solver for dimension b , up to polynomial factors. Hence the time complexity differs depending on the SVP solver used. The most efficient SVP algorithm uses the

sieving method proposed by Becker et al. in [7] which takes time $\approx 2^{0.292b+o(b)}$. The fastest known quantum variant is proposed by Chailloux and Loyer in [11] and takes time $\approx 2^{0.257b+o(b)}$.

To evaluate the concrete hardness of the underlying MLWE and MSIS problems, we utilize the Lattice Estimator library [2]. Unlike the simplified Core-SVP methodology [34] which typically ignores polynomial factors, we adopt the MATZOV cost model [40] for classical security (accounting for polynomial overheads) and the ChaLoy21 model [11] for quantum security (representing a conservative Quantum Core-SVP hardness) which correspond to the latest version of the Lattice Estimator library.

Even though the MLWE and MSIS problems have some extra algebraic structure compared to the LWE and SIS problems, we do not currently know how to exploit it to improve the best known attacks. For this reason, we estimate the concrete hardness of the MLWE and MSIS problems over the structured lattices as the concrete hardness of the corresponding binary-LWE and SIS problems over the unstructured lattices.

We summarize the costs of the known attacks in Table 3. In the table, the required block-sizes for BKZ and the costs of the attacks are estimated using the Lattice Estimator library [2] with the aforementioned models. The parameters for MLWE and MSIS problems are chosen based on the above analysis.

Primal attack. Given an LWE instance $(\mathbf{A}, \mathbf{b}) \in \mathbb{Z}_q^{k \times l} \times \mathbb{Z}_q^k$, we first define the lattices $\Lambda_m = \{\mathbf{v} \in \mathbb{Z}^{l+m+1} : \mathbf{B}\mathbf{v} = \mathbf{0} \bmod q\}$ for all $m \leq k$, where $\mathbf{B} = (\mathbf{A}_{[m]} | \mathbf{Id}_m | \mathbf{b}_{[m]}) \in \mathbb{Z}_q^{m \times (l+m+1)}$, $\mathbf{A}_{[m]}$ is the uppermost $m \times l$ sub-matrix of $\mathbf{A}$ and $\mathbf{b}_{[m]}$ is the uppermost m -dimensional sub-vector of $\mathbf{b}$. As $(\mathbf{A}, \mathbf{b}) \in \mathbb{Z}_q^{k \times l} \times \mathbb{Z}_q^k$ is an LWE instance, there exist $\mathbf{s}$ and $\mathbf{e}$ short such that $\mathbf{b} = \mathbf{A}\mathbf{s} + \mathbf{e}$. This implies that $(\mathbf{s} | \mathbf{e})$ is a short vector of Λ_m . The primal attack consists in running BKZ on Λ_m to find short vectors. The variable m is optimized to minimize the cost of the attack.

Dual attack. Given an LWE instance $(\mathbf{A}, \mathbf{b}) \in \mathbb{Z}_q^{k \times l} \times \mathbb{Z}_q^k$, we first define the lattices $\Lambda'_m = \{(\mathbf{u}, \mathbf{v}) \in \mathbb{Z}^m \times \mathbb{Z}^l : \mathbf{A}_{[m]}^\top \mathbf{u} - \mathbf{v} = \mathbf{0} \bmod q\}$ for all $m \leq k$, where $\mathbf{A}_{[m]}$ is the uppermost $m \times l$ sub-matrix of $\mathbf{A}$. If $(\mathbf{u}, \mathbf{v})$ is a short vector in Λ'_m , then $\mathbf{u}^\top \mathbf{b}_{[m]} = \mathbf{v}^\top \mathbf{s} + \mathbf{u}^\top \mathbf{e}_{[m]}$ is small if $\mathbf{b} = \mathbf{A}\mathbf{s} + \mathbf{e}$ for short vectors $\mathbf{s}$ and $\mathbf{e}$, and is uniformly distributed modulo q if $\mathbf{b}$ is uniform and independent from $\mathbf{A}$ (here $\mathbf{e}_{[m]}$ refers to the uppermost m -dimensional sub-vector of $\mathbf{e}$). This provides a distinguishing attack. The dual attack consists in finding a short non-zero vector in the lattice Λ'_m using BKZ. The variable m is optimized to minimize the cost of the attack.

Hybrid Dual Attack. Our scheme is based on binary-LWE, for which the hybrid attack approach is typically effective. A hybrid attack guesses part of the secret key and then seeks a trade-off between the cost of guessing and the cost of the dual attack. Among the guessing techniques, meet-in-the-middle (MITM)

proposed by [30] works with the best complexity. And for hybrid dual attack, [8] gets the optimal result.

SIS attack. The unforgeability of our scheme relies on the hardness of the BimodalSelfTargetMSIS problem. For concrete security estimation, we follow a conservative approach: we assess the hardness of the underlying MSIS problem, which serves as a lower bound for BimodalSelfTargetMSIS. Specifically, we leverage the classical reduction from MSIS to BimodalSelfTargetMSIS established in [13], which implies that solving BimodalSelfTargetMSIS with norm bound $B/2$ is at least as hard as solving MSIS with norm bound B . We analyze the best known attacks for SIS problem, for both MSIS and BimodalSelfTargetMSIS problems that the unforgeability of our signature scheme relies on.

Given an SIS instance $\mathbf{A} \in \mathbb{Z}_q^{k \times l}$ with a bound B , we define the lattices $\Lambda_m'' = \{\mathbf{u} \in \mathbb{Z}^m : \mathbf{B}\mathbf{u} = \mathbf{0} \bmod q\}$ for all $m \leq k + l$, where $\mathbf{B}$ is the $k \times m$ leftmost sub-matrix of $(\mathbf{A} | \mathbf{Id}_k)$. Then a short non-zero vector in the lattice Λ_m'' is a solution to the SIS problem. Once more, we use BKZ and optimize the choice of m . Note that if $B > q$, then there are trivial non-zero solutions to SIS problem. Depending on the parameters, the security could be affected by some existing attacks [4]. We choose the prime q larger than the MSIS bound B to avoid such weaknesses.

Ratio attack. The ratio attack against G+G [39] exploits a structural vulnerability: since the first part of the secret is fixed to the identity (i.e., $s_1 = 1$), the corresponding signature component z_1 follows a different distribution than the other components z_i . More crucially, $z_{1,1}$ and each $z_{i,j}$ share a common random mask $\xi_1 = x_1 + c_1$. This statistical dependency causes the ratio $z_{i,j}/z_{1,1}$ to follow a Cauchy distribution whose parameters leak information about the secret $s_{i,j}$.

In Rhyme, we break this attack vector by applying an additional, lightweight rejection sampling step to z_1 alone (Algorithm 6). The purpose of this step is not to hide the secret—as in traditional rejection sampling—but to decouple z_1 from the shared mask ξ . As formally proven in Theorem 3.1, the rejection process ensures that the distribution of z_1 becomes statistically independent of ξ . Consequently, z_1 no longer carries any information about ξ , and the statistical link between z_1 and the other z_i through ξ is completely severed. Without this link, the ratio $\frac{z_{i,j}}{z_{1,1}}$ no longer exhibits a key-dependent Cauchy distribution, rendering the ratio attack ineffective against our construction.

Security Analysis of the GA-MLWE Problem. We now analyse the concrete security of the GA-MLWE assumptions introduced in Section 3.2. Recall that in all three variants the adversary is given a standard MLWE tuple $(\mathbf{A}, \mathbf{b} = \mathbf{A}\mathbf{s}_{\text{gen}} + \mathbf{e}_{\text{gen}})$ together with a Gram matrix $\mathbf{G}$ (or a submatrix thereof) of a short unimodular matrix $\mathbf{B}$ whose first column equals the secret $\mathbf{s}_{\text{tail}} = (\mathbf{s}_{\text{gen}}, \mathbf{e}_{\text{gen}})^T$. The goal is to distinguish $\mathbf{b}$ from a uniform vector.

The Gram matrix imposes algebraic relations among the columns of $\mathbf{B}$, in particular $\mathbf{G} = \mathbf{B}\mathbf{B}^*$. A natural question is whether an attacker can exploit these relations to break the underlying MLWE problem more efficiently than by ignoring $\mathbf{G}$. We consider two generic attack strategies and explain why neither provides a feasible advantage for the parameter ranges we use. Consequently, the security of a GA-MLWE instance can be conservatively estimated by separately evaluating the hardness of the core MLWE problem and that of the additional algebraic recovery problem (which is a form of the module lattice isomorphism problem, Module-LIP).

Algebraic recovery attack. The adversary tries to factor $\mathbf{G} = \mathbf{X}\mathbf{X}^*$ to obtain a short matrix $\mathbf{X}$ whose columns generate the same lattice as $\mathbf{B}$. Because $\mathbf{G}$ is invariant under orthogonal transformations ($\mathbf{G} = (\mathbf{U}\mathbf{B})(\mathbf{U}\mathbf{B})^*$ for any unitary $\mathbf{U}$), any recovered factor $\mathbf{X}$ satisfies $\mathbf{X} = \mathbf{U}\mathbf{B}$ for an unknown unitary $\mathbf{U}$. The first column of $\mathbf{X}$ is $\mathbf{U}\mathbf{s}_{\text{tail}}$, which is an *unknown unitary transform* of the true secret; it does not directly reveal $\mathbf{s}_{\text{tail}}$. Substituting this value into the LWE verification yields no advantage. To succeed, the attacker would have to solve a structured factorisation problem: given $\mathbf{G}$, find a short factor whose first column equals the true secret. This is at least as hard as the module lattice isomorphism problem (Module-LIP) over cyclotomic rings. In Hawk, the smLIP problem with rank 2 already resists known cryptanalytic techniques, and our constructions use rank at least 4, providing an additional safety margin. For our parameters (rank ≥ 4 , $\eta \geq 0.04\sqrt{dn}+0.14$) no algorithm is known to solve this problem faster than by enumerating the secret space or performing generic lattice reduction, both of which are exponential.

Gram-assisted lattice reduction. The attacker can instead treat $\mathbf{G}$ as auxiliary data that improves lattice reduction against the MLWE sample. The secret $\mathbf{s}_{\text{tail}}$ satisfies $\mathbf{s}_{\text{tail}}^* \mathbf{B} = \mathbf{G}_{1,:}$, i.e. it is part of a short basis whose Gram matrix is known. One could attempt to embed this information into the MLWE lattice (e.g. via an augmented matrix) and run BKZ or an analogous reduction algorithm. If such an approach gave a non-trivial advantage, it would yield an algorithm that extracts structural information about a short basis solely from its Gram matrix. This would directly contradict the hardness of Module-LIP, and would also imply an unexpected speed-up for standard lattice problems such as Module-SIS. However, no such algorithm is known, and our construction ensures that the covariance matrix $4\mathbf{Id} + 4\mathbf{G}$ is well-conditioned (minimum eigenvalue ≥ 1); the secret lattice is thus highly spherical and carries no more exploitable structure than a random lattice of the same dimension. Known lattice reduction methods therefore gain no benefit from the presence of $\mathbf{G}$.

The two attack strategies discussed above both attempt to exploit the Gram matrix indirectly—either by factoring it to recover the secret, or by using it

as auxiliary input to lattice reduction. We now turn to a more direct distinguishing attack that tests an algebraic relation between the public matrices and the challenge vector. Specifically, the adversary can check whether the matrix $\mathbf{M} = \mathbf{A}'\mathbf{G}_0\mathbf{A}'^* - \mathbf{t}\mathbf{t}^*$ is positive semidefinite. As we show below, this *semidefiniteness test* is the most dangerous algebraic attack against GA-MLWE, but it can be rendered ineffective by choosing the extended dimension D appropriately.

Semidefiniteness Test against GA-MLWE_{short}. We analyse the semidefiniteness test and show that it fails with overwhelming probability as soon as the extended dimension D is chosen large enough. The test computes

$$\mathbf{M} = \mathbf{A}'\mathbf{G}_0\mathbf{A}'^* - \mathbf{t}\mathbf{t}^*, \quad \mathbf{A}' = (\mathbf{A}_{\text{gen}} \mid \mathbf{Id}) \in \mathcal{R}_q^{k \times d},$$

where $d = k+l-1$. In the real world $\mathbf{t} = \mathbf{b} = \mathbf{A}'\mathbf{s}_{\text{tail}}$, hence $\mathbf{M}_{\text{real}} = \sum_{j \geq 2} \mathbf{c}_j \mathbf{c}_j^* \succeq \mathbf{0}$. In the random world $\mathbf{t} = \mathbf{u}$ is uniform, and

$$\mathbf{M}_{\text{rand}} = \mathbf{S} + (\mathbf{b}\mathbf{b}^* - \mathbf{u}\mathbf{u}^*), \quad \mathbf{S} = \sum_{j \geq 2} \mathbf{c}_j \mathbf{c}_j^*.$$

We work in the coefficient embedding, where multiplication by a ring element becomes multiplication by an $n \times n$ circulant matrix. All dimensions are therefore multiplied by n :

- $\mathbf{A}'$ becomes a $kn \times dn$ integer matrix.
- $\mathbf{B}_{\text{short}} \in \mathcal{R}_q^{d \times D}$ becomes $dn \times Dn$.
- Deleting the first column of $\mathbf{B}_{\text{short}}$ leaves $\mathbf{B}_{\text{re}} \in \mathcal{R}_q^{d \times (D-1)}$, which becomes $dn \times (D-1)n$.

Hence $\mathbf{C}_{\text{rest}} = \mathbf{A}'\mathbf{B}_{\text{re}}$ has size $kn \times (D-1)n$, and $\mathbf{S} = \mathbf{C}_{\text{rest}}\mathbf{C}_{\text{rest}}^\top$ is a $kn \times kn$ sample covariance matrix built from $p = (D-1)n$ independent columns. Because $\mathbf{A}'$ is uniformly random modulo q , the columns of $\mathbf{C}_{\text{rest}}$ are, up to negligible statistical distance, independent and uniformly distributed in $\mathbb{Z}_q^{kn}$; viewed as real vectors they are i.i.d. subgaussian with variance $\sigma^2 \approx q^2/12$.

Set $m = kn$. Standard results on the smallest eigenvalue of a Wishart-type matrix state that for any $t \geq 0$,

$$\Pr[\lambda_{\min}(\mathbf{S}) \leq \sigma^2(\sqrt{p} - \sqrt{m} - t)^2] \leq 2e^{-ct^2},$$

where $c > 0$ depends only on the subgaussian norm of the columns. On the other hand, Hoeffding's inequality gives, for any $\delta > 0$,

$$\Pr[\|\mathbf{u}\|^2 \geq (1 + \delta)m\sigma^2] \leq e^{-c'\delta^2 m},$$

with another absolute constant $c' > 0$.

We wish to guarantee that $\lambda_{\min}(\mathbf{S}) > \|\mathbf{u}\|^2$ holds with overwhelming probability. To this end we choose parameters so that

$$(\sqrt{p} - \sqrt{m} - t)^2 > (1 + \delta)m$$

for some small $t, \delta > 0$ with $e^{-ct^2}, e^{-c'\delta^2 m} \leq \text{negl}(\lambda)$. Taking $t = \varepsilon\sqrt{m}$ and $\delta = \varepsilon$, the inequality becomes

$$(\sqrt{p} - (1 + \varepsilon)\sqrt{m})^2 > (1 + \varepsilon)m \iff \sqrt{p} > (1 + \varepsilon + \sqrt{1 + \varepsilon})\sqrt{m}.$$

As $\varepsilon \rightarrow 0$ the right-hand side tends to $2\sqrt{m}$, i.e. $p > 4m$. For any fixed n we can pick a tiny ε (e.g. $\varepsilon = 1/\sqrt{n}$) so that the tail probabilities are exponentially small in n and the condition becomes $p > 4m + O(\sqrt{m})$. Replacing $p = (D - 1)n$ and $m = kn$ yields the sufficient condition

$$D - 1 > 4k + o(1) \implies D > 4k + 1.$$

With this choice we obtain

$$\Pr[\lambda_{\min}(\mathbf{S}) > \|\mathbf{u}\|^2] \geq 1 - \exp(-\Omega(n)).$$

By Weyl's inequality,

$$\lambda_{\min}(\mathbf{M}_{\text{rand}}) \geq \lambda_{\min}(\mathbf{S}) - \max(\|\mathbf{b}\|^2, \|\mathbf{u}\|^2) > 0$$

with the same overwhelming probability. Hence $\mathbf{M}_{\text{rand}}$ is positive semidefinite and the test fails to distinguish the two worlds.

Parameter constraint. The GA-MLWE assumption is therefore plausible whenever $D > 4k + 1$. For typical compact parameters $(k, l) = (2, 3)$ we may set $D = 10$; for $(k, l) = (3, 4)$ we set $D = 14$. Under these choices the algebraic distinguisher has negligible advantage, and the security of GA-MLWE rests entirely on the module lattice isomorphism problem.

Concrete parameter selection via smLIP. To estimate the concrete security of the module-LIP instance underlying our GA-MLWE assumption, we adopt the methodology of the Hawk signature scheme [21]. For a target security level of 128/256/384/512 bits, Hawk's security estimator requires the total dimension of the module-LIP instance to be at least 1024/2048/3072/4096. Additionally, to avoid weak instances, the *expected squared Euclidean norm of each row* of the secret matrix must be at least $N \cdot \sigma_{\text{sec}}^2$, where N is the instance dimension and σ_{sec} is a reference standard deviation bounded by the experimentally validated heuristic

$$\sigma_{\text{sec}} \geq 0.04\sqrt{N} + 0.14.$$

Importantly, this condition is imposed on the row norm, not on the standard deviation of each individual entry. If the rows are longer than the instance dimension, a smaller entry-wise standard deviation can still satisfy the row-norm requirement.

In our setting, the relevant module-LIP instance has dimension $N = dn = (k+l-1)n$. The secret matrix $\mathbf{B}_{\text{short}}$ has size $dn \times Dn$, and its entries are drawn from a centred binomial distribution χ_η with variance $\eta/2$. Its rows therefore have expected squared norm $Dn \cdot (\eta/2)$, which must be at least $dn \cdot \sigma_{\text{sec}}^2$. Because we can freely increase D , this allows us to use a smaller η (e.g. $\eta = 2$) while still satisfying the condition.

Consequently, we select parameters as follows. For a given security level, we choose d, n such that

$$dn \geq N_{\text{target}} \quad \text{and} \quad \eta \geq 2 \cdot (0.04\sqrt{N_{\text{target}}} + 0.14)^2 \cdot \frac{N_{\text{target}}}{Dn},$$

where $N_{\text{target}} \in \{1024, 2048, 3072, 4096\}$ for 128/256/384/512-bit security. Taking 128-bit security as an example, with $dn = 1024$ and $Dn = 2560$ we obtain $\eta \geq 2 \cdot (0.04\sqrt{1024} + 0.14)^2 \cdot 1024/2560 \approx 1.61$, which is satisfied by taking $\eta = 2$. The resulting smLIP instance then has at least the same concrete hardness as a 1024-dimensional instance with the minimal recommended $\sigma_{\text{sec}} = 1.42$, guaranteeing that the GA-MLWE assumption inherits the desired security margin from module-LIP.

5 Performance

5.1 Parameter Sets

We provide four parameter sets which are designed to meet the 128, 256, 384 and 512-bit classical security levels (denoted as Rhyme-128/256/384/512) according to the requirements of NICCS [32] in Table 3.

The variables B_1 and B_2 denote the two verification bounds in our scheme. The numbers in parentheses for SIS security are for the strongly-unforgeable version of the signature scheme. The signature size is computed by $n \cdot \lceil \log_2(2B_1 + 1) \rceil / 8 * C_Rate_1 + (k + l - 1) * n * \lceil \log_2(2B_2 + 1) \rceil / 8 * C_Rate_2 + n * \log_2(2)/8$, where C_Rate_1, C_Rate_2 denote the compress rates of the two corresponding rANS settings, which range from 0.75 to 0.85 in our experiments. The public key size is computed by $\lceil k * n * \lceil \log_2(q) \rceil / 8 \rceil + n * \log_2(2)/8$ and the secret key size is $(k + l - 1) * D * n * \lceil \log_2(2\eta + 1) \rceil / 8 + n * \log_2(2)/8$. Due to the fact that private keys are usually not used for transmission, their NTT form can be saved simultaneously in practical implementation to reduce computation latency..

Table 3. Parameters for Rhyme_NGCC

Security	128	256	384	512
n	256	512	512	1024
q	3329	9473	11777	18433
$\sigma/(2\sqrt{\pi})$	1.295870	1.316036	1.326374	1.335898
(k, l)	(2, 3)	(2, 3)	(3, 4)	(2, 3)
D	10	10	14	10
τ	30	58	113	115
η	2	4	5	6
σ_y	122.0	180.0	182.0	266.0
B_1	388	598	605	916
B_2	581	1219	1654	2209
B_{L_2}	4685.06	12842.68	20258.77	31350.08
$\Pr(\ \mathbf{z}_1\ _\infty < B_1)$	0.5	0.5	0.5	0.5
$\Pr(\ \mathbf{z}_{\text{bottom}}\ _\infty < B_2)$	> 0.999	> 0.999	> 0.999	> 0.999
$\Pr(\ \mathbf{z}\ _2 < B_{L_2})$	0.7	0.7	0.7	0.7
repetitions ($1/\Pr_{\text{succ}}$)	2.73	2.99	2.95	2.69
Forgery				
Classical BKZ block-size b	357(282)	812(671)	1263(1059)	1738(1472)
Classical hardness	130.25(109.48)	257.99(218.58)	384.94 (327.69)	518.97(444.10)
Quantum BKZ block-size b	357(282)	812(671)	1263(1059)	1738(1472)
Quantum hardness	91.75(72.47)	208.68(172.45)	324.59(272.16)	446.67(378.30)
Key-Recovery				
Classical BKZ block-size b	363	802	1280	1824
Classical hardness	132.84	256.24	390.93	516.40
Quantum BKZ block-size b	376	813	1293	1739
Quantum hardness	96.64	208.95	332.60	446.95
LIP				
Classical BKZ block-size b	456	932	1412	1893
Classical hardness	133.15	272.14	412.30	552.76
Quantum BKZ block-size b	456	932	1412	1893
Quantum hardness	120.84	246.98	374.18	501.65
Signature size	1483	3258	4743	7002
Public key size	800	1824	2720	3872
Secret key size	11072	22336	45760	44864

5.2 Performance Analysis

In this subsection, we report on the implementation performance. The reference implementation and the security estimation scripts are available at <https://github.com/pq-CUC/Rhyme>, where we provide a C-language reference implementation for the four sets of parameters mentioned above.

At the theoretical level, the main overhead of key generation in Rhyme consists of expanding $\mathbf{A}_{\text{gen}}$ from a seed, one matrix-vector NTT multiplication modulo q (to compute $\mathbf{b}_{\text{gen}}$), and the **GenUnimod** operation. During signing, the main overhead involves expanding $\mathbf{A}_{\text{gen}}$, one matrix-vector NTT multiplication modulo $2q$ (for $\mathbf{w}$), one lightweight coefficient-wise rejection sampling, and one polynomial-vector NTT multiplication modulo $2q$ (for $\mathbf{z}_{\text{bottom}}$), with possible repetitions of the entire loop to meet the norm bound. During verification, the main overhead is expanding $\mathbf{A}_{\text{gen}}$ and performing a single matrix-vector NTT multiplication modulo $2q$. It can be seen that **GenUnimod** accounts for a significant fraction of the key-generation time. This overhead can be reduced by replacing the existing deterministic decision algorithm with a probabilistic one, which reduces the asymptotic complexity from $n^2 \log^2 n$ to $n \log n$.

Table 4 compares the performance and sizes of Rhyme (SHAKE version) and Dilithium. The experimental results show that our bandwidth is 30%–40% smaller than that of Dilithium. In terms of efficiency, signing takes about twice as long as Dilithium, while verification takes about half the time; however, key generation is much slower (over 300 $\times$), which is caused by **GenUnimod** as analyzed above. Replacing it with a probabilistic algorithm can improve key-generation efficiency at the cost of a small loss in the security reduction. Besides, we summarize the performance of the SM3 version with the experiment results shown in Table 5 by utilizing the function interface provided by NGCC.

The reference implementation of the Rhyme signature algorithm (using SM3 with no architecture-specific optimizations) was compiled and executed on an x86_64 machine running Ubuntu 20.04.6 LTS (Linux kernel 5.15.0-139-generic) with GCC 9.4.0. The CPU clock speed was 2.9 GHz. For each security level (Rhyme-SM3-128, Rhyme-SM3-256, Rhyme-SM3-384, and Rhyme-SM3-512), the build process consists of navigating to the corresponding subdirectory under `Implementations/Reference.Implementation/Rhyme-SM3/` (e.g., Rhyme-SM3-128) and running `make clean && make && ./KAT_SIG`. The compiler flags used for all builds are `-O3 -DNDEBUG -DRHYME_NO_AES -D_GNU_SOURCE -std=gnu11 -D_POSIX_C_SOURCE=199309L`. Deterministic known-answer tests (KAT) are generated using the API_PKC support files `KAT_SIG.c`, `drng.c`, and `auxfunc.c`; the output must match the test vectors located in `Test_Vectors/Rhyme-SM3/KAT_SIG.Rhyme-SM3-xxx.txt`. This reference implementation does not include dedicated performance benchmarking; timing evaluation of the signature primitives (keygen, sign, verify) can be carried out separately using platform-specific cycle counters.

Although the machine is equipped with an NVIDIA GeForce RTX 3090 GPU, all operations are CPU-only; the GPU was not used.

The optimized implementation of the Rhyme signature algorithm (using SM3 with AVX2 acceleration, no GPU usage) was compiled and executed on an x86_64 machine running Ubuntu 20.04.6 LTS (Linux kernel 5.15.0-139-generic) with GCC 9.4.0. The CPU clock speed was 2.9 GHz and the processor supports the AVX2 instruction set. For each security level (Rhyme-SM3-128, Rhyme-SM3-256, Rhyme-SM3-384, and Rhyme-SM3-512), the build process consists of navigating to the corresponding subdirectory under `Implementations/Optimized_Implementation/Rhyme-SM3/` (e.g., `Rhyme-SM3-128`) and running `make clean && make && ./KAT_SIG`. The compiler flags used for all builds are `-O3 -DNDEBUG -mavx2 -DRHYME_NO_AES -D_GNU_SOURCE -std=gnu11 -D_POSIX_C_SOURCE=199309L`. The AVX2 optimizations are concentrated in the SM3 compression layer: the file `simd/sm3_avx2.c` provides two acceleration paths—a batched `rhyme_sm3_xof_batch8` that processes eight XOF output blocks in parallel when at least eight are needed, and a single-hash `rhyme_sm3_hash_avx2` fallback for fewer blocks. The SM3-XOF state machine in `sm3_xof.c` dispatches to these SIMD routines when `USE_AVX2_SM3` is defined; all other source files are identical to the reference implementation. Deterministic known-answer tests (KAT) are generated using the APLPKC support files `KAT_SIG.c`, `drng.c`, and `auxfunc.c`; the output is byte-identical to the reference implementation test vectors located in `Test_Vectors/Rhyme-SM3/ KAT_SIG_Rhyme-SM3-xxx.txt`. Performance benchmarking of the signature primitives (key generation, signing, verification) was performed on the same system, pinning the process to core 0, with 1,000 warm-up iterations and measuring the median of 50–100 keygen runs and 1,000 sign+verify runs per security level, using the `rdtsc_x86` cycle counter. Although the machine is equipped with an NVIDIA GeForce RTX 3090 GPU, all cycle measurements are CPU-only; the GPU was not used. Compared with the reference implementation, the AVX2-accelerated version provides modest speedups: approximately $1.05\times$ for key generation, $1.11\times$ – $1.17\times$ for signing, and $1.40\times$ – $1.62\times$ for verification, depending on the security level.

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Parameter set		KeyGen	Sign	Verify	SIG	PK	Bandwidth
Rhyme-128	med	94509600	2112650	171040	1483	800	2283
	avg	160877270	2622907	175515			
Rhyme-256	med	366124710	4295166	387072	3258	1824	5082
	avg	615745885	4944023	395830			
Dilithium-120	med	294822	1008660	313772	2420	1312	3732
	avg	338308	1267208	316207			
Dilithium-260	med	777396	2067854	799102	4627	2592	7219
	avg	816470	2498907	803060			

Table 4. Efficiency (Median and average cycle counts of 1000 executions) and Sizes (bytes) for Rhyme (SHAKE version) and Dilithium on x86_64 platform

Parameter set		KeyGen	Sign	Verify
Rhyme-128(ref)	med	579643332	5964664	403476
	avg	1184133792	7280252	405759
Rhyme-128(avx2)	med	585214818	4819214	253088
	avg	1196162408	5903877	260761
Rhyme-256(ref)	med	2425409156	12267064	962226
	avg	4644364756	18121447	966673
Rhyme-256(avx2)	med	2343818636	9794910	613860
	avg	4481319590	14358769	621899
Rhyme-384(ref)	med	13159955550	18296200	1616744
	avg	11392405947	22756732	1620526
Rhyme-384(avx2)	med	12795285560	14384960	963200
	avg	11085814499	17834035	976007
Rhyme-512(ref)	med	18769719464	24460332	2141014
	avg	18351416310	34077677	2147683
Rhyme-512(avx2)	med	18910685086	18719130	1268350
	avg	18484158517	25980999	1279235

Table 5. Median and average cycle counts of 1000 executions for Rhyme (SM3 version) on x86_64 platform

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A Proof of the Lightweight Rejection Sampling Process

We use a simplified variant of the rejection sampling method from [24], adapted to operate over $\mathbb{Z}$. Let σ_y be the parameter of the masking Gaussian, and let D_{σ_y} denote the discrete Gaussian distribution over $\mathbb{Z}$ with probability mass function proportional to $\rho_{\sigma_y}(x) = \exp(-\pi x^2/\sigma_y^2)$. For each $i \in [n]$, the value y_i is sampled from D_{σ_y} conditioned on $|y_i| \leq B_1 + R$, where R is a small integer (e.g., $R = \lceil 5\sigma_{tar} \rceil$). Let σ_{tar} be the parameter of the target discrete Gaussian. Define, for each challenge bit $c \in \{0, 1\}$, the set

$$\mathcal{V}_c = \{v \in \mathbb{Z} : |v| \leq R, v \equiv c \pmod{2}\}$$

and the discrete Gaussian distribution $\chi_{\sigma_{tar}, R}^{(c)}$ on $\mathcal{V}_c$ with probability mass function

$$p_c(v) = \frac{\rho_{\sigma_{tar}}(v)}{\sum_{u \in \mathcal{V}_c} \rho_{\sigma_{tar}}(u)}, \quad \rho_{\sigma_{tar}}(v) = \exp(-\pi v^2/\sigma_{tar}^2).$$

On input (y_i, c_i) the algorithm outputs $(x_i, z_i = y_i + c_i + 2x_i)$ satisfying the following three properties:

1. The distribution of z_i is D_{σ_y} conditioned on $|z_i| \leq B_1$.
2. When $c_i = 1$, $v_i = 2x_i + 1$ is distributed as $\chi_{\sigma_{tar}, R}^{(1)}$ (a discrete Gaussian over odd integers in $[-R, R]$); when $c_i = 0$, $v_i = 2x_i$ is distributed as $\chi_{\sigma_{tar}, R}^{(0)}$ (a discrete Gaussian over even integers in $[-R, R]$).
3. x_i is independent of z_i .

The details are given in Algorithms 3 and 4, with the proof of Theorem 3.1 followed. Because both the masking and the target distributions are Gaussians, the acceptance probability admits a clean closed-form estimate, and the constant M can be tightly controlled.

Proof. The inputs y_i are drawn independently from the conditional distribution $\Pr(y) \propto \rho_{\sigma_y}(y)$ for $|y| \leq B_1 + R$. Fix $c \in \{0, 1\}$ and write $\mathcal{V} = \mathcal{V}_c$, $p = p_c$.

In Algorithm 3, given y , each $v \in \mathcal{V}$ with $|y + v| \leq B_1$ is chosen with probability

$$\frac{p(v) \rho_{\sigma_y}(y + v)}{M \rho_{\sigma_y}(y)}.$$

An output pair $(z, 2x)$ uniquely determines $v = 2x + c$ and $y = z - v$. Hence the joint probability of $(z, 2x)$ is proportional to

$$\Pr(y) \cdot \frac{p(v) \rho_{\sigma_y}(y+v)}{M \rho_{\sigma_y}(y)} \propto \rho_{\sigma_y}(y) \cdot \frac{p(v) \rho_{\sigma_y}(z)}{M \rho_{\sigma_y}(y)} = \frac{p(v) \rho_{\sigma_y}(z)}{M},$$

where we used $\rho_{\sigma_y}(y+v) = \rho_{\sigma_y}(z)$. The boundary condition explicitly drops any v for which $|z| > B_1$; therefore the above relation holds for all $|z| \leq B_1$ and $v \in \mathcal{V}$.

Summing over $v \in \mathcal{V}$ gives the marginal distribution of z :

$$\Pr(z) \propto \frac{\rho_{\sigma_y}(z)}{M} \sum_{v \in \mathcal{V}} p(v) = \frac{\rho_{\sigma_y}(z)}{M}, \quad |z| \leq B_1,$$

which is exactly D_{σ_y} conditioned on $|z| \leq B_1$. This proves property 1.

The conditional distribution of $v = 2x + c$ given z is

$$\Pr(v | z) = \frac{\Pr(z, 2x)}{\Pr(z)} \propto p(v),$$

which is independent of z and matches $\chi_{\sigma_{tar}, R}^{(c)}$. Consequently, x is independent of z and follows the required shifted distribution, establishing properties 2 and 3.

Acceptance probability. For a single coordinate, the probability that Algorithm 3 does not return $\perp$ is

$$\begin{aligned} \Pr_{\text{acc}} &= \sum_{|y| \leq B_1 + R} \Pr(y) \sum_{\substack{v \in \mathcal{V} \\ |y+v| \leq B_1}} \frac{p(v) \rho_{\sigma_y}(y+v)}{M \rho_{\sigma_y}(y)} \\ &= \frac{1}{M} \cdot \frac{1}{\sum_{|y| \leq B_1 + R} \rho_{\sigma_y}(y)} \sum_{|y| \leq B_1 + R} \sum_{\substack{v \in \mathcal{V} \\ |y+v| \leq B_1}} p(v) \rho_{\sigma_y}(y+v) \\ &= \frac{1}{M} \cdot \frac{1}{\sum_{|y| \leq B_1 + R} \rho_{\sigma_y}(y)} \sum_{v \in \mathcal{V}} p(v) \sum_{\substack{|y| \leq B_1 + R \\ |y+v| \leq B_1}} \rho_{\sigma_y}(y+v). \end{aligned}$$

Substituting $z = y + v$, the inner sum becomes $\sum_{|z| \leq B_1} \rho_{\sigma_y}(z)$ because $|z| \leq B_1$ automatically implies $|z - v| \leq B_1 + R$ for all $|v| \leq R$. Since $\sum_v p(v) = 1$, we obtain

$$\Pr_{\text{acc}} = \frac{1}{M} \frac{\sum_{|z| \leq B_1} \rho_{\sigma_y}(z)}{\sum_{|y| \leq B_1 + R} \rho_{\sigma_y}(y)}.$$

Thus the acceptance rate is inversely proportional to M and is governed by the tail mass of the masking Gaussian.

Choice of M . The constant M must satisfy

$$M \geq \max_{|y| \leq B_1 + R} \sum_{\substack{v \in \mathcal{V}_c \\ |y+v| \leq B_1}} p_c(v) \frac{\rho_{\sigma_y}(y+v)}{\rho_{\sigma_y}(y)}.$$

Using $p_c(v) = \rho_{\sigma_{tar}}(v)/S_c$ with $S_c = \sum_{u \in \mathcal{V}_c} \rho_{\sigma_{tar}}(u)$ and $\rho_{\sigma_y}(y+v)/\rho_{\sigma_y}(y) = \exp(-\frac{2\pi yv + \pi v^2}{\sigma_y^2})$, the right-hand side becomes

$$\frac{1}{S_c} \sum_{\substack{v \in \mathcal{V}_c \\ |y+v| \leq B_1}} \exp\left(-\frac{\pi v^2}{\sigma_{tar}^2} - \frac{2\pi yv + \pi v^2}{\sigma_y^2}\right).$$

To obtain a clean analytic bound, we first drop the restriction $|y+v| \leq B_1$ and extend the sum to all integers:

$$A(y) \leq \frac{1}{S_c} \sum_{v \in \mathbb{Z}} \exp\left(-\frac{\pi v^2}{\sigma_{tar}^2} - \frac{2\pi yv + \pi v^2}{\sigma_y^2}\right).$$

Define

$$\alpha = \pi\left(\frac{1}{\sigma_{tar}^2} + \frac{1}{\sigma_y^2}\right), \quad \beta = \frac{2\pi y}{\sigma_y^2}.$$

The exponent rewrites as $-\alpha v^2 - \beta v$. Completing the square gives

$$-\alpha v^2 - \beta v = -\alpha\left(v + \frac{\beta}{2\alpha}\right)^2 + \frac{\beta^2}{4\alpha}, \quad \frac{\beta^2}{4\alpha} = \frac{\pi y^2 \sigma_{tar}^2}{\sigma_y^2(\sigma_{tar}^2 + \sigma_y^2)}.$$

Because summing $\exp(-\alpha(v + \delta)^2)$ over $\mathbb{Z}$ is translation invariant, we obtain

$$\sum_{v \in \mathbb{Z}} e^{-\alpha(v + \beta/(2\alpha))^2} = \sum_{v \in \mathbb{Z}} e^{-\alpha v^2} \triangleq \vartheta(\alpha),$$

and therefore

$$A(y) \leq \frac{\vartheta(\alpha)}{S_c} \exp\left(\frac{\pi y^2 \sigma_{tar}^2}{\sigma_y^2(\sigma_{tar}^2 + \sigma_y^2)}\right).$$

The exponential factor is increasing with $|y|$, so the maximum over $|y| \leq B_1 + R$ is attained at $|y| = B_1 + R$, leading to the universal bound

$$M \leq \frac{\vartheta(\alpha)}{S_c} \exp\left(\frac{\pi(B_1 + R)^2 \sigma_{tar}^2}{\sigma_y^2(\sigma_{tar}^2 + \sigma_y^2)}\right).$$

In practice, a much tighter value is obtained by evaluating the original sum directly on the boundary. Let R_c be the largest integer in $\mathcal{V}_c$ with $R_c \leq R$ and $R_c \equiv c \pmod{2}$; explicitly,

$$R_c = \begin{cases} R, & \text{if } R \equiv c \pmod{2}, \\ R-1, & \text{otherwise.} \end{cases}$$

Consider $y = B_1 + R_c$. The condition $|y + v| \leq B_1$ forces $v \leq -R_c$, while $v \in \mathcal{V}_c \cap [-R, R]$ forces $v \geq -R_c$; hence the only admissible v is $v = -R_c$. The sum at this y therefore reduces to the single term

$$p_c(-R_c) \frac{\rho_{\sigma_y}(y - R_c)}{\rho_{\sigma_y}(y)} = \frac{\rho_{\sigma_{tar}}(R_c)}{S_c} \frac{\rho_{\sigma_y}(B_1)}{\rho_{\sigma_y}(B_1 + R_c)}.$$

By symmetry, the same value is attained at $y = -B_1 - R_c$. The exponential growth of the ratio with $|y|$ guarantees that this boundary point dominates all other y . Consequently we can simply set

$$M = \frac{\rho_{\sigma_{tar}}(R_c)}{S_c} \frac{\rho_{\sigma_y}(B_1)}{\rho_{\sigma_y}(B_1 + R_c)} = \frac{\rho_{\sigma_{tar}}(R_c)}{S_c} \exp\left(\frac{2\pi B_1 R_c + \pi R_c^2}{\sigma_y^2}\right).$$

This closed form provides a valid (and typically tight) upper bound for M , as the boundary point dominates the sum for typical parameter choices.

B Proof of Construction 1

Observation. Let $\mathcal{R}_q = \mathbb{Z}_q[x]/(x^n + 1)$, where $n = 2^\kappa$ and $q \equiv 1 \pmod{2n}$ is a prime. Denote by $\text{Gal} := \text{Gal}(\mathcal{R}_q/\mathbb{Z}_q)$ the Galois group of the cyclotomic ring; its elements act as automorphisms $\tau \in \text{Gal}$. For an element $a \in \mathcal{R}_q$, its *trace* is $\text{Tr}(a) = \sum_{\tau \in \text{Gal}} \tau(a)$, which always evaluates to a constant polynomial (i.e., a scalar in $\mathbb{Z}_q$).

Construction B.1 (Full Automorphism Isotropic Construction). *Let $\mathcal{R}_q = \mathbb{Z}_q[x]/(x^n + 1)$ with $n = 2^\kappa \geq 2$ and $q \equiv 1 \pmod{2n}$ a prime. Let $\text{Gal} = \{\tau_j : x \mapsto x^j \mid j \in (\mathbb{Z}_{2n})^\times\}$ be the Galois group of $\mathcal{R}_q$ over $\mathbb{Z}_q$; its order is n . Fix integers $a, b, c, d \in \mathbb{Z}$ with $a^2 + b^2 + c^2 + d^2 = 4(m - 1)$. Let a secret vector $\mathbf{s} = (s_2, \dots, s_m)^\top \in \mathcal{R}_q^m$ have all coefficients in $\{\pm 1\}$. For each component $i = 2, \dots, m$, the raw output polynomial is*

$$z_i = 2s_i \cdot \left(\sum_{j < i} E_{j,i} - \sum_{j > i} E_{i,j} \right) + \sum_{\tau \neq \text{id}} \tau(s_i) \cdot (aU_{i,\tau}^{(1)} + bU_{i,\tau}^{(2)} + cU_{i,\tau}^{(3)} + dU_{i,\tau}^{(4)}) + 2s_i \cdot X_0 + e'_i.$$

All noise polynomials have independent coefficients drawn from the one-dimensional discrete Gaussian $D_{\mathbb{Z},\sigma}$ with Gaussian parameter $\sigma > 0$. The noises are:

- $E_{i,j}$ for $2 \leq i < j \leq m$, and $E_{j,i} = E_{i,j}$;
- X_0 ;
- $U_{i,\tau}^{(1)}, \dots, U_{i,\tau}^{(4)}$ for each i and $\tau \neq \text{id}$;
- e'_i for each i .

Theorem B.1 (Statistical Hiding with Tight Parameters). *For the Construction B.1, let $m \geq 2$ and $n = 2^\kappa$. Choose the discrete Gaussian parameter σ such that*

$$\sigma \geq 2\sqrt{(m-1)n} \cdot \sqrt{\frac{\ln(N - (m-1)n - 1 + 2(N - (m-1)n)/\varepsilon)}{\pi}}$$

for a negligible ε , where $N = \binom{m-1}{2} + 4(m-1)(n-1) + m)n$ is the total number of scalar noise coefficients extracted from all independent noise polynomials. Then the distribution of the final coefficient vector $\tilde{\mathbf{z}}$ is statistically indistinguishable (within max-log distance 4ε) from a rounded spherical continuous Gaussian $\mathcal{N}(\mathbf{0}, \frac{(4(m-1)n^2+1)\sigma^2}{2\pi} \mathbf{I}_{(m-1)n})$. In particular, the output reveals no information about the secret $\mathbf{s}$ beyond the public constant n .

Proof. Flatten all polynomials into coefficient vectors under the power basis. Collect the N noise coefficients into $\mathbf{w} \sim D_{\mathbb{Z}^N, \sigma}$. Set

$$\tilde{\mathbf{B}} = [\mathbf{B} \mathbf{I}_{(m-1)n}],$$

where $\mathbf{B} \in \mathbb{Z}^{(m-1)n \times (N - (m-1)n)}$ is the integer matrix of the first $N - (m-1)n$ columns of the construction. Then the output vector is $\tilde{\mathbf{z}} = \tilde{\mathbf{B}}\tilde{\mathbf{w}}$.

We now verify that $\mathbf{B}\mathbf{B}^\top = 4(m-1)n^2\mathbf{I}$. Consider the contributions of each noise source:

- For each pair (i, j) with $i < j$, the column $E_{i,j}$ contributes $-4\mathbf{M}(\mathbf{s}_i\mathbf{s}_j^*)$ to the (i, j) off-diagonal block and $4\mathbf{M}(\mathbf{s}_i\mathbf{s}_i^*)$, $4\mathbf{M}(\mathbf{s}_j\mathbf{s}_j^*)$ to the i -th and j -th diagonal blocks respectively.
- The shared noise X_0 contributes $4\mathbf{M}(\mathbf{s}_i\mathbf{s}_j^*)$ to every off-diagonal block (i, j) and $4\mathbf{M}(\mathbf{s}_i\mathbf{s}_i^*)$ to each diagonal block.
- Thus the off-diagonal blocks cancel completely: for $i \neq j$, the sum is zero.
- Summing the diagonal contributions from cancellation columns and X_0 , each index i receives $4(m-1)\mathbf{M}(\mathbf{s}_i\mathbf{s}_i^*)$.
- The automorphism columns add $4(m-1)\sum_{\tau \neq \text{id}} \mathbf{M}(\tau(\mathbf{s}_i\mathbf{s}_i^*))$, so the total diagonal block becomes $4(m-1)\sum_{\tau \in \text{Gal}} \mathbf{M}(\tau(\mathbf{s}_i\mathbf{s}_i^*))$. Since $\sum_{\tau} \tau(a) = \text{Tr}(a)$ is a scalar matrix, and $\text{Tr}(\mathbf{s}_i\mathbf{s}_i^*) = n^2$ for $\{\pm 1\}$ -coefficients, this block equals $4(m-1)n^2\mathbf{I}_n$.

Adding the independent noises e'_i (which contribute $\mathbf{I}_n$ each) completes the computation of $\tilde{\mathbf{B}}\tilde{\mathbf{B}}^\top$. Therefore

$$\tilde{\mathbf{B}}\tilde{\mathbf{B}}^\top = \mathbf{B}\mathbf{B}^\top + \mathbf{I}_{(m-1)n} = (4(m-1)n^2 + 1)\mathbf{I}_{(m-1)n}.$$

According to Lemma 2.2 and 2.3, we should set $\sigma \geq \eta_\varepsilon(\Lambda^\perp(\tilde{\mathbf{B}}))$ to satisfies the demand for smoothing parameter.

Step 1: Explicit short basis for the new kernel lattice. The kernel lattice of the construction is

$$\Lambda^\perp(\tilde{\mathbf{B}}) = \{(\mathbf{x}, \mathbf{y}) \in \mathbb{Z}^{N-(m-1)n} \times \mathbb{Z}^{(m-1)n} \mid \mathbf{B}\mathbf{x} + \mathbf{y} = \mathbf{0}\}.$$

For each original noise coordinate $j = 1, \dots, N - (m-1)n$, let $\mathbf{b}_j$ be the j -th column of $\mathbf{B}$. Define

$$\mathbf{v}_j = (\mathbf{e}_j, -\mathbf{b}_j) \in \mathbb{Z}^N,$$

where $\mathbf{e}_j$ is the j -th standard basis vector. Clearly $\mathbf{B}\mathbf{e}_j + (-\mathbf{b}_j) = \mathbf{b}_j - \mathbf{b}_j = \mathbf{0}$, so each $\mathbf{v}_j \in \Lambda^\perp(\tilde{\mathbf{B}})$. Moreover, the matrix $[\mathbf{I}_{N-(m-1)n}; -\mathbf{B}]$ has full column rank, hence the $N - (m-1)n$ vectors $\{\mathbf{v}_j\}$ are linearly independent and span the whole $(N - (m-1)n)$ -dimensional lattice $\Lambda^\perp(\tilde{\mathbf{B}})$. Their Euclidean lengths are

$$\|\mathbf{v}_j\|_2 = \sqrt{1 + \|\mathbf{b}_j\|_2^2}.$$

From the construction, every column $\mathbf{b}_j$ falls into one of three categories:

- E -noises: the column contains exactly two $\pm 2\mathbf{s}_i, \pm 2\mathbf{s}_j$ for some $i \neq j$; $\|\mathbf{s}_i\|_2 = \|\mathbf{s}_j\|_2 = \sqrt{n}$, so $\|\mathbf{b}_j\|_2 = 2\sqrt{2n}$.
- Base noise X_0 : the column contains exactly all $2\mathbf{s}_i$, so $\|\mathbf{b}_j\|_2 = 2\sqrt{(m-1)n}$.
- U -noises: the column is $c_t \cdot \tau(\mathbf{s}_i)$ with $c_t \in \{a, b, c, d\}$ and $\|\tau(\mathbf{s}_i)\|_2 = \sqrt{n}$; because $a^2 + b^2 + c^2 + d^2 = 4(m-1)$, the longest such column has $\|\mathbf{b}_j\|_2 = \sqrt{4(m-1)}\sqrt{n} = 2\sqrt{(m-1)n}$.

Hence every basis vector satisfies $\|\mathbf{v}_j\|_2 \leq \sqrt{1 + 4(m-1)n} = 2\sqrt{(m-1)n} + o(1)$. For concreteness, set $L = 2\sqrt{(m-1)n}$.

Step 2: Smoothing parameter of the kernel lattice. The dimension of $\Lambda^\perp(\tilde{\mathbf{B}})$ is $d = N - (m-1)n$. With a negligible ε , the standard smoothing parameter bound [19, 43] yields

$$\eta_\varepsilon(\Lambda^\perp(\tilde{\mathbf{B}})) \leq \sqrt{\frac{\ln(d-1+2d/\varepsilon)}{\pi}} \cdot \lambda_d \leq \sqrt{\frac{\ln(2d/\varepsilon)}{\pi}} \cdot L,$$

because λ_d is bounded by the length of any full set of linearly independent lattice vectors, in particular our basis $\{\mathbf{v}_j\}$.

Inserting $d = N - (m-1)n$ and $L = 2\sqrt{(m-1)n}$ reproduces the condition stated in the theorem; under that condition, $\sigma \geq \eta_\varepsilon(\Lambda^\perp(\tilde{\mathbf{B}}))$.

Step 3: From discrete noise to a spherical discrete Gaussian. When $\sigma \geq \eta_\varepsilon(\Lambda^\perp(\tilde{\mathbf{B}}))$, the linear transformation theorem for discrete Gaussians (Lemma 2.3) implies that the distribution of $\tilde{\mathbf{z}} = \tilde{\mathbf{B}}\tilde{\mathbf{w}}$ is statistically close (within 4ε) to the discrete Gaussian $D_{\Lambda', \tilde{\mathbf{B}}\sigma}$, where $\Lambda' = \tilde{\mathbf{B}}\mathbb{Z}^N$ is the image lattice. Since $\tilde{\mathbf{B}}\tilde{\mathbf{B}}^\top = (4(m-1)n^2 + 1)\mathbf{I}_{(m-1)n}$, all singular values of $\tilde{\mathbf{B}}$ are $\sqrt{4(m-1)n^2 + 1}$, and thus $D_{\Lambda', \tilde{\mathbf{B}}\sigma} = D_{\Lambda', \sqrt{4(m-1)n^2 + 1}\sigma}$. Because $\Lambda' = \mathbf{B}\mathbb{Z}^{N-(m-1)n} + \mathbb{Z}^{(m-1)n} =$

$\mathbb{Z}^{(m-1)n}$, the image lattice is simply the standard integer lattice $\mathbb{Z}^{(m-1)n}$. So the output distribution is statistically close (within max-log distance at most 4ε) to the distribution obtained by sampling from $\mathcal{N}(\mathbf{0}, \frac{4(m-1)n^2+1}{2\pi} \sigma^2 \mathbf{I}_{(m-1)n})$ and rounding each coordinate to the nearest integer. This rounded distribution is manifestly spherically symmetric and depends only on n, m, σ , not on the secret $\mathbf{s}$. $\square$

Construction B.2 (Isotropic Construction with Random Subset of Automorphisms). *Let $\chi_\sigma^\mathcal{R}$ be the distribution over $\mathcal{R}_q$ where each of the n coefficients is drawn independently from the one-dimensional discrete Gaussian $D_{\mathbb{Z}, \sigma}$ (so the corresponding continuous distribution has variance $\sigma^2/(2\pi)$). Choose an integer $w \geq 2$ (e.g. $w = \lfloor \sqrt{n} \rfloor$) and let $G_w \subseteq \text{Gal} \setminus \{\text{id}\}$ be a uniformly random subset of size w , subject to the condition that $G_w \cup \{\text{id}\}$ is **not** a subgroup of Gal . Fix integers $a, b, c, d \in \mathbb{Z}$ such that $a^2 + b^2 + c^2 + d^2 = 4(m-1)$. Given a secret vector $\mathbf{s} = (s_2, \dots, s_m)^\top \in \mathcal{R}_q^{m-1}$ with all coefficients in $\{\pm 1\}$, the output vector $\mathbf{z} = (z_2, \dots, z_m)^\top \in \mathcal{R}_q^{m-1}$ is defined componentwise as*

$$\boxed{z_i = 2s_i \cdot \left(\underbrace{\sum_{j < i} E_{j,i} - \sum_{j > i} E_{i,j}}_{Z_i \in \mathcal{R}_q} \right) + \sum_{\tau \in G_w} \tau(s_i) \cdot (aU_{i,\tau}^{(1)} + bU_{i,\tau}^{(2)} + cU_{i,\tau}^{(3)} + dU_{i,\tau}^{(4)}) + 2s_i \cdot X_0 + e'_i,}$$

where the noise terms are sampled as follows:

- $E_{i,j}$ for $2 \leq i < j \leq m$, and $E_{j,i} = E_{i,j}$;
- X_0 ;
- $U_{i,\tau}^{(1)}, \dots, U_{i,\tau}^{(4)}$ for each i and $\tau \in G_w$;
- e'_i for each i .

All noise terms are sampled independently from $\chi_\sigma^\mathcal{R}$.

Compared with the full automorphism Construction B.1, here the sum runs only over a random subset G_w of the non-trivial automorphisms. A direct coefficient-domain computation gives

$$\tilde{\mathbf{B}}\tilde{\mathbf{B}}^\top = \text{diag}\left(4(m-1) \sum_{\tau \in G_w \cup \{\text{id}\}} \tau(s_2 s_2^*) + \mathbf{I}_n, \dots, 4(m-1) \sum_{\tau \in G_w \cup \{\text{id}\}} \tau(s_m s_m^*) + \mathbf{I}_n\right),$$

which is block-diagonal. Although this matrix is no longer a scalar multiple of the identity, recovering s_i from its Gram block amounts to solving a structured factorization problem that we formalize below.

Assumption 1 (GA-MLWE_{diag}). Let $\mathcal{R} = \mathbb{Z}[x]/(x^n + 1)$ with $n = 2^\kappa$, $\mathcal{R}_q = \mathcal{R}/q\mathcal{R}$ for a prime $q \equiv 1 \pmod{2n}$. Let $k, l \geq 1$ be integers and set $m =$

$k + l + 1$. Let $\chi_{\pm 1}$ be the distribution over $\mathcal{R}_q$ where each of the n coefficients is uniformly chosen from $\{-1, 1\}$. Let Gal be the Galois group of $\mathcal{R}_q$ over $\mathbb{Z}_q$, and let $G_w \subseteq \text{Gal} \setminus \{\text{id}\}$ be a subset of size w chosen uniformly at random, subject to the condition that $G_w \cup \{\text{id}\}$ is *not* a subgroup of Gal . For a vector $\mathbf{s} = (1, \mathbf{s}_{\text{gen}}, \mathbf{e}_{\text{gen}}) = (s_1, \dots, s_m)^\top \in \mathcal{R}_q^m$, define the *Gram block* of $s_i, i \geq 2$ (under G_w) as the integer matrix

$$\mathcal{G}_{G_w}(s_i) = 4(m-1) \sum_{\tau \in G_w \cup \{\text{id}\}} \mathbf{M}_\tau(s_i s_i^*) + \mathbf{I}_n \in \mathbb{Z}^{n \times n},$$

where $\mathbf{M}_\tau(a)$ denotes the $n \times n$ circulant matrix representing multiplication by $\tau(a)$ in the power basis.

The $\text{GA-MLWE}_{\text{diag}}$ *assumption* (an instance of the $\text{GA-MLWE}_{\mathcal{F}}$ framework of Section 3.2) states that for any PPT adversary $\mathcal{A}$ the following advantage is negligible in the security parameter:

$$\begin{aligned} \text{Adv}_{\mathcal{A}}^{\text{GA-MLWE}_{\text{diag}}}(\lambda) = & \left| \Pr[\mathcal{A}(\mathbf{A}, G_w, \mathbf{A}\mathbf{s}_{\text{gen}} + \mathbf{e}_{\text{gen}}, \{\mathcal{G}_{G_w}(s_i)\}_{i=2}^m) = 1] \right. \\ & \left. - \Pr[\mathcal{A}(\mathbf{A}, G_w, \mathbf{u}, \{\mathcal{G}_{G_w}(s'_i)\}_{i=2}^m) = 1] \right|, \end{aligned}$$

where

- $\mathbf{A} \leftarrow \mathcal{R}_q^{k \times l}$ uniformly,
- $\mathbf{s}_{\text{gen}}, \mathbf{s}'_{\text{gen}} \leftarrow \chi_{\pm 1}^l$,
- $\mathbf{e}_{\text{gen}}, \mathbf{e}'_{\text{gen}} \leftarrow \chi_{\pm 1}^k$,
- $\mathbf{s} = (1, \mathbf{s}_{\text{gen}}, \mathbf{e}_{\text{gen}}) = (s_1, \dots, s_m)^\top \in \mathcal{R}_q^m, \mathbf{s}' = (1, \mathbf{s}'_{\text{gen}}, \mathbf{e}'_{\text{gen}}) = (s'_1, \dots, s'_m)^\top \in \mathcal{R}_q^m$
- $\mathbf{u} \leftarrow \mathcal{R}_q^k$ uniformly,
- $G_w \subseteq \text{Gal} \setminus \{\text{id}\}$ is chosen uniformly at random among all subsets of size w that satisfy the non-subgroup condition stated above.

The probability is taken over the random choices of $\mathbf{A}, \mathbf{s}, \mathbf{s}', \mathbf{u}, G_w$ and the coins of $\mathcal{A}$.

Plausibility of $\text{GA-MLWE}_{\text{diag}}$. The $\text{GA-MLWE}_{\text{diag}}$ assumption interpolates between two extremes whose hardness is well understood, while explicitly avoiding known algebraic weaknesses.

The case $w = 0$. When $G_w = \emptyset$, each Gram block reduces to $\mathcal{G}_{\emptyset}(s_i) = 4(m-1) \mathbf{M}_{\text{id}}(s_i s_i^*) + \mathbf{I}_n$, i.e. the full circulant multiplication matrix of $s_i s_i^*$. In cyclotomic rings, recovering a ring element a from its self-adjoint product aa^* is known to be polynomial-time via algebraic factorization techniques [25, 28]. For our setting with secret coefficients in $\{\pm 1\}$, an attacker can factor the polynomial corresponding to $s_i s_i^*$ and efficiently reconstruct s_i . Hence $w = 0$ is **insecure** and must be excluded.

Small w and the limits of algebraic attacks. For $w = 1$, the Gram block reveals $s_i s_i^* + \tau(s_i s_i^*)$ (up to scaling). While Gentry–Szydło-style techniques can factor a single aa^* by exploiting the strong algebraic properties of cyclotomic rings, it is unclear whether they extend to a sum involving an independent automorphic conjugate. The sum destroys the simple product structure, and no known algorithm efficiently recovers s_i from such a sum without brute-forcing the $\{\pm 1\}$ coefficients. Nevertheless, because the theoretical understanding of this regime is still developing, very small values of w remain a gray area that warrants caution.

This situation is reminiscent of the Hawk signature scheme [21], which bases its security on the module LIP (lattice isomorphism problem). In Hawk, the adversary is given a *full* rank-2 Gram matrix $Q = B^*B$ of a 2×2 polynomial matrix B , complete with cross-correlation terms. The hardness of that problem, even for rank as low as 2, has resisted significant cryptanalytic effort, underlining how little is understood about structured Gram factorization in cyclotomic rings. Our $\text{GA-MLWE}_{\text{diag}}$ assumption can be viewed as a “rank- $(w + 1)$ ” variant where the Gram blocks are diagonal and cross-terms are absent.

Given these considerations, we recommend $w = \lfloor \sqrt{n} \rfloor$. With this choice, the Gram block is a sum of $\sqrt{n} + 1$ conjugates. Even if an attacker could factor small- w sums, the number of conjugates here is large enough to make the algebraic system highly over-determined yet non-linear in the unknown coefficients of s_i . Solving such a system in the $\{\pm 1\}$ -constrained setting has no known subexponential algorithm. Combined with the explicit exclusion of subfield-forming subsets G_w , this choice provides a comfortable security margin. In practice this reduces the number of automorphism columns from $4(m - 1)n$ to $4(m - 1)\sqrt{n}$, substantially decreasing the signature size and sampling cost.

C Proof of Construction 2

Theorem C.1. *If the $\text{GA-MLWE}_{\text{full}}$ assumption holds, the $\text{MSIS}_{m+1,k,2q,\beta}$ problem is hard, and $\sigma \geq \eta_\varepsilon(\mathbb{Z}^{nd})$ for some negligible ε , then Π_{th} is EUF-CMA secure in the random oracle model.*

Proof. We proceed via a sequence of games. Let $\mathcal{A}$ be an EUF-CMA adversary that makes at most q_s signing queries and q_h random oracle queries.

Game 0. This is the real EUF-CMA game.

Game 1 (Simulated signatures with real $\mathbf{G}$). Instead of using the secret key $\mathbf{B}$ to compute $\mathbf{z}_{\text{bottom}} = 2\mathbf{B}\mathbf{X}_{\text{full}} + \mathbf{e}_{\text{bottom}}$, the signing oracle now simulates the signature distribution using only the public Gram matrix $\mathbf{G} = \mathbf{B}\mathbf{B}^*$. Concretely:

1. Sample $z_1 \in \mathbb{Z}^n$ from the truncated discrete Gaussian $D_{\sigma_y}^n$ conditioned on $\|z_1\|_\infty \leq B_1$, identically to the real scheme.
2. For the bottom part $\mathbf{z}_{\text{bottom}}$, generate two independent components:
 - $\mathbf{V}_{\text{sim}} \sim D_{\mathbb{Z}^{nd}, 2\sigma}$. In the real protocol the masking noise is $\mathbf{e}_{\text{bottom}} \sim D_{\mathbb{Z}^{nd}, 2\sigma}$; hence this simulation is perfect (zero statistical distance). The choice of parameter 2σ guarantees

$$2\sigma \geq 2\eta_\varepsilon(\mathbb{Z}^{nd}) = \eta_\varepsilon(2\mathbb{Z}^{nd}),$$

so the modulo-2 distribution of $\mathbf{V}_{\text{sim}}$ is 2ε -close to uniform over all parity patterns. Consequently, any fixed shift added to $\mathbf{V}_{\text{sim}}$ also becomes 2ε -close to uniform modulo 2. This property is critical to absorb the secret-dependent parity information from the structured component.

- To simulate the structured component $\mathbf{U} = 2\mathbf{B}\mathbf{X}_{\text{full}}$ with $\mathbf{X}_{\text{full}} \sim D_{\mathbb{Z}^{nd}, \sigma}$, note that $\mathbf{B} \in \mathbb{Z}^{nd \times nd}$ is a unimodular integer matrix ($\det(\mathbf{B}) = 1$). Hence the kernel lattice $\Lambda^\perp(2\mathbf{B})$ is trivial, and the linear transformation lemma (Lemma 2.3) applies as soon as $\sigma \geq \eta_\varepsilon(\mathbb{Z}^{nd})$. Under this condition, the distribution of $2\mathbf{B}\mathbf{X}_{\text{full}}$ is statistically close (within $4\varepsilon + O(\varepsilon^2)$) to the rounded continuous Gaussian $\mathcal{N}(\mathbf{0}, \frac{4\sigma^2}{2\pi}\mathbf{G})$. The simulator therefore samples $\mathbf{U}_{\text{sim}}$ from this continuous Gaussian and rounds each coordinate to the nearest integer.

Set $\mathbf{z}_{\text{bottom}} = \mathbf{U}_{\text{sim}} + \mathbf{V}_{\text{sim}}$.

3. Assemble $\mathbf{z} = (z_1, \mathbf{z}_{\text{bottom}})$. If $\|\mathbf{z}\|_\infty > B$, restart.
4. Choose a random challenge $c \in \mathcal{C}$ and compute $\mathbf{w} = \mathbf{A}\mathbf{z} - q\mathbf{j}c \bmod 2q$. Program the random oracle so that $\text{Hash}(\mathbf{w}, \mu) = c$.
5. Output $(\mathbf{z}, c)$.

The component $\mathbf{V}_{\text{sim}}$ is perfectly simulated (zero statistical distance), and its parameter 2σ independently satisfies the parity-masking condition $2\sigma \geq \eta_\varepsilon(2\mathbb{Z}^{nd})$. The component $\mathbf{U}_{\text{sim}}$ deviates from the true $\mathbf{U}$ by at most $4\varepsilon + O(\varepsilon^2)$ under the condition $\sigma \geq \eta_\varepsilon(\mathbb{Z}^{nd})$, which also controls the rounding error of the continuous Gaussian approximation. Because $\mathbf{U}$ and $\mathbf{V}$ are independent in the real protocol, the statistical distance between the simulated and real $\mathbf{z}_{\text{bottom}}$ is bounded by the sum of the individual distances, i.e. $4\varepsilon + O(\varepsilon^2)$. The parity-masking property of $\mathbf{V}_{\text{sim}}$ ensures that the adversary cannot exploit any residual parity correlation; the 4ε bound already absorbs the contribution of the parity classes. The first component z_1 is identical, and the random oracle programming is consistent since c is uniform and independent. Hence

$$|\Pr[\mathcal{A} \text{ wins in Game 1}] - \Pr[\mathcal{A} \text{ wins in Game 0}]| \leq q_s \cdot \text{negl}(\lambda).$$

Game 2 (Replace LWE sample with uniform). (Identical to the original proof; omitted for brevity.)

Game 3 (Extracting an MSIS solution). (Identical to the original proof.)

Conclusion. Combining the three game transitions,

$$\text{Adv}_{\Pi_{\text{th}}}^{\text{EUF-CMA}}(\mathcal{A}) \leq q_s \cdot \text{negl}(\lambda) + \text{Adv}_{\text{GA-MLWE}_{\text{full}}}(\lambda) + \text{Adv}_{\text{MSIS}}(\lambda),$$

which is negligible. This completes the proof.

Remark on the practical scheme Π_{real} . Since Π_{real} omits $\mathbf{G}$ from the public key, an adversary against it has strictly less information. Any EUF-CMA adversary against Π_{real} can be trivially transformed into an adversary against Π_{th} by forwarding queries and ignoring the extra $\mathbf{G}$. Therefore Π_{real} inherits the same security guarantee.

Practical scheme Π_{real} (without Gram matrix). For deployment we simply omit $\mathbf{G}$ from the public key; the signing and verification algorithms remain identical.

$\text{KeyGen}_{\text{real}}(1^\lambda)$: run $\text{KeyGen}_{\text{th}}(1^\lambda)$ and output $\text{pk} = (\mathbf{A})$, $\text{sk} = (\mathbf{B}, \mathbf{s})$.

Theorem C.2. *Any EUF-CMA adversary against Π_{real} can be transformed into an EUF-CMA adversary against Π_{th} with the same success probability and a tight running-time overhead. Consequently, Π_{real} is EUF-CMA secure under the same GA-MLWE assumption.*

Proof. Given an adversary $\mathcal{A}$ against Π_{real} , we construct $\mathcal{B}$ against Π_{th} :

- $\mathcal{B}$ receives $\text{pk}_{\text{th}} = (\mathbf{A}, \mathbf{G})$ and forwards $(\mathbf{A})$ to $\mathcal{A}$.
- Signing queries are forwarded to the Π_{th} oracle; the responses are identically distributed.
- The forgery is forwarded directly.

The reduction is perfect because $\mathbf{G}$ is never used by $\mathcal{A}$.